

Unintended Risks of Bank Capital Regulation

Dongkeun Choi*

University of California, San Diego

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Abstract

I show that tighter bank capital regulation can increase aggregate credit risk by shifting credit supply toward nonbank financial intermediaries (NBFIs) that operate outside the scope of regulation. Using U.S. mortgage data, I document that NBFIs approve observably similar applicants at higher rates than banks and that NBFI loans have higher default rates. I show that tighter bank capital regulation contributed to a shift in credit toward riskier borrowers with higher default rates. I develop a general equilibrium model in which banks and NBFIs choose loan rates and underwriting standards. In the model, tighter regulation raises bank loan rates and tightens underwriting standards, shifting applicants toward NBFIs. NBFIs' looser underwriting standards extend credit to applicants who would otherwise remain unfunded, increasing aggregate credit risk. The model further shows that tighter regulation can amplify credit losses from adverse financial shocks.

*University of California, San Diego, email: dochoi@ucsd.edu

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1. Introduction

Banks fund their lending activities through deposits and equity capital. To enhance the resilience of credit intermediation, capital regulations require banks to hold sufficient equity buffers relative to their assets to absorb unexpected losses and protect depositors.

However, recent empirical evidence identifies a channel through which capital regulation affects the economy by reshaping credit markets. Buchak et al. (2018) and Irani et al. (2021) document that capital regulations contributed to the rise of nonbank lenders in the U.S. residential mortgage and syndicated corporate loan markets, respectively. The intuition is that when capital requirements constrain bank lending, credit can shift toward less-regulated intermediaries, in particular, nonbank financial intermediaries (henceforth, NBFIs), which provide credit without taking deposits and operate outside the scope of bank capital regulation.

The expansion of NBFIs has prompted a discussion of the effectiveness of capital regulation when banks and NBFIs coexist.¹ Existing work has focused on balance-sheet responses to tighter capital requirements, such as the leverage of banks and NBFIs (Begenau and Landvoigt 2022) and banks' failure risk (Dempsey 2025). However, this leaves open how the migration of credit toward NBFIs affects the risk composition of aggregate credit. If NBFIs apply looser lending standards and originate riskier credit, tighter capital regulation does not simply reallocate credit across intermediaries but can also increase aggregate credit risk, undermining its original goal of maintaining the resilience of credit intermediation.

This paper shows that tighter bank capital regulation can *increase* aggregate credit risk when banks and NBFIs differ in lending standards, contributing to the literature in three ways. First, I show that loan approval is a key margin that generates differences between bank and NBFI loans. Using the U.S. residential mortgage loan-level data, I document that NBFIs are more likely than banks to approve applications conditional on applicant characteristics, particularly among riskier applicants. Second, I show that tighter regulation contributed to a shift toward riskier credit and higher subsequent defaults. Using regional variation in exposure to the Basel III capital requirement revisions, measured by the pre-Basel III period market share of regulated banks, I document that regions more exposed to the regulation experienced a credit shift toward riskier borrowers and higher subsequent default rates. Third, I develop a general equilibrium model to suggest a mechanism that explains the empirical findings and, importantly, to conduct counterfactual analyses of the effectiveness of regulation under financial stress. The key innovation

¹General equilibrium analyses of capital requirements are extensive in settings where banks and NBFIs coexist, including Plantin (2015), Huang (2018), Ordonez (2018), Martinez-Miera and Repullo (2019), Donaldson et al. (2021), Farhi and Tirole (2021), Gebauer and Mazelis (2023), Darst et al. (2025), Lyonnet and Chretien (2025), and Xu (2025).

of the model is to allow financial intermediaries to adjust their underwriting standards. Tighter bank capital requirements lead banks to increase loan rates and tighten underwriting standards to reduce expected credit losses. These changes shift marginal applicants toward NBFIs, resulting in applicants who would not be approved by banks receiving credit from NBFIs under their looser underwriting standards. This reallocation brings otherwise unfunded risk into the aggregate loan pool and increases aggregate credit risk. In addition, the model shows that tighter capital requirements amplify the effects of exogenous increases in defaults, leading to greater credit losses and greater erosion of NBFI equity.

I start by establishing empirical evidence on differences in lending behavior between banks and NBFIs using U.S. residential mortgage data. This setting is particularly well suited to this analysis because I can trace the lending process from application and approval to origination and subsequent performance, allowing me to identify at which stage differences between banks and NBFIs emerge. I use two main datasets. The first is Home Mortgage Disclosure Act (HMDA) data, which contain mortgage applications along with borrower and loan characteristics. I complement these data with loan-level performance data provided by Government-Sponsored Enterprises (GSEs) and Ginnie Mae, which allow me to track subsequent loan performance.

I first show that NBFI loans exhibit higher default rates than bank loans and that most of this gap reflects differences in the borrower risk composition of originated loans. Buchak et al. (2018) find a small default gap among GSE mortgages originated between 2010 and 2015 and limited differences in borrower characteristics. Bosshardt et al. (2025) document a wider default gap in 2016 and 2017 GSE loans and greater NBFI borrower risk when characteristics are considered jointly. I use a broader and longer-term sample than those used in prior studies, covering 2010 to 2024 and including Ginnie Mae loans. This broader coverage captures post-regulation differences between the two sectors more comprehensively, including stress periods. I find that NBFI loans exhibit default rates that are 2 percentage points higher than those of bank loans. I further show that 87 percent of the gap can be explained by differences in observable borrower risk.

I next show that approval decisions are a key margin generating the differences between bank and NBFI loans. I find that NBFIs are around 5 percentage points more likely to approve applications than banks, relative to an average approval rate of 79 percent, after controlling for observable borrower and loan characteristics. More importantly, the approval gap increases with applicant risk, as NBFIs are increasingly more likely than banks to approve applicants as their risk rises. I provide further evidence by comparing applicant pools before and after screening. Bank and NBFI applicant pools do not differ significantly in income or a principal-component measure of borrower risk before screening. Among approved applications, however, the observable risk

gap is wider, suggesting that differences in borrower risk emerge through the approval process.²

As further corroboration, I document that tighter bank capital regulation contributes to a shift in the composition of newly originated mortgages toward riskier borrowers and higher subsequent default rates. Following the approach in Buchak et al. (2018), I exploit regional heterogeneity in exposure to the tighter capital requirements introduced under Basel III and proxy regulatory exposure using the pre-Basel III period market share of regulated banks. Using a dynamic difference-in-differences design across MSAs, the resulting estimates indicate a shift toward riskier newly originated credit in more exposed MSAs, reflected in both higher observable borrower risk and higher subsequent default rates. I further show that regulated banks exhibit lower approval rates following tighter capital regulation, while NBFI approval rates remain broadly unchanged. At the same time, the NBFI share of *applications* increases in more exposed regions. This indicates that the reallocation toward NBFIs operates through both a shift in applications and tighter bank lending standards.

I develop a model consistent with the empirical facts to study the effects of tighter bank capital regulation on aggregate credit risk in a counterfactual financial stress scenario when banks and NBFIs jointly supply credit. I build on the saver–borrower general equilibrium framework of Curdia and Woodford (2010) and Eggertsson et al. (2024) by allowing banks and NBFIs to supply credit and jointly choose loan rates and underwriting stringency. The most closely related theoretical framework is provided by Martinez-Miera and Repullo (2019). In their model, financial intermediaries choose screening effort, which affects project success. The capital held by a lender commits it to screen, and the intensity of screening determines loan risk. In my model, underwriting instead determines whether an application becomes a loan and whether the underlying credit risk can materialize. It also affects the volume of loan applications, since borrowers choose where to apply based on approval probabilities as well as loan rates.

In my model, banks and NBFIs differ along two institutional dimensions. The first is their funding structures. As Drechsler et al. (2017) show, banks obtain deposit funding at rates below the risk-free rate because they exercise market power in deposit markets. By contrast, NBFIs rely on money-market funding, which is more expensive than deposit funding. I additionally introduce a reduced-form wedge that governs the extent to which each financial intermediary internalizes its default risk. This wedge captures differences arising from their funding sources and franchise values. Banks place greater weight on avoiding default because they have more franchise value associated with their deposit funding. NBFIs, in contrast, place less weight on these losses. In the

²This evidence does not rule out sorting on omitted or unobserved characteristics, which Agarwal et al. (2024) and Bhutta et al. (2026) highlight. The evidence instead shows that differences in observable borrower risk become more pronounced after approval decisions are made.

calibrated equilibrium, NBFIs therefore internalize default risk less than banks, leading them to apply looser lending standards. The second difference is regulatory exposure. Capital requirements apply only to banks and enter the model as an additional cost of lending, following Balloch and Koby (2023) and Buchak et al. (2025).

On the borrowing side, the representative borrower household consists of a continuum of members who face idiosyncratic application preferences that capture non-pricing factors (e.g., convenience and familiarity). After these preferences are realized, each member decides where to apply based on loan rates and approval probabilities, without knowing whether the application will be approved.³ In the calibrated equilibrium, banks offer lower loan rates but exhibit lower approval probabilities, whereas NBFIs offer higher loan rates but easier access to credit. Because neither offer dominates along both dimensions, each member chooses where to apply by considering both borrowing costs and approval probability against application preferences. Once applications are submitted to either sector, each intermediary evaluates its applicant pool according to its own underwriting policy, and rejected applicants fail to obtain credit. At the repayment date, each borrower decides whether to repay or default after the income shock is realized.

Tighter capital requirements create two opposing forces that affect aggregate credit risk. The first operates through banks. Higher capital requirements raise the marginal cost of bank lending, leading banks to increase loan rates. Higher loan rates increase banks' payoff when borrowers repay, but they also make defaults more costly. Banks therefore tighten their lending standards to reduce expected default. As a result, banks receive fewer applications and make fewer but safer loans. This reduces their contribution to the aggregate credit risk.

The second force operates through how applications are allocated across intermediaries. As capital requirements tighten, banks raise loan rates and lower approval probabilities, making bank loans less attractive. This leads some applicants who chose banks under the low-regulation regime to shift to NBFIs. Among these applicants, the key group is borrowers whose risk falls between the lending standards of the two sectors. Under the low-regulation regime, these borrowers applied to banks but were rejected.⁴ After tighter regulation, the bank offer becomes less attractive, so they shift to NBFIs and obtain credit because they satisfy the NBFI underwriting rule. As a result, risks that would otherwise remain unfunded enter the originated loan pool. This force not only raises the NBFI share of credit but also the aggregate credit risk.

³In the baseline model, application friction shocks are independent of creditworthiness of each member. Therefore, application volume responds to lenders' loan rates and approval probabilities, while the underlying distribution of borrower creditworthiness remains invariant. Section 6.4 relaxes this assumption.

⁴The model abstracts from within-period reapplication. Section 4.3.3 presents empirical evidence suggesting that reapplication is costly rather than frictionless. Section 6.4 shows that allowing costly reapplication weakens but does not overturn the main results.

I calibrate the model to match the credit allocation and performance of banks and NBFIs in the U.S. mortgage market. The calibrated model predicts that tighter capital regulation increases the aggregate default rate, implying that the increase in risk from the shift of applicants toward NBFIs is large enough to offset the reduction in risk from tighter bank underwriting. This result also holds when aggregate defaults are measured as defaulted principal, which takes into account changes in total credit. Although aggregate credit declines because NBFI loans do not fully replace bank loans, the decline in credit quality is large enough to outweigh this reduction in lending.

I then evaluate whether tighter bank capital regulation dampens shock amplification during a crisis using the model. I subject the model to a financial stress scenario in which default rates rise and the net worth of both banks and NBFIs is exogenously eroded. In this experiment, tighter bank capital requirements initially achieve their intended goal by limiting the erosion of bank equity. However, aggregate credit losses increase, and NBFI equity erosion rises and remains persistent. The erosion and credit losses are amplified as capital regulation becomes tighter. This amplification of the shock on credit losses is due to the initially larger share of NBFI lending under looser lending standards. The shock is also amplified as funds flow from money markets into deposits as the banking sector becomes safer, while the erosion of NBFI net worth remains persistent. This implies that while capital regulation can achieve its goal when the evaluation focuses on regulated banks, credit losses could be amplified and the failure probability of unregulated NBFIs could increase when we focus on their equity erosion.

This unintended deterioration in credit outcomes under tighter bank capital regulation arises from differences in underwriting standards between banks and NBFIs in the first place. Tighter capital requirements lead banks to tighten underwriting, while NBFI standards remain unchanged, widening the gap between the two sectors. At the same time, more applicants shift to NBFIs, so more credit is evaluated under looser standards. These results suggest that tighter bank regulation should be accompanied by underwriting requirements for NBFIs. Under the counterfactual policy, NBFI underwriting standards become stricter as bank capital requirements tighten. The policy reduces the gap between the two sectors' lending standards and limits the increase in risky lending that would otherwise follow from the shift of borrowers toward NBFIs. Importantly, the complementary policy does not materially change the reduction in bank equity losses achieved by tighter bank capital regulation. The policy implication is that bank capital regulation should be complemented by a policy that directly reduces the underwriting gap between banks and NBFIs.

Roadmap. The remainder of the paper proceeds as follows. Section 2 documents differences between banks and NBFIs in mortgage applications, approval decisions, originated borrower risk, and subsequent loan performance, and examines how these differences changed following tighter bank capital regulation. Section 3 develops a static intermediary problem in which a

financial intermediary jointly chooses its underwriting stringency and loan rate. Section 4 embeds banks and NBFIs in a general equilibrium framework with borrowers and savers. Section 5 calibrates the model to sector-level moments. Section 6 quantifies the effects of tighter bank capital requirements in steady state, and Section 7 quantifies them under financial stress. Section 8 evaluates a complementary minimum screening requirement for NBFIs. Section 9 briefly discusses why credit risk still matters even after securitization is taken into account, and Section 10 concludes.

2. NBFi Lending and Its Association with Capital Regulation

While credit intermediation has shifted from banks to NBFIs across both household and nonfinancial business credit markets in the U.S.,⁵ this paper focuses on the residential mortgage market. Mortgage data offer several advantages over other credit markets data. First, the data cover the credit creation process, from applications and approvals through subsequent performance and default, which allows me to identify at which stage banks' and NBFIs' loans exhibit differences. Second, they provide detailed information on borrowers and loan products as well as the lender information.

Data. The data used in this paper consist of publicly available Home Mortgage Disclosure Act (HMDA) data and loan-level performance data from Fannie Mae, Freddie Mac, and Ginnie Mae. HMDA data cover roughly 90 percent of U.S. mortgage originations and are used to analyze the application stage, including approval and origination outcomes. The loan-level datasets from Fannie Mae, Freddie Mac, and Ginnie Mae together cover roughly 65 percent of U.S. first-lien mortgage originations, provide monthly loan performance histories, and are used to analyze the relationship between borrower characteristics and ex post loan performance. The sample period runs from 2010 through 2024.

2.1. NBFi Loans Exhibit Higher Ex Post Default Rates

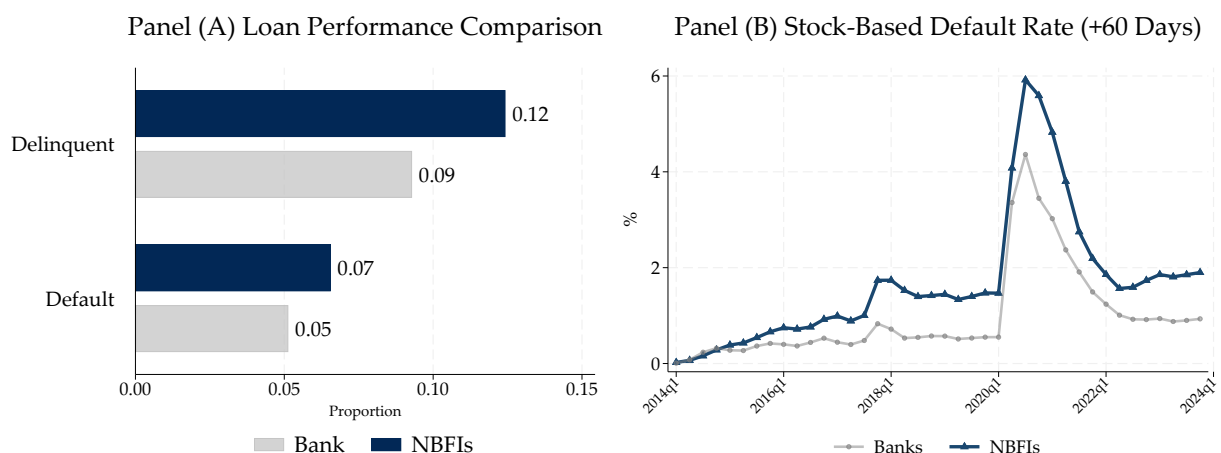
First, I examine whether loans originated by banks and NBFIs differ in terms of ex post performance. To compare ex post loan performance across originator types, I use loan-level performance data from Freddie Mac, Fannie Mae, and Ginnie Mae. These datasets track mortgage outcomes at a

⁵This shift extends beyond mortgage markets. In consumer credit, Fintech lenders and finance companies account for large shares of personal lending, with delinquency rates exceeding those of banks. In 2022, delinquency rates on unsecured personal loans were 1.89 percent for depository institutions, 3.38 percent for Fintech lenders, and 7.01 percent for finance companies. In business credit, private credit, defined as debt provided by NBFIs to private businesses, has grown rapidly to a market comparable in size to leveraged loans and high-yield bonds.

monthly frequency that allows me to construct measures of subsequent loan performance. The main sample includes loans originated from 2014 onward, since Ginnie Mae loan-level performance data are publicly available beginning in 2014.

The two primary measures of ex post loan performance are delinquency rate and default rate. Following Agarwal et al. (2012), delinquency is defined as a loan becoming 30 or more days past due within two years of origination, while default is defined as a loan becoming 60 or more days past due within two years of origination.

FIGURE 1. Ex Post Loan Performance by Lender Type



Notes. Delinquency is defined as becoming 30 or more days past due within two years of origination, while default is defined as becoming 60 or more days past due within two years of origination (Agarwal et al. 2012). The left panel reports the average loan-level probability of default. The right panel reports monthly default rates by originator type for mortgages originated from 2014Q1 onward, tracking loan performance over time among loans observed in each month. The sample consists of loans sold to Freddie Mac, Fannie Mae, or placed into Ginnie Mae pools. *Sources.* Single-Family Loan-Level Datasets provided by Freddie Mac, Fannie Mae, and Ginnie Mae.

The left panel of Figure 1 shows that NBFIs loans exhibit higher delinquency and default rates than bank loans on average. Specifically, the delinquency and default rates of NBFIs loans are 3 and 2 percentage points higher, respectively, than those of bank loans. The right panel of Figure 1 reports calendar-time default rates. The sample includes loans originated in the first quarter of 2014 or later, and the default rate in this panel measures the share of loans that are 60 or more days past due among those that remain outstanding and observed in the panel. Because prepaid loans exit the performance panel, differences in prepayment between bank loans and NBFIs loans can change the composition of loans observed over time. Nevertheless, the measure helps show when the performance gap between bank loans and NBFIs loans becomes particularly large. Default rates rise gradually after 2014, increase sharply around 2020, and decline thereafter. The gap widens again during the policy rate hike period. This highlights that NBFIs loans perform worse, especially during recessionary episodes. The same pattern emerges when I use 30-day delinquency instead

of 60-day default.

2.1.1. Where Does the Default Gap Come From?

NBFI loans may exhibit higher default rates either because NBFIs extend credit to riskier borrowers or because their loans perform worse conditional on the same observable borrower characteristics. To assess whether the higher default rate of NBFI loans reflects differences in borrower composition or worse performance conditional on observed risk, I decompose the NBFI-bank default gap using a Blinder–Oaxaca decomposition (Blinder 1973; Oaxaca 1973).

Let R_i denote the risk index for loan i . I construct R_i as the first principal component of credit score, loan-to-value (LTV) ratio, and debt-to-income (DTI) ratio, all of which are known to be strong observable predictors of loan performance.⁶ The index is normalized so that higher values correspond to riskier underwriting profiles, lower credit scores and higher LTV and DTI ratios jointly. Then, let $D_i \in \{0, 1\}$ indicate whether loan i defaults. For each lender type $j \in \{B, NB\}$, where B denotes banks and NB denotes NBFIs, define

$$F_j(r) = \Pr(R_i \leq r \mid j), \quad d_j(r) = \mathbb{E}[D_i \mid R_i = r, j].$$

Here, $F_j(r)$ is the distribution of the underwriting risk index for loans originated by lender type j , and $d_j(r)$ is the default probability conditional on risk index r and lender type j . Finally, the average default rate for lender type j , denoted by δ_j , can be expressed as

$$\delta_j = \mathbb{E}[D_i \mid j] = \int d_j(r) dF_j(r).$$

Then, the difference in average default rates between the two lender types can be decomposed into two components as follows.

$$\delta_{NB} - \delta_B = \underbrace{\left(\int d_{NB}(r) dF_{NB}(r) - \int d_B(r) dF_{NB}(r) \right)}_{\text{Conditional-performance Effect}} + \underbrace{\left(\int d_B(r) dF_{NB}(r) - \int d_B(r) dF_B(r) \right)}_{\text{Composition Effect}} \quad (1)$$

The conditional-performance effect captures differences in default probabilities conditional on the same observed risk, holding the risk distribution fixed at the NBFI distribution. The composition

⁶Bosshardt et al. (2025) regress a default indicator on bins of credit scores, LTV ratios, and DTI ratios and construct their risk index using the fitted values. In contrast, I use principal component analysis to construct an index directly from the joint variation in these underwriting characteristics without relying on ex post loan outcomes.

effect captures the portion of the default-rate gap attributable to differences in the risk-index distributions of loans originated by NBFIs and banks.

The decomposition indicates that borrower composition accounts for most of the NBFI-bank default gap. Differences in the underwriting risk-index distribution explain 86.8 percent of the gap, while the conditional-performance component accounts for the remaining 13.2 percent. This residual is small, suggesting that NBFI loans do not perform substantially worse than bank loans once observed risk is held fixed. This finding is consistent with Fuster et al. (2019), who show that differences in screening on unobservables between banks and NBFIs are limited, as differences in default rates become small after controlling for observable borrower characteristics. In other words, the higher default rate of NBFI loans is driven primarily by the fact that NBFIs' portfolio consists of a larger share of riskier loans.

2.2. NBFI Loans Have Riskier Observable Profiles

Given the evidence above that differences in the quality of approved loans account for the majority of the default-rate gap between NBFI loans and bank loans, I provide direct evidence by comparing the distributions of observable risk of the loans originated by each type of lender.

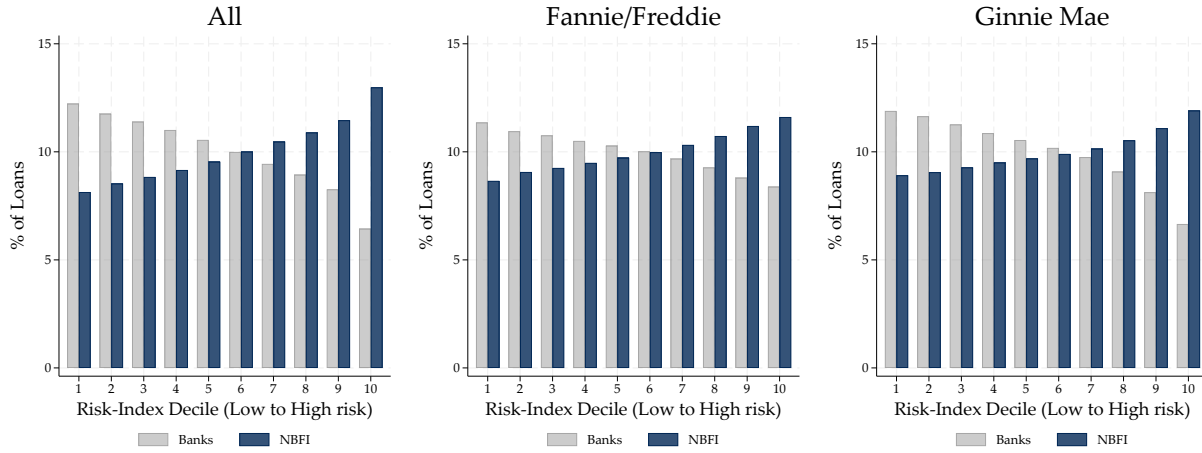
Previous evidence on differences in observable risk between bank and NBFI loans varies across periods and market segments. Demyanyk and Loutskina (2016) show that, before the financial crisis, mortgage-company affiliates of banks originated loans to riskier borrowers than their depository affiliates and experienced higher subsequent default rates. In the post-crisis market, Fuster et al. (2019) compare banks and NBFIs using HMDA data and report that NBFIs originate loans to borrowers with lower income and higher loan-to-income ratios than banks. Buchak et al. (2018) use Fannie Mae and Freddie Mac conforming loans and find that NBFI borrowers have slightly lower FICO scores and higher DTI, but also lower LTV, making the overall risk comparison ambiguous when underwriting variables are examined separately.

Examining each characteristic separately is informative, but it may miss differences in borrowers' overall underwriting profiles. For example, a borrower with a lower credit score, a higher DTI ratio, and a higher LTV ratio may have little effect on the average of any one characteristic. However, when these characteristics are considered together, the same borrower may have a riskier underwriting profile.

To maintain consistency with the default gap decomposition analysis in Section 2.1, I use the same PCA-based risk index, constructed as the first principal component of credit score, DTI, and LTV, and compare the distribution of originated loans across risk levels between banks and NBFIs.

Figure 2 reports the distribution of originated loans across risk-index deciles, separately for

FIGURE 2. Share of Lending by Risk Decile Within Banks and NBFIs



Notes. The risk index is constructed using principal component analysis of credit scores, DTI, and LTV, and is normalized so that higher values correspond to riskier underwriting profiles. Loans are sorted into deciles of the risk index. The figure shows the distribution of originated loans across risk-index deciles, separately for banks and NBFIs. The shares are computed within lender type, so each series sums to one across deciles. Each title represents the loans used to construct the figure. *Sources.* Single-Family Loan-Level Datasets provided by Freddie Mac, Fannie Mae, and Ginnie Mae, and author’s calculations.

banks and NBFIs. The first panel shows the results using the full sample, while the remaining two panels report results separately for GSEs and Ginnie Mae loans. Bank loans are more concentrated in lower-risk deciles, whereas NBFIs loans are more concentrated in higher-risk deciles. As the risk decile increases, the share of bank loans declines steadily, while the share of NBFIs loans rises. This divergence is not readily apparent from a comparison of average borrower characteristics. It shows that NBFIs originate relatively more loans to borrowers with lower credit scores, higher DTI ratios, and higher LTV ratios. As the rest of the panels show, the main results are not driven by Ginnie Mae loans, which target low-income applicants. After dividing the sample into two groups, GSE loans and Ginnie Mae loans, the same pattern can be identified.

Robustness Checks. To connect the literature that examines individual underwriting characteristics separately across lender types, Appendix A reports univariate comparisons of credit scores, DTI ratios, and LTV ratios. These distributional comparisons show that NBFIs loans are more concentrated among borrowers with lower credit scores, higher DTI ratios, and higher LTV ratios in both the GSE and Ginnie Mae samples. This pattern is particularly pronounced in the Ginnie Mae sample.

2.3. NBFIs Approve Applications at Higher Rates Conditional on Applicant Observables

A natural next question is at which stage the riskier composition of NBFI loans emerges. One possibility is that NBFIs are more likely to approve an application than banks conditional on observable borrower and loan characteristics. Another is that NBFIs attract a riskier pool of applicants *before* approval decisions are made. Distinguishing between these two channels is important because they imply different mechanisms behind the riskier composition of NBFI lending. If the difference arises at the approval stage, lenders' underwriting decisions are key margins. If it is already present in the applicant pool, however, borrowers' sorting matters to understand how the credit risks are realized in the economy.

I first examine whether otherwise similar applicants face different approval probabilities depending on whether they apply to a bank or an NBFI. I then analyze the distribution of *applicants* processed by each lender prior to approval to assess the extent to which borrower sorting further contributes to the riskier composition of NBFI loans.

2.3.1. Approval Rate Gap between Banks and NBFIs

To quantify differences in approval propensities, I estimate an application-level linear probability model in which an approval indicator is regressed on lender type and application characteristics, using HMDA data from 2018 to 2024.⁷ The estimated regression model is

$$\mathbb{I}(\text{Approved}_{l,t}) = \beta \cdot \mathbb{I}(\text{NBFI}_l) + \Gamma' X_{l,t} + \alpha_{c(l),t} + \varepsilon_{l,t}, \quad (2)$$

where $\mathbb{I}[\text{Approved}_{l,t}]$ equals one if application l in year t is approved,⁸ and $\mathbb{I}[\text{NBFI}_l]$ equals one if the lender processing the application is an NBFI. The coefficient of interest is β . It measures the NBFI approval differential relative to banks after conditioning on borrower, loan, and local-market characteristics. The vector $X_{l,t}$ includes LTV, DTI, income, loan amount, loan purpose, and applicant demographic characteristics, including gender, ethnicity, and age. County-by-year fixed effects $\alpha_{c(l),t}$ absorb local demand conditions and time-varying regional shocks common to lenders operating in the same county and year. I separately define high-risk applications as those

⁷Credit scores are not available in HMDA, and detailed borrower characteristics are available only starting in 2018. Including LTV in the risk index yields similar results. I also obtain similar results for 2010–2017 using an index based only on income and loan size, which are available throughout that earlier period.

⁸An application enters the approval denominator when the HMDA action taken code equals 1, 2, or 3, where 1 denotes a loan originated, 2 denotes an application approved but not accepted by the applicant, and 3 denotes an application denied. The approval indicator equals one when the action taken code is 1 or 2. The results are nearly unchanged when I restrict the sample to action taken codes 1 and 3, following Gete and Reher (2021).

with DTI ratios of at least 45 percent and LTV ratios of at least 70 percent to examine whether approval rates differ across lender types within the higher-risk applicant pool.⁹

TABLE 1. NBFIs Exhibit Higher Approval Rates

	Conforming						All	
	Conventional		FHA		All		(7) Full	(8) High-Risk
	(1) Full	(2) High-Risk	(3) Full	(4) High-Risk	(5) Full	(6) High-Risk		
$\mathbb{I}[NBF]$	0.049*** (0.009)	0.067*** (0.011)	0.035*** (0.009)	0.055*** (0.010)	0.038*** (0.009)	0.114*** (0.012)	0.048*** (0.009)	0.081*** (0.011)
Log Loan-to-Value	-0.024*** (0.003)	-0.264*** (0.036)	0.034** (0.017)	-0.194*** (0.065)	-0.037*** (0.005)	-0.091*** (0.021)	-0.029*** (0.004)	-0.261*** (0.033)
Log Debt-to-Income	-0.227*** (0.009)	-2.685*** (0.031)	-0.117*** (0.011)	-1.099*** (0.026)	-0.210*** (0.008)	-1.706*** (0.034)	-0.213*** (0.009)	-1.928*** (0.033)
Log Income	0.027*** (0.003)	0.031*** (0.004)	0.071*** (0.012)	0.118*** (0.014)	0.037*** (0.003)	0.074*** (0.005)	0.027*** (0.003)	0.070*** (0.005)
Average Approval Rate	0.781	0.526	0.791	0.781	0.789	0.670	0.791	0.669
Fixed Effect	✓	✓	✓	✓	✓	✓	✓	✓
Observations	30,758,023	3,567,007	5,122,992	2,359,968	38,645,573	6,792,373	40,533,494	6,989,249
Adj. R-squared	0.160	0.495	0.130	0.177	0.144	0.328	0.141	0.351

Notes. Standard errors are two-way clustered at the county and lender levels and are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns (1)–(6) restrict the sample to conforming loans, split into conventional loans, FHA loans, and all loan types pooled. Columns (7)–(8) use all loans without the conforming restriction. Full refers to all applications. High-Risk refers to applications with $DTI \geq 45$ and $LTV \geq 70$. Fixed Effect captures loan purpose, applicant demographic characteristics (gender, ethnicity, age), and county $\times$ year. *Sources.* HMDA and author's calculations.

Table 1 shows that NBFIs approve observably comparable applications at higher rates than banks. In the conventional conforming sample, NBFI applications are 4.9 percentage points more likely to be approved than otherwise similar bank applications. The approval gap increases to 6.7 percentage points in the high-risk subsample. This suggests that even in segments of the mortgage market subject to secondary-market purchase rules, NBFIs exhibit higher approval rates than banks. For FHA loans, shown in columns (3) and (4), the corresponding gaps are 3.5 and 5.5 percentage points. When FHA and other government-insured loans are included in the conforming sample, reported in columns (5) and (6), the gap is 3.8 percentage points in the full sample and rises to 11.4 percentage points among high-risk applicants. The pattern remains when jumbo loans are also included, with approval gaps of 4.8 and 8.1 percentage points in the full and high-risk samples, respectively.

These results indicate that lender-type differences are already visible at the approval stage,

⁹DTI and LTV capture two important dimensions of mortgage credit risk, repayment capacity and borrower leverage, both of which are closely related to mortgage default risk. The 45 percent DTI cutoff also corresponds to a salient threshold in GSE underwriting guidelines.

after conditioning on observed applicant characteristics and local market conditions. The evidence therefore points to approval behavior as an important margin through which the riskier composition of NBFIs loans is formed. In particular, the higher approval rates for observably similar applications imply that the risk differences in originated loans are not driven solely by differences in the applicant pool.

One concern is that HMDA does not report credit scores. However, if NBFIs applicants have lower credit scores than bank applicants conditional on the included observables, omitting credit scores would bias the estimated approval gap downward because lower credit scores reduce approval probabilities. Thus, this type of unobserved sorting cannot explain the positive approval gap.

NBFIs instead appear more willing than banks to translate comparable observable applicant risk into approved credit. This difference becomes larger among high-risk applicants, suggesting that NBFIs are particularly more likely to approve riskier applications. To examine this pattern more closely, Appendix B shows that the NBFIs approval gap generally increases across observable risk deciles and is largest among riskier applicants.

2.3.2. Observable Applicant Composition and Approval Stage Divergence

An additional margin that may contribute to the riskier composition of NBFIs loans is differences in the characteristics of applicants across the two sectors. Beyond NBFIs' higher approval propensity, their originated loans may appear riskier simply because riskier borrowers are more likely to apply to NBFIs in the first place.

To examine this margin, I use the same HMDA data as in the approval analysis and compare the observable characteristics of applicants across lender types *before* approval decisions are made. These characteristics correspond to the borrower and loan controls used in the approval rate regressions. If NBFIs applicants are riskier before approval, differences in who applies to NBFIs and banks may account for part of the higher-risk composition of NBFIs originations, in addition to differences in approval decisions.

Observable applicant risk is summarized using principal component analysis, following the same approach used for originated loans. The index is based on income, DTI, and loan amount, and is normalized so that higher values indicate riskier applicant profiles. I standardize the index using the full application sample, including both bank and NBFIs applications and both approved and rejected applications. I then use this same measure to compare applicants in the full sample and among approved applications.

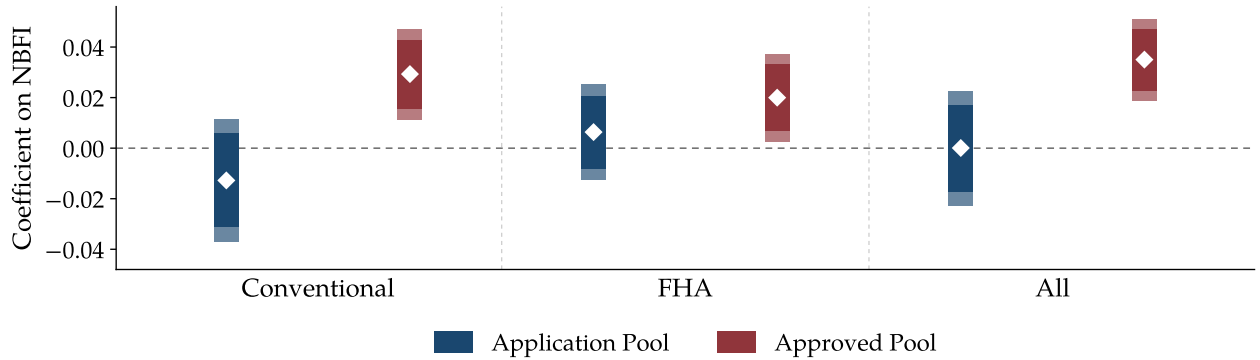
I test whether applicant pools differ statistically after accounting for geography, loan type,

and time. For each applicant risk proxy, I estimate

$$Proxy_{l,c,t} = \beta \times \mathbb{I}[\text{NBFI}_l] + \alpha_{c,t} + \varepsilon_{l,c,t}, \quad (3)$$

where $Proxy_{l,c,t}$ denotes the applicant risk proxy for application l in county c , and year t . $\mathbb{I}[\text{NBFI}_l]$ equals one for applications submitted to NBFIs and zero for applications submitted to banks. The specification absorbs county-by-year fixed effects and standard errors are clustered at the county level. The coefficient β measures whether applicants to NBFIs differ systematically from applicants to banks along observable risk, conditional on these controls.

FIGURE 3. Differences in Applicant Risk by Lender Type



Notes. The figure reports the coefficient on the NBFI indicator from regressions of the applicant risk proxy on the NBFI indicator. Blue estimates are based on all applications, while red estimates are based on approved applications. Vertical bars indicate confidence intervals, with darker and lighter shades corresponding to the 95% and 99% confidence intervals, respectively. All regressions absorb county-by-year fixed effects, and standard errors are clustered at the county level. *Sources.* HMDA and author's calculations.

Figure 3 plots the estimated coefficient on $\mathbb{I}[\text{NBFI}_l]$ across market segments, including conventional loans, FHA loans, and the full market. The results are similar across segments. At the application stage, which includes both approved and rejected applications, the estimated NBFI-bank difference in applicant risk is close to zero. By contrast, among *approved* applications, the difference shifts toward higher risk for NBFIs. This pattern suggests that the risk gap emerges at the approval stage rather than from observable sorting in the applicant pool.

Robustness Checks. As robustness checks, I examine the full distribution of the applicant risk proxy by lender type and year across mortgage market segments, including the conventional loans and FHA loans. To extend the time horizon, I repeat the exercise using real income as the risk proxy, for which the underlying data are available over the full 2010–2024 sample. In both exercises and across all market segments, the distribution of NBFI applicants largely overlaps with that of bank applicants, indicating that NBFI applicants do not appear observably riskier than bank applicants at the application stage. Results are provided in Appendix B.

Taken together, the application-stage evidence shows limited differences in observable applicant characteristics across banks and NBFIs and points to approval decisions as an important margin. This evidence does not rule out sorting on omitted or unobserved characteristics. Rather, it highlights that, along the characteristics observed in HMDA, the two applicant pools look similar before screening, while differences become more pronounced after approval decisions.

2.4. Capital Requirements and NBFIs' Lending

The preceding sections show that NBFIs extend credit to riskier borrowers under more lenient underwriting standards and that their loans subsequently perform worse. Prior work by Buchak et al. (2018) and Irani et al. (2021) shows that tighter bank capital regulation contributed to the growth of NBFI lending. This raises a natural question of whether the expansion of NBFI lending following tighter capital regulation merely changes who supplies credit or also changes the credit risk of borrowers receiving loans and the subsequent performance of those loans. Such changes are not mechanical. If NBFIs simply take the borrowers who would otherwise have borrowed from banks before banks' lending capacity becomes constrained, tighter capital regulation may simply shift lending from banks to NBFIs without changing borrower credit quality or subsequent loan performance. If, instead, the expansion of NBFI lending is accompanied by a deterioration in borrower credit quality and subsequent loan performance, then tighter capital regulation generates a composition effect, shifting credit toward riskier borrowers rather than merely reallocating credit across lenders and the provided total credit.

I estimate changes around the regulatory tightening in NBFI market share, borrower risk, and subsequent default rates among newly originated loans. I then examine how approval rates at regulated banks and NBFIs respond.

2.4.1. Tighter Capital Regulation Contributed to Higher Borrower Risk and Default Rates

I examine whether tighter capital regulation also affects borrower risk and subsequent default rates, building on Buchak et al. (2018) by exploiting regional variation in exposure to Basel III capital regulation.¹⁰ I use the pre-Basel III local market shares of banks exposed to the Basel III capital requirements to measure regional exposure to the regulatory tightening. I then estimate dynamic difference-in-differences specifications comparing MSAs with greater and lesser exposure.

¹⁰Buchak et al. (2018) and Fuster et al. (2019) highlight fintech innovation as a driver of nonbank growth, while Sarto and Wang (2026) point to the secular decline in bank profitability as another contributing factor.

Regulation Exposure Measure. To measure local exposure to the U.S. Basel III capital reform, I define $\mathbb{I}[\text{Regulated}_b]$ as an indicator equal to one if bank b was subject to the Basel III capital requirements announced by the U.S. banking agencies on June 7, 2012, and zero otherwise. I classify banks according to the applicability of the rules specified in the June 7, 2012 proposal and then aggregate this bank-level indicator using each bank's pre-reform mortgage market share within an MSA. Let $s_{mb,2008}$ denote bank b 's share of mortgage originations in MSA m in 2008 following Buchak et al. (2018).¹¹ I define MSA-level regulatory exposure as

$$\text{Regulation}_m = \sum_{b \in \mathcal{B}} s_{mb,2008} \times \mathbb{I}[\text{Regulated}_b]. \quad (4)$$

Thus, Regulation_m is the share of mortgage originations in MSA m in 2008 accounted for by banks classified as exposed to the tighter capital requirements and captures cross-sectional variation in local mortgage markets' pre-Basel III reliance on these banks.

Dynamic Difference-in-Difference Specification. I then estimate the following event-study specification

$$Y_{m,t} = \sum_{\tau=2010Q1, \tau \neq 2012Q2}^{2019Q4} \beta_{\tau} \times \mathbb{I}[t = \tau] \times \text{Regulation}_m + \mathbf{X}'_{m,t} \Gamma + \alpha_m + \delta_t + \varepsilon_{m,t}, \quad (5)$$

The dependent variable $Y_{m,t}$ is measured at the MSA-quarter level and includes the NBFi market share, the average borrower risk index, and the default rate. I construct these measures using GSE data for newly originated loans containing information on borrower characteristics, loan amounts, and subsequent loan performance. The reason for using GSE data is that borrower and loan characteristics at origination can be consistently linked to subsequent loan performance, allowing the three measures to be constructed for the same loan cohorts and matched over time and across regions. The NBFi market share is defined as the share of GSE loans originated by NBFis in MSA m during quarter t . The borrower risk index is the average risk index among loans originated in the MSA-quarter, where the index is constructed as the first principal component of credit score, DTL, and LTV, as in the previous section, and normalized so that higher values indicate greater ex ante borrower risk. The loading on the index is negative for credit score and positive for all other metrics, such that a higher value of the index captures the joint characteristics associated with greater borrower risk. Finally, the default rate is measured based on newly originated loans. Specifically,

¹¹I use 2008 market shares to mitigate concerns about anticipation effects. Discussions of tighter capital regulation began around 2010, and the Basel III capital rules were formally proposed in June 2012. Market shares measured closer to these dates may therefore already reflect lenders' responses to the anticipated regulatory changes. As a robustness check, I re-estimate the model using market shares from 2010, 2011, and 2012 and get similar results.

for loans originated in MSA m during time t , I calculate the share that subsequently defaults. Thus, the default rate reflects the subsequent performance of each quarter’s loan originations rather than the performance of the outstanding loans in time t .

The specification includes MSA fixed effects, α_m , to absorb differences across local mortgage markets, and quarter fixed effects, δ_t , which control for aggregate macroeconomic shocks common to all regions. The identifying assumption is that, conditional on the controls, MSAs with different pre-Basel III exposure would have followed similar outcome trends in the absence of the regulatory change. In particular, the local factors that led regulated banks to have larger mortgage market shares before the reform should not themselves generate differential changes in NBF lending, borrower risk of approved loans, or subsequent default rates after 2012. The vector X_{mt} includes pre-Basel III local market characteristics interacted with quarter indicators. This allows areas with different initial conditions to follow different trends. These controls also address the concern that regulatory exposure may be correlated with banks’ geographic sorting and local housing market dynamics. In particular, Oberfield et al. (2024) shows banks sort across locations and operate in dense urban markets. Chodorow-Reich et al. (2024) shows local housing boom cycles may also affect the pre-Basel III market share of regulated banks. To account for differences in fintech demand across regions, I control for the share of the population aged 18 to 44, which captures regional differences in the use of fintech channel, highlighted by Fuster et al. (2019). X_{mt} , therefore, includes house price growth during the 2004–2007 boom, the homeownership rate, density, the unemployment rate, and the share of the population aged 18–44, each interacted with time indicators. I examine whether regions with greater regulatory exposure systematically differ in these pre-regulation characteristics. Appendix C.1 reports the standardized differences across the two groups.

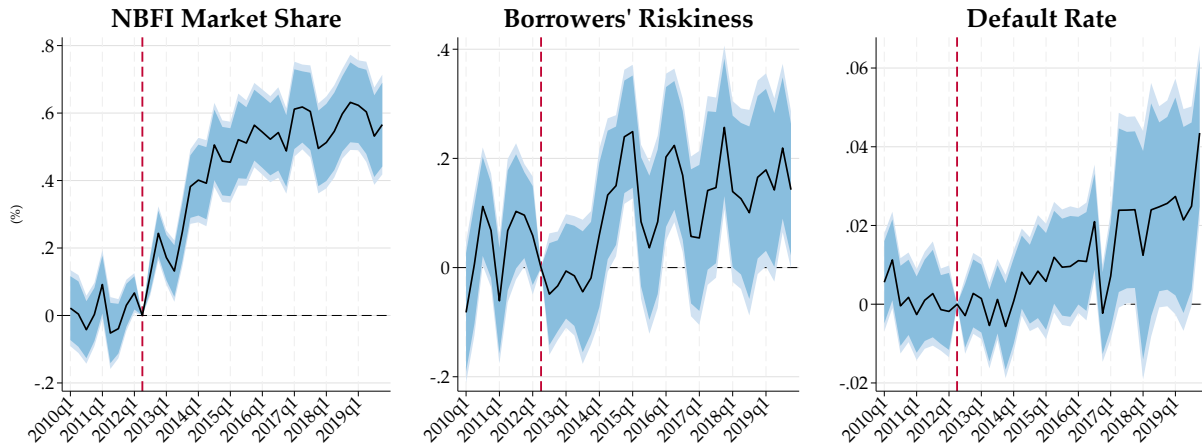
Finally, to address concerns that regulated banks may also have been involved in significant legal disputes and that other mortgage-related assets, such as mortgage servicing rights, may be an important channel driving the results, I additionally control for regional-level Legal Issue/Asset and MSR/Equity measures, each constructed as the market-share-weighted average of the corresponding bank-level ratio using banks’ 2008 mortgage market shares as weights.¹² In Section 2.5.2, I discuss these channels in more detail.

The omitted period is 2012Q2, as the proposed rules were announced on June 7, 2012. Given the time between application, lending decisions, and recorded originations, loans appearing in the data in 2012Q2 largely reflect applications and lending decisions made before the announcement. I

¹²I compile major mortgage-related settlements and enforcement actions reported in public DOJ, Federal Reserve, OCC, HUD, SEC, and FHFA documents during 2010–2012, assign each event to the ultimate parent bank, and aggregate the publicly reported dollar amounts through 2012.

therefore use 2012Q2 as the last quarter plausibly unaffected by the announcement and normalize all β_τ relative to that quarter. As a robustness check, I also use 2012Q1 as the omitted quarter and obtain similar results.

**FIGURE 4. Dynamic Effects of Bank Regulation on
NBFI Share in Origination, Borrowers' Riskiness, and Defaults**



Note. This figure reports the event-study estimates $\{\beta_\tau\}$ from equation (5), where MSA-level regulatory exposure, $Regulation_m$, is interacted with quarter indicators. The omitted period is 2012Q2. The dependent variables are NBFI mortgage market share, the lending standard proxy, and the mortgage default rate. The lending standard proxy is the first principal component of FICO score, debt-to-income ratio, and loan-to-value ratio. The specification includes MSA fixed effects, quarter fixed effects, and pre-Basel III local characteristics interacted with quarter indicators. The solid lines represent the point estimates, and the shaded areas denote 90 and 95 percent confidence bands clustered at the MSA level. *Sources:* HMDA, Single-Family Loan-Level Datasets provided by Freddie Mac and Fannie Mae

Estimation Results. The first panel of Figure 4 shows the effects of regulatory exposure on NBFI market share. MSAs with greater exposure to regulated banks experience a persistent increase in the NBFI share of origination. The effect grows gradually, which is consistent with the main finding of Buchak et al. (2018) that examines the growth rate of market share between 2008 and 2015.

The new findings are based on the borrower risk index and subsequent default rates. The second panel of Figure 4 reports the estimated coefficients on the borrower risk index. The results show that regions with greater exposure to the regulation experienced a credit shift toward riskier borrowers, represented by the index that combines lower credit scores with higher DTI and LTV ratios. Because the index is normalized so that higher values indicate riskier borrowers with lower creditworthiness and higher DTI and LTV ratios, positive coefficients indicate a shift toward riskier borrowers. The borrower-risk coefficients remain close to zero in the immediate post-announcement period and become persistently positive around 2014. Thus, the increase in

the NBFi origination share precedes the persistent deterioration in observable borrower risk.

The default rate estimates provide additional consistent evidence. As shown in the third panel of Figure 4, the estimated coefficients begin to rise around 2014 and become more pronounced. This timing coincides with the persistent increase in borrower risk, suggesting that the deterioration in realized loan performance is associated with the shift toward riskier borrowers. This interpretation is also consistent with [Bosshardt et al. \(2025\)](#), who show, using loans originated in 2016 and 2017, that the default rate increases more steeply with observable borrower risk for NBFi loans. In particular, their positive interaction between borrower risk and NBFi lending implies that the default gap between NBFis and banks widens as borrower risk increases. Therefore, in regions experiencing a larger shift of credit toward both NBFis and riskier borrowers, these two forces can reinforce each other.

Robustness Checks I conduct robustness checks by changing the covariates to exclude regulatory measures and by excluding all covariates, as suggested in [Altonji et al. \(2005\)](#) and [Oster \(2019\)](#). In addition, I change the treatment by measuring the market share of regulated banks in other years, including 2010, 2011, and 2012. The results are robust to changes in the covariates and the timing used to measure the market share of regulated banks. In addition, I estimate the response of average credit scores separately and the amount of defaulted principal. I show that average credit scores decline in regions with greater exposure to the regulation and that defaulted principal in those regions also increases after the regulation. Results are reported in Appendix C.2.

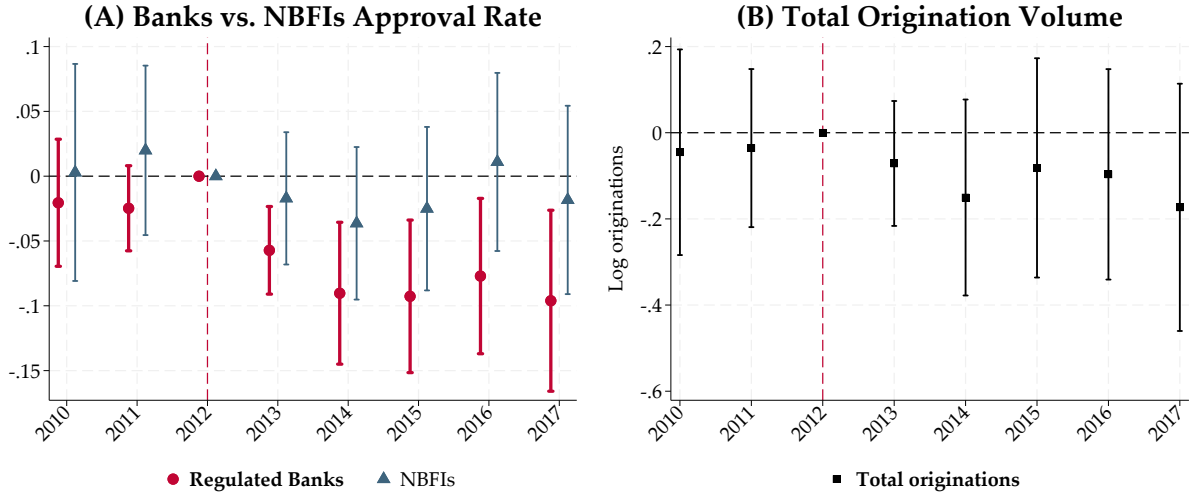
Finally, there can be a concern that, rather than the market share of regulated banks, the actual tightness of the capital ratio matters. I discuss alternative measures and other potential confounding effects in more detail in Section 2.5.2.

2.4.2. Tighter Capital Regulation Contributed to a Decline in Bank Approval Rates

In the previous section, the results show that regions more exposed to tighter capital requirements experienced an increase in the NBFi share of GSE originations together with a shift toward riskier borrowers and higher subsequent default rates within the observed GSE cohorts. These results could arise if banks and NBFis exhibit different approval rates for their respective applicant pools. Therefore, I examine whether regulatory exposure affected the approval decisions of regulated banks and NBFis differently. Furthermore, I test whether regulatory exposure affected the total amount of loans originated in a given region. To do so, I reestimate the same empirical specification using HMDA data and examine approval rates separately for regulated banks and NBFis.

Figure 5 reports the results. Panel (A) shows that the approval rate of regulated banks declines by approximately 10 percentage points as regulatory exposure increases from zero to one. In

FIGURE 5. Dynamic Effects of Bank Regulation on Approval Rates and Total Credit



Note. This figure reports the annual event-study estimates β_t from equation (5), where MSA-level regulatory exposure, $Regulation_m$, is interacted with year indicators. The omitted year is 2012. In Panel (A), the dependent variables are the approval rates of banks and NBFIs within each MSA. In Panel (B), the dependent variable is the MSA-level total credit originated. The specification includes MSA fixed effects, year fixed effects, and pre-Basel III local characteristics interacted with year indicators. Vertical bars denote 95% confidence intervals based on standard errors clustered at the MSA level. *Source.* HMDA

contrast, the approval rate of NBFIs does not exhibit statistically significant changes. This aligns with the institutional fact that NBFIs are not subject to bank capital requirements. These results show that regulated banks tightened their lending standards, reflected in lower approval rates, whereas NBFIs maintained their approval rates. While Panel (A) reports the response of the average approval rate at the MSA level for each sector, I also calculate the residual of each application approval decision after controlling for borrower characteristics, including income, loan size, purpose, sex, and ethnicity, and recompute the response using the average residual approval rate. This residualized measure helps isolate changes in lending standards from changes in the observable composition of applicants and the results are robust. The results are presented in Appendix C.3.

Then, I examine the total amount of loans originated in a given region. Panel (B) shows that regions with greater exposure to this regulation experienced a decrease in total loan supply, although the decrease lacks statistical significance. This is consistent with Buchak et al. (2018), who show that NBFIs increase their market share and partially, but not perfectly, substitute for bank lending.

Finally, I test the response of the NBFI application share and find that it also rises after the regulation. Regions with greater exposure to the regulation exhibit faster growth in the NBFI

application share, although the increase is smaller than that in the NBFIs origination share. This difference can be partly explained by the widening approval-rate gap between the two sectors. The results are reported in Appendix C.3.

Key Takeaway. Tighter bank capital requirements shifted mortgage credit from regulated banks toward NBFIs. In more exposed markets, regulated banks reduced their approval rates, while NBFIs maintained their approval rates and expanded their market share. This reallocation was accompanied by a shift toward riskier borrowers and a delayed increase in mortgage defaults. The findings suggest that tighter capital regulation did not simply reduce credit supply but redirected lending toward less regulated intermediaries, with adverse consequences for lending standards and subsequent loan performance.

2.5. Discussion of Mortgage Market and Capital Regulation

I analyze mortgage market data, where detailed application-level data and performance data shed light on how lenders' decisions about which applicants to approve differ across sectors and how those decisions affect the realization of credit risk. However, the mortgage market raises two concerns for interpreting the empirical results. The first concern is why capital requirements affect banks' mortgage lending when mortgages can be sold through a secondary market. Once banks sell their loans, their exposure to capital requirements may decline because the loans are replaced on their balance sheets by assets with lower risk weights. The second concern is whether other mortgage market factors, such as large litigation costs and regulatory channels operating through mortgage servicing rights, rather than banks' capital constraints, could explain the observed retreat of banks and expansion of NBFIs.

2.5.1. Why Do Capital Requirements Affect Bank Mortgage Lending Despite Securitization?

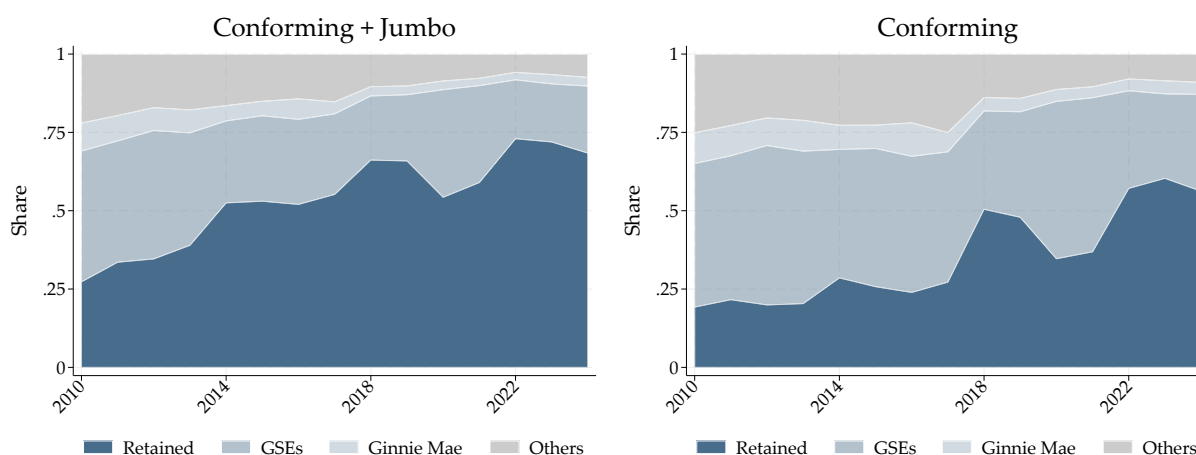
The first concern is that capital requirements may not constrain banks' mortgage lending because mortgages can be sold or securitized. Once banks sell the loans to third parties, their exposure to capital requirements may decline as the loans are replaced on their balance sheets by other assets with lower risk weights. However, securitization does not eliminate the relevance of capital requirements for banks' mortgage lending for three reasons.

First, selling the underlying loans does not necessarily eliminate the capital requirement exposure associated with mortgage origination. The assets that banks receive after selling the loans, such as mortgage-backed securities, are also subject to capital requirements, although they

carry lower risk weights. In addition, while selling loans removes the underlying loans from the balance sheet, banks commonly retain mortgage servicing rights (MSRs) after selling the loans (Hamdi et al. 2023; Zheng 2024). MSRs represent the asset value of the right to service mortgage loans, including collecting principal and interest payments from borrowers and transferring them to investors, after the originating banks sell the loans to third parties. Importantly, these assets are subject to relatively high risk weights under capital regulation.¹³ In other words, mortgage origination remains relevant to capital requirements even when the underlying loans are sold, because banks can remain exposed through the assets and servicing rights associated with those loans.

Second, in practice, mortgages that are eligible for sale to third parties still remain on bank balance sheets. Figure 6 shows that a non-negligible number of loans are still retained on the bank balance sheet.

FIGURE 6. Bank Retention Rate



Note. This figure shows the disposition of mortgage loans originated by banks from 2010 to 2024 using HMDA data. Loans are classified based on the HMDA loan purchaser code into four categories. “Retained” includes loans that are not reported as sold to these entities and remain with the originating bank. “Bank to GSE” includes loans sold to Freddie Mac and Fannie Mae, while “Bank to Ginnie Mae” includes loans sold through Ginnie Mae. “Others” captures the case where the loans are sold to other institutions. The left panel reports the shares for all mortgage originations, while the right panel restricts the sample to conforming loans. *Source.* HMDA

After 2018, banks have retained around 30 ~ 60% of loans that are sellable (i.e., conforming loans), and this share rises to 50 ~ 70% when including loans that are not eligible for sale (i.e., jumbo loans). These retained loans remain subject to risk-based capital requirements.¹⁴ Importantly,

¹³MSRs are the rights to collect mortgage payments and provide other servicing functions in exchange for servicing fees. Capital rules require banks to deduct part of their MSRs from common equity tier 1 capital, while the remaining portion is subject to a 250 percent risk weight. See 12 C.F.R. § 217.22(d) and § 217.32(l)(4).

¹⁴Qualifying first-lien residential mortgages generally receive a 50 percent risk weight, while other first- and

selling loans is not a costless way to avoid capital requirements. Fuster et al. (2013) emphasize that the originating lender remains exposed to put-back risk. A delinquency may prompt a review of the loan file, and the lender may be required to repurchase the loan if the GSE identifies a violation of its underwriting or eligibility representations and warranties. They also argue that a larger cost may arise from the additional underwriting and legal resources used to prevent or contest repurchase claims. Therefore, banks face a trade-off between retaining loans that are subject to capital requirements and selling them at a cost. The substantial retention rate in Figure 6 shows that mortgage origination remains exposed to bank capital regulation even when loans are eligible for sale.

Lastly, even for mortgages that are ultimately sold, there is pipeline risk between origination and sale because the loans are held on banks' balance sheets during this period and are therefore relevant to capital regulation, highlighted by Kim et al. (2022). Capital requirements can therefore affect banks' origination incentives during the period in which loans remain on their balance sheets.

Therefore, securitization does not eliminate the relevance of capital requirements for banks' mortgage lending decisions. It is noteworthy that the results showing the association between capital regulation and credit allocation to riskier borrowers, as well as higher default rates, are based on GSE loans. This evidence is consistent with capital regulation affecting mortgage lending even for loans that are ultimately sold. Nevertheless, securitization can mitigate the relevance of capital regulation for bank lending, and I explicitly incorporate this margin in an extension of the model in Section 6.4.

2.5.2. Could Other Mortgage Market Factors Drive the Results?

The second concern is that factors specific to the mortgage market, rather than banks' capital constraints, could generate the empirical results. Frame et al. (2026) highlighted that large banks faced substantial litigation exposure following the financial crisis, which may have reduced their lending. The Basel III revisions also tightened the capital treatment of servicing assets, and this provides another channel through which capital regulation could affect mortgage lending, as Buchak et al. (2018) and Benson et al. (2026) highlight. Although the revision to MSRs is itself part of the effect of capital requirements and provides a channel through which the concern about securitization discussed above can be addressed, the results could still be interpreted as being driven mainly by the MSR rule change rather than by how well capitalized banks were to

junior-lien mortgages receive a 100 percent risk weight. See 12 C.F.R. §217.32(g). Even mortgages intended for sale remain subject to capital requirements while they are held on the bank's balance sheet before sale. See 12 C.F.R. §§217.31(a), 217.32(g).

begin with. In other words, the question is whether the estimated effect reflects the capital charge associated with servicing rights alone or the bank's overall capital position, and this distinction matters for the modeling choice.

To address concerns that the increase in NBFI lending may reflect other post-crisis changes affecting banks, I conduct two exercises. I first change the treatment from the market share of regulated banks to the regional-level capital ratio of regulated banks. In other words, within each region, I replace the treatment variable with the capital ratios of regulated banks, aggregated to the regional level using their market shares as weights, which captures banks' initial capitalization. Then, I augment the specification with exposures to mortgage-related legal overhang and exposure to the MSR rule change, as I do in the baseline. When these exposures are included jointly, counties more exposed to weakly capitalized banks experience a larger increase in the NBFI origination share after 2013. Appendix C.4 provides the construction of these measures, the specification, and the event-study results.

Second, I conduct a bank-level analysis using variation within regulated banks. I restrict the sample to banks that are subject to stress tests and use variation in capital ratios, MSR exposure measured as MSR holdings relative to equity, legal overhang, and stress capital buffers, which are proxied by projected stress losses relative to assets. By doing so, I can examine to what extent each exposure can explain the reduction in originations by each bank, and I show that the capital ratio can explain the reduction in originations along with the MSR holdings relative to equity. Appendix C.5 provides the details on the regression and the results.

3. A Static Financial Intermediary Problem

I develop a general equilibrium model in which banks and NBFIs coexist to explain the empirical findings in Section 2 and to examine how tighter bank capital requirements affect aggregate credit risk under financial shocks. The key features of the model are that financial intermediaries choose their underwriting standards, which determine which borrowers can obtain loans, and that capital regulation affects not only the loan rate but also which borrowers receive credit.

I first introduce a static intermediary problem to show the channel through which capital regulation affects banks' underwriting standards. The intermediary jointly chooses the loan rate and underwriting stringency. This differs from Dell'Ariccia et al. (2014) and Martinez-Miera and Repullo (2019), where costly effort directly changes borrowers' default probabilities. In my model, underwriting instead changes which borrowers are approved and therefore the risk composition of originated loans. The static problem shows how a change in the marginal cost of lending affects both the loan rate and underwriting choice. This provides the main mechanism through

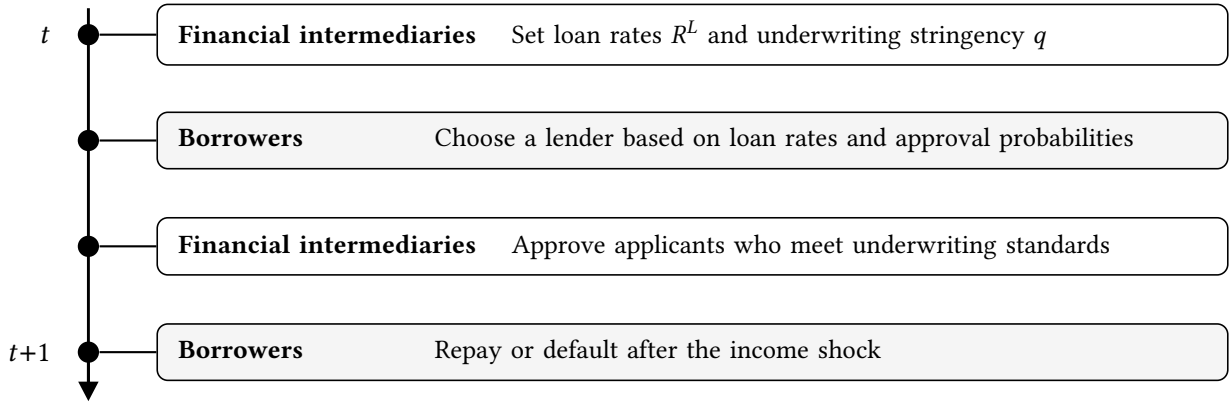
which bank capital regulation, which is introduced as an additional marginal cost of bank lending, affects lending standards in the full model. I then discuss the key modeling assumptions before introducing banks and NBFIs into the general equilibrium model.

3.1. Timing and Setup

Time is discrete, and each intermediary b operates under monopolistic competition, taking the loan rates posted by other lenders as given. The intermediary originates one-period nominal loans and finances them with external funding at the gross rate R^f .

At the beginning of period t , the intermediary jointly sets the loan rate R^L and its underwriting stringency q to maximize its profit. q determines which borrowers, based on observables, can be approved and determines the approval probability $p(q)$. Applicants observe the approval probability before submitting an application, but do not directly observe q or know whether their application will be approved. Instead, they choose where to apply based on the loan rate and the observed approval probability. Once applications are submitted to each lender, the lender decides which applicants receive loans based on the underwriting stringency q and the distribution of applicants' creditworthiness. Approved applicants receive loans and, on the repayment date $t + 1$, decide whether to default or repay after their income shock is realized. Figure 7 summarizes the timing of the intermediary's lending decisions and borrowers' application and repayment choices.

FIGURE 7. Timing of intermediary and borrower decisions



The stringency q shapes the intermediary's profit through three distinct channels, an origination volume channel, an underwriting cost channel, and a default channel. The first channel through which q affects profit is origination volume. q can be interpreted as a cutoff on applicants' observable creditworthiness. A higher q therefore requires greater creditworthiness and lowers

the approval probability, implying $p'(q) < 0$. Because applicants observe the approval probability rather than q itself, and the approval probability affects their lender choice, as I will discuss in Section 4.3, stricter underwriting also reduces application demand, $L^{app}(R^L, p(q))$, by making the lender less attractive to applicants. The resulting volume of originated loans $L^{orig}(R^L, q)$ is

$$L^{orig}(R^L, q) = p(q) \cdot L^{app}(R^L, p(q)),$$

which is the product of the approval probability and application volume.

The second channel is through underwriting costs. To maintain underwriting stringency q , the intermediary needs to pay an underwriting cost of $s(q)$ per dollar of originated lending, with $s'(q) > 0$. This cost captures the resources required to implement stricter verification and underwriting procedures.

The third channel is through default. At the repayment date, idiosyncratic income shocks are realized for borrowers with approved loans. Borrowers with higher creditworthiness have greater repayment capacity and are therefore less likely to default when realized income falls below a fixed threshold. Under underwriting stringency q , the default rate among approved loans can be summarized by $d(q)$, as discussed in detail in Section 4.3. Because a higher q shifts the composition of approved loans toward safer borrowers, it lowers the expected default rate of originated loans, implying $d'(q) < 0$.

Each dollar of originated lending requires one dollar of external funding at a cost of R^f , and therefore the intermediary faces the funding requirement $F = L^{orig}(R^L, q)$. The intermediary's profit can therefore be written as

$$\Pi(R^L, q) = \left[(1 - d(q)) R^L - s(q) - R^f \right] \cdot p(q) \cdot L^{app}(R^L, p(q)). \quad (6)$$

The first term is the revenue per dollar of lending. A fraction $d(q)$ of originated loans defaults, so the intermediary collects $(1 - d(q))R^L$ per dollar of originated lending. In this benchmark, the financial intermediary bears the full losses from defaults on the loans it originates. This assumption is discussed in Section 3.4 and relaxed in the general equilibrium model. The intermediary also pays the underwriting cost $s(q)$ and the funding cost R^f for each dollar originated. Multiplying the resulting per-dollar margin by $p(q)L^{app}(R^L, p(q))$ gives total profit.

3.2. First Order Conditions

Given that application demand is differentiable in the loan rate and has a constant loan-rate elasticity $\varepsilon > 1$, the first-order condition with respect to R^L implies

$$R^L = \frac{\varepsilon}{\varepsilon - 1} \frac{1}{1 - d(q)} (R^f + s(q)). \quad (7)$$

The intermediary applies the markup factor $\varepsilon/(\varepsilon - 1)$ to the per-dollar marginal cost of origination, $R^f + s(q)$, adjusted by the expected repayment share $1 - d(q)$.

The first-order condition with respect to q can be derived as

$$\underbrace{-d'(q)R^L}_{\text{default-risk margin}} = \underbrace{s'(q)}_{\text{underwriting-cost margin}} - \underbrace{\left((1 - d(q))R^L - R^f - s(q) \right) \left(\frac{p'(q)}{p(q)} + \frac{d \log L^{app}(R^L, p(q))}{dq} \right)}_{\text{volume margin}}. \quad (8)$$

The left-hand side of equation (8) captures the marginal benefit of tightening underwriting stringency. Raising q lowers the default probability at the marginal rate $-d'(q)$, so the resulting reduction in expected default losses is $-d'(q)R^L$. The right-hand side captures the costs of tightening underwriting stringency. The first term denotes the direct cost of operating under a stricter underwriting standard. The second term captures the margin lost from originating fewer loans as both the approval rate and application demand decline. The term $p'(q)/p(q)$ captures the change in the share of applications that are approved, while $d \log L^{app}(R^L, p(q))/dq$ captures the change in application demand.

3.3. Complementarity between Loan Rates and Underwriting Stringency

In this section, I show how an increase in the cost of funding R^f changes both the underwriting standard and the loan rate. The reason is that, once the financial intermediary problem is extended to a bank problem, the capital requirement is introduced as an additional cost that the bank needs to internalize when maximizing profits. The proposition below therefore provides a channel through which capital requirements affect banks' choices.

PROPOSITION 1. *Suppose $s'(q) > 0$ and the interior optimum satisfies the second-order condition. Then*

$$\frac{dq^*(R^f)}{dR^f} > 0.$$

PROOF. See Appendix D. □

The mechanism is a complementarity between the loan rate and underwriting. A higher funding cost R^f first raises the posted loan rate through equation (7). Using the envelope condition, a higher loan rate increases the amount the intermediary loses when a borrower defaults. This raises the marginal benefit of lowering the default probability and induces the intermediary to tighten q . According to (8), however, the intermediary does not tighten underwriting indefinitely. A higher q is costly to implement and reduces the number of originated loans by lowering the approval rate and application demand. Therefore, underwriting becomes stricter until the additional reduction in default losses is offset by the higher underwriting costs and lost lending margins.

This result connects capital regulation to banks' underwriting decisions in the full model. Capital requirements enter as an additional marginal cost of lending, which corresponds to an increase in R^f in the partial-equilibrium. Banks respond to the higher cost by raising loan rates. As shown above, a higher loan rate increases the benefit of reducing default risk and therefore leads banks to tighten underwriting. Capital regulation thus affects both the price of bank credit and the risk composition of bank originations. This prediction is also consistent with the empirical evidence in Section 2.4.2, where bank approval rates decline as capital regulation tightens.

3.4. Discussion of Model Assumptions

3.4.1. Source of Default

In this model, default occurs after idiosyncratic income shocks are realized. More specifically, a borrower defaults when realized income falls below a fixed threshold. As shown in Appendix F, borrowers with higher creditworthiness are less likely to cross this threshold, and tighter q shifts the composition of approved loans toward borrowers with safer observables. Integrating over the income shock process and the conditional distribution of approved borrowers under q generates a default rate that can be expressed as a function of q .

The default mechanism in this model differs from the costly state verification mechanism. In Townsend (1979), only the borrower observes the realized state, and the lender must pay a cost to verify it. Bernanke et al. (1999) and Durdu and Zhong (2023) embed this friction in macroeconomic models in which a low idiosyncratic return triggers default and the lender incurs a monitoring cost in the default state. In this model, however, the approval cutoff determines which applicants enter the loan pool, and tightening this cutoff changes both the risk composition of originated loans and the default rate.

The mechanism also differs from the screening channel in Martinez-Miera and Repullo (2019) in the margin through which default risk is affected. In their model, screening changes the default probability of a project that is already financed. In my model, the underwriting standard does not

change each applicant's default risk. Rather, it determines which applicants' risks become part of the aggregate credit risk. Tightening the underwriting standard excludes applicants with higher default risk, making the originated loan pool safer and lowering its default rate.

3.4.2. Default Loss Internalization

In the static model, I assume that the lender fully internalizes default losses when maximizing profit. In practice, however, the extent to which intermediaries internalize default losses can differ across intermediaries. Limited liability implies that intermediary owners need not absorb the full losses in sufficiently adverse states. Such incentives also depend on their business models, franchise value, and incentives to preserve continuation value. This motivates introducing a wedge that captures the share of default losses internalized by the intermediary, which may be less than one.

The wedge is important for differentiating banks and NBFIs. Banks derive franchise value from operating across multiple business lines. In particular, Kim et al. (2022) emphasize that banks tend to operate in several business areas, while NBFIs are more likely to focus on a single line of business. As a result, the costs that banks perceive from defaults are higher than those perceived by NBFIs, as banks need to take into account their other business lines and have stronger incentives to internalize default losses rather than divert or shift them to funding providers. However, NBFIs have fewer additional business profits and less franchise value to protect, so the costs of defaults they internalize are smaller. This difference becomes even larger when losses are severe because banks stand to lose more franchise value if they are forced to stop operating.

In the general equilibrium model, I capture these differences through κ_j . The parameter κ_j measures the share of the default loss, $d_j(q_j)R_j^\ell$, absorbed by intermediary j . The expected payoff per dollar of lending is therefore $(1 - \kappa_j d_j(q_j))R_j^\ell$. A higher κ_j means that borrower default has a larger effect on the intermediary's profits and net worth, strengthening its incentive to tighten underwriting and increase the loan rate. The differences between banks and NBFIs described above therefore imply $\kappa_B > \kappa_{NB}$, which, together with other key features including exposure to capital regulation, can generate differences in underwriting standards.¹⁵

¹⁵This interpretation is also consistent with Bosshardt et al. (2025), who find that banks charge more for observable default risk than NBFIs even for loans sold into the same GSE market.

4. The Quantitative Model

In this section, I build a general equilibrium model to formalize the mechanism through which bank capital requirements affect the allocation of credit across banks and NBFIs. To this end, I use the model as a laboratory to quantify the effects of tighter regulation on credit losses and financial intermediation in counterfactual financial stress scenarios.

I extend the partial-equilibrium intermediary framework by embedding banks and NBFIs as credit suppliers, with each sector jointly choosing loan rates and underwriting standards. The two sectors differ along two key dimensions. First, banks are funded through deposits, while NBFIs are funded through money market funds. Second, capital requirements apply only to banks, whereas NBFIs operate outside the regulatory perimeter. I embed these intermediaries in a saver–borrower general equilibrium framework of [Curdia and Woodford \(2010\)](#) and [Eggertsson et al. \(2024\)](#), in which borrowers allocate applications across the two sectors in response to their lending terms.¹⁶ Savers provide funding to both intermediary sectors by holding deposits and money market funds, as in [Begenau and Landvoigt \(2022\)](#). The production side consists of intermediate-good producers subject to nominal rigidities, and a monetary authority sets the policy rate. For tractability, loans are modeled as one-period nominal claims.¹⁷ The timeline follows the order described in the partial-equilibrium section.

I present the model in the following order: banks, NBFIs, borrowers, savers, and the remaining general-equilibrium blocks. The financial intermediaries’ balance sheet items and profits are expressed in nominal terms.

4.1. Banks

I extend the partial-equilibrium intermediary block to a banking sector, in which a continuum of banks indexed by b fund lending with deposits, hold liquid securities, and internalize regulatory-capital costs when setting loan rates and underwriting standards. For notational simplicity I suppress the index b and mark bank specific objects with the subscript B . Banks post the loan rate

¹⁶Another class of models represents households as savers and firms as the borrowers that finance their capital ([Ulate 2021](#); [Abadi et al. 2023](#); [Wang et al. 2022](#)), or has banks directly hold firms’ working capital ([Begenau 2020](#); [Elenev et al. 2021](#); [Begenau and Landvoigt 2022](#)). Because my focus is the substitution between banks and NBFIs in the provision of loan credit, I instead adopt a framework with two household types, building on [Curdia and Woodford \(2010\)](#) and [Eggertsson et al. \(2024\)](#).

¹⁷A natural extension separates new originations from outstanding stocks. [Garriga et al. \(2017\)](#) and [Berger et al. \(2021\)](#) suggest an extension where each intermediary has a flow of newly originated loans, an outstanding principal stock, a promised payment stock, and a risk stock embedded in the outstanding pool. The stock default rate then evolves as new originations with current default risk are added to inherited stocks.

$R_{B,t}^\ell$ and choose underwriting stringency $q_{B,t}$. As in the partial-equilibrium block, originations are the product of the approval rate and loan applications,

$$L_{B,t} = p_B(q_{B,t}) L_B^{app}(R_{B,t}^\ell, p_B(q_{B,t})).$$

Similarly, $d_B(q_{B,t})$ and $s_B(q_{B,t})$ denote the corresponding default rate and per-dollar underwriting cost. The bank originates loans $L_{B,t}$ and holds liquid securities $S_{B,t}$, financed by deposits $D_{B,t}$ and predetermined equity $E_{B,t}$,

$$L_{B,t} + S_{B,t} = D_{B,t} + E_{B,t}.$$

Liquid securities earn the gross policy rate R_t . Following Drechsler et al. (2017), banks exercise market power in the deposit market and face the deposit supply schedule

$$D_{B,t} = D_B(R_{B,t}^D).$$

Given these balance-sheet constraints, the bank chooses the loan rate $R_{B,t}^\ell$, underwriting stringency $q_{B,t}$, the amount of security holdings $S_{B,t}$, and deposit rate $R_{B,t}^D$ at date t . Loans, securities, and deposits chosen at date t settle at date $t + 1$. Then, I can express the pre-dividend profit as

$$\text{Profit}_{B,t} = \left((1 - \kappa_B d_B(q_{B,t})) R_{B,t}^\ell - s_B(q_{B,t}) \right) L_{B,t} + R_t S_{B,t} - R_{B,t}^D D_{B,t} - c_B(L_{B,t}, E_{B,t}; \nu).$$

Banks internalize a share $\kappa_B < 1$ of default losses, as discussed in Section 3.4.2. Limited liability places a limit on the losses that are absorbed through bank profits and net worth. The remaining share is not absorbed within the bank and is instead borne directly by savers. In particular, the fraction $1 - \kappa_B$ enters the saver budget constraint as a lump-sum charge. The capital regulations are captured by $c_B(L_{B,t}, E_{B,t}; \nu)$ as an adjustment cost, which is discussed in the next paragraph.

At date $t + 1$, after the payoff from the date- t portfolio is realized and dividends are paid, bank equity evolves according to

$$E_{B,t+1} = \text{Profit}_{B,t} - \gamma_B E_{B,t}.$$

The term $\gamma_B E_{B,t}$ captures dividends, assumed to be a constant fraction $\gamma_B \in (0, 1)$ of beginning-of-period bank equity.

Capital Requirement. Gerali et al. (2010), Ulate (2021), and Balloch and Koby (2023) motivate modeling capital requirements as a balance-sheet adjustment cost that reflects the resources associated with external equity issuance or portfolio restructuring as banks approach the regulatory threshold. The functional form of the capital-requirement cost follows Balloch and Koby (2023)

$$c_B(L, E; \nu) = \xi \frac{E}{E/L - \nu}. \quad (9)$$

This specification is taken for two reasons. First, [Begenau et al. \(2026\)](#) document that regulatory capital constraints rarely bind strictly for U.S. bank holding companies. The vast majority of banks maintain capital buffers above the regulatory minimum.¹⁸ This evidence suggests that banks generally adjust before violating the regulatory threshold, which motivates representing capital regulation as a marginal cost that rises sharply as the bank approaches the threshold rather than as a constraint that is routinely violated. Second, [Balloch and Koby \(2023\)](#) highlight that equation (9) provides a smooth approximation to a hard risk-based capital constraint, such as $E \geq \nu L$.

Accordingly, the cost function is defined for $E/L > \nu$. In the model, loans receive a unit regulatory weight, while the liquid securities considered here are assigned a zero regulatory weight. The model counterpart of the regulatory capital ratio is therefore E/L .

First Order Conditions. At the symmetric equilibrium, the optimal loan rate satisfies

$$R_{B,t}^\ell = \frac{\epsilon_B^\ell}{\epsilon_B^\ell - 1} \frac{1}{1 - \kappa_B d_B(q_{B,t})} (R_t + s_B(q_{B,t}) + c_{B,L}(L_{B,t}, E_{B,t}; \nu)), \quad (10)$$

where ϵ_B^ℓ is the loan-rate elasticity of application demand and $c_{B,L} \equiv \partial c_B / \partial L_{B,t}$ is the marginal regulatory-capital cost of originating an additional dollar of loans. The optimal underwriting standard satisfies

$$\begin{aligned} -\kappa_B d'_B(q_{B,t}) R_{B,t}^\ell &= s'_B(q_{B,t}) \\ &- \left((1 - \kappa_B d_B(q_{B,t})) R_{B,t}^\ell - R_t - s_B(q_{B,t}) - c_{B,L}(L_{B,t}, E_{B,t}; \nu) \right) \left(\frac{p'_B(q_{B,t})}{p_B(q_{B,t})} + \chi_B^q \right), \end{aligned} \quad (11)$$

where $\chi_B^q \equiv d \log L_B^{app}(R_{B,t}^\ell, p_B(q_{B,t})) / dq_{B,t}$. The last two terms capture the effects of underwriting stringency on origination volume through the approval rate and bank application demand, respectively.

Finally, the optimal deposit rate satisfies

$$(R_t - R_{B,t}^D) D'_B(R_{B,t}^D) = D_B(R_{B,t}^D).$$

Defining the deposit-supply elasticity as $\epsilon^D \equiv \partial \log D_B(R^D) / \partial \log R^D$, the symmetric-equilibrium

¹⁸Even at the height of the global financial crisis, more than 90 percent of banks were classified as well capitalized according to their Tier 1 capital ratios, while approximately 5 percent fell below the regulatory minimum.

deposit rate is

$$R_{B,t}^D = \frac{\epsilon^D}{1 + \epsilon^D} R_t. \quad (12)$$

4.2. NBFIs

NBFIs are monopolistically competitive loan intermediaries that originate loans but cannot issue insured deposits. Instead, they finance lending through wholesale funding and operate outside the regulatory perimeter. The NBFI sector consists of a continuum of intermediaries indexed by n . As with banks, I focus on a symmetric equilibrium, suppress the individual NBFI index n , and use the subscript NB to denote NBFI-sector variables. The representative NBFI posts the loan rate $R_{NB,t}^\ell$ and chooses an underwriting stringency $q_{NB,t}$, and originations are again the product of the approval rate and loan applications,

$$L_{NB,t} = p_{NB}(q_{NB,t}) L_{NB,t}^{app}(R_{NB,t}^\ell, p_{NB}(q_{NB,t})). \quad (13)$$

The functions $d_{NB}(q_{NB,t})$ and $s_{NB}(q_{NB,t})$ are the associated default rate and per-dollar underwriting cost. Unlike banks, the NBFI holds no securities and issues no deposits. It funds loans with wholesale funding $F_{NB,t}$ and predetermined equity $E_{NB,t}$,¹⁹

$$L_{NB,t} = F_{NB,t} + E_{NB,t}. \quad (14)$$

The NBFI chooses the loan rate $R_{NB,t}^\ell$, underwriting stringency $q_{NB,t}$, and wholesale funding $F_{NB,t}$ at date t . Loans and wholesale funding chosen at date t settle at date $t + 1$. I express the associated pre-dividend payoff from the date- t portfolio as

$$\text{Profit}_{NB,t} = \left((1 - \kappa_{NB} d_{NB}(q_{NB,t})) R_{NB,t}^\ell - s_{NB}(q_{NB,t}) \right) L_{NB,t} - R_t^F F_{NB,t}.$$

NBFIs take R_t^F as given. The parameter κ_{NB} denotes the share of default losses absorbed through NBFI profits and net worth based on the discussion in Section 3.4.2.²⁰ The remaining share, $1 - \kappa_{NB}$, is borne by savers, who hold money market funds for liquidity purposes, and enters their budget constraint as a lump-sum charge. Because $F_{NB,t} = L_{NB,t} - E_{NB,t}$, the marginal funding cost of an additional dollar of NBFI lending is R_t^F . The underwriting cost associated with the date- t portfolio

¹⁹As in Buchak et al. (2025), NBFIs can also be financed by banks through secondary markets. In other words, NBFIs originate loans and issue securities that banks hold, so that banks indirectly fund NBFI lending. Banks also support NBFIs through credit lines (Jiang 2023; Acharya et al. 2026). This extension is discussed in Section 6.4.

²⁰Because the model abstracts from the explicit sale and securitization of individual mortgages, κ_{NB} should not be interpreted as the fraction of originated loans retained by the originating NBFI.

is expressed in the same date- t units and is settled with the portfolio at date $t + 1$. At date $t + 1$, after the payoff from the time t portfolio is realized and dividends are paid, NBFI equity evolves according to

$$E_{NB,t+1} = \text{Profit}_{NB,t} - \gamma_{NB} E_{NB,t}.$$

where $\gamma_{NB} E_{NB,t}$ captures dividends paid by NBFIs to savers, with $\gamma_{NB} \in (0, 1)$.

First Order Conditions. At the symmetric equilibrium, the optimal loan rate satisfies

$$R_{NB,t}^\ell = \frac{\epsilon_{NB}^\ell}{\epsilon_{NB}^\ell - 1} \frac{1}{1 - \kappa_{NB} d_{NB}(q_{NB,t})} \left(R_t^F + s_{NB}(q_{NB,t}) \right), \quad (15)$$

where ϵ_{NB}^ℓ is the loan-rate elasticity of application demand, with $\epsilon_{NB}^\ell > 1$ under monopolistic competition. Relative to the bank pricing condition (10), the NBFI marginal cost replaces the policy rate and regulatory-capital cost with the wholesale funding rate R_t^F , reflecting its position outside the regulatory perimeter. The optimal underwriting standard satisfies

$$\begin{aligned} -\kappa_{NB} d'_{NB}(q_{NB,t}) R_{NB,t}^\ell &= s'_{NB}(q_{NB,t}) \\ &- \left((1 - \kappa_{NB} d_{NB}(q_{NB,t})) R_{NB,t}^\ell - R_t^F - s_{NB}(q_{NB,t}) \right) \left(\frac{p'_{NB}(q_{NB,t})}{p_{NB}(q_{NB,t})} + \chi_{NB}^q \right), \end{aligned} \quad (16)$$

where $\chi_{NB}^q \equiv d \log L_{NB}^{app} (R_{NB,t}^\ell, p_{NB}(q_{NB,t})) / dq_{NB,t}$. The last two terms capture the effects of underwriting stringency on origination volume through the approval rate and NBFI application demand, respectively.

4.3. Borrowers

The borrower consists of a continuum of ex ante identical households of measure χ . At date t , the representative borrower household chooses consumption C_t^b , labor supply N_t^b , and total loan applications L_t^{app} . Before applications are submitted, each lender posts loan rates and chooses an underwriting standard. Household members observe the posted rates and each sector's approval probabilities, but not their individual approval outcomes or the exact underwriting standard. They choose either a bank or an NBFI based on these terms and idiosyncratic application preference. Appendix E provides the aggregation procedure.

4.3.1. Preferences and total applications

The representative borrower household solves

$$\max_{\{C_t^b, N_t^b, L_t^{app}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} (\beta^b)^t \mathcal{U}(C_t^b, N_t^b),$$

with GHH preferences (Greenwood et al. 1988). For $\sigma \neq 1$, utility is given by

$$\mathcal{U}(C_t^b, N_t^b) = \frac{1}{1-\sigma} \left[C_t^b - \Xi_b \frac{N_t^{b^{1+\phi}}}{1+\phi} \right]^{1-\sigma}.$$

When $\sigma = 1$, utility takes the logarithmic form. Borrowing is subject to a loan-to-income constraint at the application stage,

$$L_t^{app} \leq \theta W_t N_t^b.$$

The variable L_t^{app} denotes the aggregate face value submitted at the application stage, and originated borrowing is determined after lender choice and underwriting. The budget constraint is explained after describing the allocation of loan applications and their settlement.

4.3.2. Application allocation

At date t , given the loan application amount L_t^{app} , each member i of a continuum of household members, whose creditworthiness x_i follows a distribution $F(x)$, selects one lender and submits a single application. Before submitting an application, applicant i observes each lender's posted loan rate $R_{k,j}^\ell$ and approval probability $p_{k,j}(q_{k,j})$, but not the lender's internal underwriting standard $q_{k,j}$ or the applicant's individual approval outcome. The applicant also draws lender-specific application preference shocks $e_{i,k,j} > 0$ from a nested Fréchet distribution as in Herreño (2023). This captures non-price attributes such as accessibility, convenience, and prior experience with a lender. The upper level of the nested structure distinguishes the bank and NBFI sectors, while the lower level differentiates lenders $k \in \mathcal{K}_j$ within each sector $j \in \{B, NB\}$.

In the model, application preference shocks are independent of creditworthiness. Therefore, application allocation preserves the underlying creditworthiness distribution at every lender.

ASSUMPTION 1. *Application preference shocks are independent of applicant creditworthiness:*

$$\{e_{i,k,j}\}_{k \in \mathcal{K}_j, j \in \{B, NB\}} \perp x_i, \quad x_i \sim F(x).$$

This restriction provides a baseline consistent with the limited differences in observable

applicant characteristics documented in Section 2.3. This assumption is relaxed in Section 6.4 that introduces heterogeneous borrower types that differ in their sensitivity to approval standards and in their performance under the same screening technology.

In a sector-symmetric equilibrium, where $R_{k,j,t}^\ell = R_{j,t}^\ell$ and $q_{k,j,t} = q_{j,t}$, the resulting sector- j application share is

$$\omega_{j,t} = \frac{\psi_j [p_j(q_{j,t})]^\eta (R_{j,t}^\ell)^{-\varphi}}{\sum_{m \in \{B, NB\}} \psi_m [p_m(q_{m,t})]^\eta (R_{m,t}^\ell)^{-\varphi}}. \quad (17)$$

Here, $\psi_j > 0$, with $\psi_B + \psi_{NB} = 1$, captures the non-price attractiveness of sector j . The parameter φ governs substitution across sectors in response to loan rates, while η governs the response of applications to approval probabilities. Appendix E provides the full distributional specification and derives the application choice probabilities. Sector j therefore receives

$$L_{j,t}^{app} = \omega_{j,t} L_t^{app}$$

amount of applications to process. When tighter bank capital requirements raise the bank loan rate and reduce the bank approval probability, the bank option becomes less attractive, leading more borrowers to direct their initial applications to NBFIs with higher ω_{NB} .

4.3.3. Application to Origination

After applications are assigned, lender k applies its underwriting standard, and its origination amount is $L_{k,j,t}^{orig} = p_j(q_{k,j,t}) L_{k,j,t}^{app}$. At the symmetric equilibrium, sector-level originations are $L_{j,t}^{orig} = p_j(q_{j,t}) L_{j,t}^{app} = p_j(q_{j,t}) \omega_{j,t} L_t^{app}$.

The model assumes that denied applicants do not resubmit another application within the same period. The reason is twofold. First, if within-period reapplication is allowed under both the loose- and tight-regulation regimes, the model predicts that the aggregate NBFI approval rate increases as bank regulation tightens. When banks tighten their underwriting standards, the pool of bank-denied applicants expands to include borrowers with relatively higher creditworthiness who still satisfy the NBFI cutoff. If these borrowers reapply to NBFIs, the approval rate among reapplicants increases. At the same time, tighter bank regulation shifts more initial applications directly to NBFIs. Since these initial NBFI applicants have a higher approval probability than bank-denied reapplicants, their larger share also increases the aggregate NBFI approval rate. However, the empirical analysis does not show such an increase in the NBFI approval rate. Moreover, if rejected bank applicants can subsequently reapply to NBFIs, the NBFI underwriting cutoff effectively becomes the final underwriting standard for these borrowers. In the absence of a decline in the

NBFI approval rate, this mechanism would not imply a deterioration in the credit quality of originated borrowers or an increase in their subsequent default rates. However, the empirical analysis shows that borrower credit quality declines and subsequent default rates increase after bank regulation tightens.

Second, there is empirical evidence that a rejection decision generates meaningful delays in subsequent credit applications, either to the same lender or to other institutions. Han et al. (2009) shows that borrowers can be discouraged from applying for credit when they perceive a high probability of rejection, while Albertazzi et al. (2017) documents that previous rejection increases the probability that borrowers stop searching for credit. Finally, Freddie Mac (2022) provides direct evidence in the context of mortgage applications that denied borrowers do not reapply and instead delay subsequent applications while addressing credit-score, credit-history, or down-payment constraints.

4.3.4. Settlement and Budget Constraint

Loans originated at date t settle at date $t + 1$. Approved borrowers receive idiosyncratic income shocks, and each borrower decides whether to default after the income shock is realized. A borrower defaults when realized income falls below a threshold. Aggregating these decisions over the distribution of approved borrowers' creditworthiness and the income process, the default rate under underwriting stringency $q_{j,t}$ can be derived as $d_{j,t} = d_j(q_{j,t})$, with $d'_j(q_{j,t}) < 0$. The derivation is provided in Appendix F. The reason the default rate is determined by each sector's underwriting stringency is the assumption that application preference shocks are drawn independently of applicants' creditworthiness. As a result, whenever the application volume processed by each sector changes, the underlying creditworthiness distribution remains invariant, and underwriting stringency determines the distribution of approved borrowers.

After taking into account the default, the borrower budget constraint can be derived as

$$P_t C_t^b + (1 - d_{t-1}) R_{t-1}^\ell L_{t-1}^{orig} = W_t N_t^b + L_t^{orig} \quad (18)$$

where R_t^ℓ is the aggregate loan-rate index and the repayment wedge d_t is defined implicitly by

$$(1 - d_t) R_t^\ell L_t^{orig} = \sum_j (1 - d_{j,t}) R_{j,t}^\ell L_{j,t}^{orig}. \quad (19)$$

Current borrowing resources are the principal of approved and originated loans. Rejected applications provide no resources and create no repayment obligations, so only L_t^{orig} enters the budget.

Lastly, default appears in two ways. First, the defaulted repayment represents a financial loss borne by the financial intermediaries and the savers. Given the amount of default, it is allocated between intermediaries and savers through κ_j and $1 - \kappa_j$. Second, it entails an additional deadweight resource cost equal to $\zeta \sum_{j \in \{B, NB\}} d_{j,t} L_{j,t}$. This term represents real resources destroyed in default resolution, including legal and recovery costs. This reduced-form measure is distinct from the default that is transferred between borrowers and savers in that this cost is a direct destruction of aggregate resources rather than a financial transfer between private agents. This deadweight cost is borne directly by the saver as a lump-sum charge.

4.4. Savers

Savers constitute a continuum of infinitely lived households of measure $1 - \chi$ in the economy. A representative saver maximizes expected lifetime utility

$$\max_{\{C_t^s, N_t^s, B_t, D_t, F_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} (\beta^s)^t \left[\mathcal{U}(C_t^s, N_t^s) + \Xi_L \mathcal{L} \left(\frac{D_t}{P_t}, \frac{F_t}{P_t} \right) \right], \quad (20)$$

subject to the budget constraint

$$P_t C_t^s + B_t + D_t + F_t + T_t + \Psi_t^D = W_t N_t^s + R_{t-1} B_{t-1} + R_{t-1}^D D_{t-1} + R_{t-1}^F F_{t-1} + \text{Div}_t^s. \quad (21)$$

by choosing consumption C_t^s , labor N_t^s , and financial vehicles. B_t , D_t , and F_t denote holdings of bonds, bank deposits, and money market fund claims that finance NBFIs, respectively. W_t denotes the nominal wage and Div_t^s collects profit income from intermediate-good firms, banks, and NBFIs.

T_t captures the lump-sum charges associated with defaults by the two financial intermediaries that are not internalized by the financial sector.

$$T_t = \sum_{j \in \{B, NB\}} (1 - \kappa_j) d_{j,t-1} R_{j,t-1}^\ell L_{j,t-1}^{\text{orig}}. \quad (22)$$

For loans originated by intermediary j , the fraction κ_j is absorbed by intermediary equity, while the remaining fraction $1 - \kappa_j$ is borne directly by savers through claims they hold.²¹ Because savers also own banks and NBFIs, losses absorbed by intermediary equity can still reduce their dividend income and wealth. Ψ_t^D captures the resource costs associated with default, such as legal and recovery costs, and is defined as

²¹If the government guarantees transferred claims, it compensates claim holders and finances the guarantee with taxes on saver households. The resulting burden on savers is equivalent to their absorbing the losses on the transferred claims directly.

$$\Psi_t^D \equiv \zeta \left[d_{B,t-1} L_{B,t-1} + d_{NB,t-1} L_{NB,t-1} \right].$$

I assume GHH preferences (Greenwood et al. 1988). For $\sigma \neq 1$, utility is given by

$$\mathcal{U}(C_t^s, N_t^s) = \frac{1}{1-\sigma} \left[C_t^s - \Xi_s \frac{(N_t^s)^{1+\phi}}{1+\phi} \right]^{1-\sigma}. \quad (23)$$

When $\sigma = 1$, utility takes the logarithmic form. Real liquidity services are provided by a CES aggregate of deposits and MMF claims. Feenstra (1986) shows that introducing liquidity services directly into utility is isomorphic to specifying money demand in the presence of transaction costs. Following Begenau and Landvoigt (2022), deposits and MMF claims are imperfect substitutes because both provide low-risk and liquid transaction services. I capture this imperfect substitutability using the following CES liquidity aggregator

$$\mathcal{L}\left(\frac{D_t}{P_t}, \frac{F_t}{P_t}\right) = \left[\alpha \left(\frac{D_t}{P_t}\right)^{\frac{\vartheta-1}{\vartheta}} + (1-\alpha) \left(\frac{F_t}{P_t}\right)^{\frac{\vartheta-1}{\vartheta}} \right]^{\frac{\vartheta}{\vartheta-1}}, \quad \alpha \in (0, 1), \quad \vartheta > 0, \quad (24)$$

where ϑ denotes the elasticity of substitution between deposits and MMF claims. The aggregate quantity of liquid claims is determined by intermediary balance sheets, while savers choose their composition conditional on equilibrium liquidity availability. Combining the first-order conditions for D_t and F_t gives

$$\frac{D_t}{F_t} = \left(\frac{\alpha}{1-\alpha} \frac{R_t - R_t^F}{R_t - R_t^D} \right)^{\vartheta}. \quad (25)$$

Equation (25) describes the household's portfolio allocation between two liquid liabilities, bank deposits and the short-term funding of NBFIs. Each spread against the policy rate measures the financial return forgone in exchange for the liquidity services provided by the corresponding instrument. When the deposit spread $R_t - R_t^D$ widens relative to the MMF spread $R_t - R_t^F$, deposits become more costly to hold and households shift toward MMFs.

4.5. Rest of the Standard New Keynesian Block

Intermediate-good producers are monopolistic competitors with a linear production technology, so real marginal cost is equal to the real wage divided by aggregate productivity. Nominal rigidities are introduced following Calvo (1983). The production block then provides the New Keynesian Phillips Curve, whose log-linearized version is

$$\pi_t = \kappa_\pi mc_t + \beta_s \mathbb{E}_t \pi_{t+1}. \quad (26)$$

where mc_t denotes the log deviation of real marginal cost from its steady state and π_t denotes the log-linearized gross inflation rate, where $\Pi_t = P_t/P_{t-1}$. The discount factor β_s is used because intermediate-good firms are owned by savers, and therefore future profits in the firms' price-setting problem are discounted using the savers' discount factor. The steady-state inflation rate is zero, so the steady-state gross inflation rate is one.

Monetary policy sets the nominal interest rate using a Taylor-type rule with interest-rate smoothing and an exogenous policy shock,

$$\frac{R_t}{\bar{R}} = \left(\frac{R_{t-1}}{\bar{R}} \right)^{\rho_R} \left(\left(\frac{\Pi_t}{\bar{\Pi}} \right)^{\phi_\pi} \left(\frac{Y_t}{\bar{Y}} \right)^{\phi_y} \right)^{1-\rho_R} \exp(\varepsilon_t^R). \quad (27)$$

Market clearing. The labor market clears when total labor demand equals the sum of saver and borrower labor supplies. The goods market clears when output covers total consumption together with the real resources spent on intermediation. Define the resource costs of financial intermediation associated with the portfolio as

$$\Psi_t^I \equiv s_B(q_{B,t-1})L_{B,t-1} + s_{NB}(q_{NB,t-1})L_{NB,t-1} + c_B(L_{B,t-1}, E_{B,t-1}; \nu),$$

where Ψ_t^I collects the resources used for underwriting and the capital-regulation friction.

The goods market clearing condition can then be written compactly as

$$Y_t = (1 - \chi)C_t^s + \chi C_t^b + \frac{\Psi_t^I + \Psi_t^D}{P_t}. \quad (28)$$

Underwriting standards consume resources in proportion to originated lending in each sector, while the capital-regulation friction also absorbs resources as in [Ulate \(2021\)](#). These costs are summarized by Ψ_t^I . Default entails additional resource costs summarized by Ψ_t^D as discussed in [Section 4.3.4](#).

5. Calibration

This section describes the calibration strategy. I calibrate the model to the U.S. economy at a quarterly frequency. I first discuss the parameters that are set externally. I then explain the remaining parameters that are relevant to sector-level moments for banks and NBFIs, with particular

emphasis on approval rates, default rates, and verification costs.

5.1. Externally Calibrated Parameters

Household. The saver discount factor β_s is calibrated to match 2% annual policy rate and the borrower discount factor β_b is matched with 6.3% annual discount rate following Gatt (2025). Both agents' intertemporal elasticities are set to one, $\sigma_s = \sigma_b = 1$. The inverse labor supply elasticity is set to $\phi = 2/3$ as in Herreño and Pedemonte (2025). The labor-disutility weights, Ξ_s and Ξ_b , are chosen so that steady-state labor equals 1/3 for each type, and the borrower population share is $\chi = 0.39$ following Iacoviello (2005).

Saver Liquidity Preference. The saver deposit weight α matches a steady-state deposit-to-money-fund ratio of 3.58 in 2019 (Im et al. 2025). The elasticity of substitution between deposits and money-market funds is calibrated as $\vartheta = 3.17$ following d'Avernas and Vandeweyer (2024).

Borrower Application. The response of applications to approval probabilities, governed by η , is disciplined by the search logic of Agarwal et al. (2024). In their model, the expected benefit of an application is the approval probability times the surplus conditional on approval. I carry this proportionality into the allocation rule by setting $\eta = 1$.²²

The bank–NBFI loan substitution elasticity is set to $\varphi = 6$. Buchak et al. (2025) calibrate the substitution between information-sensitive loans and information-insensitive loans to be around 4. Paul et al. (2024) calibrate the analogous elasticity between bank loans and corporate bonds to 5 and suggest a feasible range of 1 to 11.8 implied by their model. These values provide a broad range for the degree of substitution across credit intermediaries. I further discipline this parameter using the application-level evidence in my sample. Reduced-form regressions of relative bank–NBFI application volumes on relative approval probabilities and relative loan rates imply values of φ between approximately 6 and 11 across alternative specifications, reported in Appendix G.2. Based on both the existing literature and this empirical evidence, I choose $\varphi = 6$ as the baseline value, which lies at the lower end of the range implied by my estimates. A sensitivity analysis over φ , together with η , confirms that the key results are robust across the range that includes the values suggested by Buchak et al. (2025) and Paul et al. (2024) (See Appendix H).

²²This is a reduced-form mapping rather than a derivation from the application-allocation block. In the search model of Agarwal et al. (2024), the expected benefit of an application is the approval probability times the surplus conditional on approval, so it scales in proportion to approval rate p_j . In my allocation rule, approval access enters as the separate demand shifter $[p_j(q_j)]^\eta$, and setting $\eta = 1$ makes application flows scale one-for-one with approval access, matching that proportionality.

I set $\epsilon_B^\ell = \epsilon_{NB}^\ell = 360$, following Ulate (2021) and Abadi et al. (2023).²³ The loan-to-income limit is $\theta = 1.9$ (Campbell and Cocco 2015), adjusted with HMDA. The bank loan-preference weight, $\psi_B = 0.43$, matches the share of bank applications in HMDA after accounting for the steady-state approval rates and loan rates in each sector.

Financial Intermediaries. The equity-payout parameters γ_B and γ_{NB} match the sectoral equity-to-loan ratios, 0.164 for banks and 0.249 for NBFIs (Jiang 2023), conditional on the capital-requirement cost scale $\xi = 0.0012$ (Ulate 2021) and the requirement $\nu = 0.06$. For deposit market power, I set $\epsilon^D = 269.5$ following Ulate (2021).

Non-Financial Intermediaries. For the New Keynesian block, the elasticity of substitution among intermediate goods, θ_{prod} , is set to 3.9, following Abadi et al. (2023), and the Phillips-curve slope, κ_π , is set to 0.0062, as estimated by Hazell et al. (2022). The Taylor-rule parameters are based on Hofmann et al. (2024), $\rho_R = 0.72$, $\phi_\pi = 2.11$, and $\phi_y = 0.26$.

Externally calibrated parameters are summarized in Table 2.

5.2. Internally Calibrated Parameters

The remaining parameters govern each lender’s approval decision, default, screening costs, and the share of default losses internalized by each intermediary.

Because underwriting standards q are not directly observed, I first construct an applicant quality index x from income, the debt-to-income ratio, and loan amount using the HMDA dataset, as in Section 2.3.2. These variables are standardized in the entire pooled sample of bank and NBFI applications, including both approved and denied applications, and the first principal component is used to form the index. The index is normalized so that higher values indicate safer observable applicant profiles. While the index omits credit scores due to the limitation of the HMDA dataset, I merge HMDA with GSE data that report credit scores following Buchak and Jørring (2026) and show a strong positive correlation between the measure I construct and credit scores (See Figure 8). Using this index, I first calibrate the parameters relevant to approval and default. Given these parameters, I then back out the screening cost and default-loss retention parameters from the optimality conditions of the lender’s decision.

²³Note that ϵ (within-sector) and φ (bank–NBFI) are both defined over *gross* interest rates but capture different objects, substitution within sectors and substitution between sectors, respectively. Their magnitudes are therefore not comparable. Single-digit values for substitution within sector calculated using net rates, such as 5 in Gerali et al. (2010), are likewise not directly comparable. For context, evaluated at the steady-state loan rate, the baseline gross-rate elasticity of 360 implies a local net-rate elasticity of approximately 5.4. This confirms that the baseline calibration is in the same ballpark as the single-digit net-rate elasticities typically used in the literature Frame et al. (2026)

TABLE 2. Externally Set Parameters and Targets

Parameter	Description	Value	Target or Source
<i>Panel A. Saver preferences and portfolios</i>			
β_s	Saver discount factor	0.995	Annual policy rate 2%
$1/\sigma_s, 1/\sigma_b$	Intertemporal elasticities of substitution	1	Standard
ϕ	Labor-supply curvature	2/3	Herreño and Pedemonte (2025)
α	Deposit weight in liquidity	0.734	$D/F = 3.58$
ϑ	Deposit–MMF substitution	3.17	d’Avernas and Vandeweyer (2024)
Ξ_L	Liquidity–service weight	0.023	Saver money–fund level FOC
<i>Panel B. Borrower preferences and credit demand</i>			
β_b	Borrower discount factor	0.985	Gatt (2025)
χ	Borrower population share	0.39	Iacoviello (2005)
θ	Loan-to-income limit	1.90	Campbell and Cocco (2015), HMDA
ψ_B	Bank loan–preference weight	0.43	Bank application share, HMDA
φ	Bank–NBFI loan substitution	6	Buchak et al. (2025), Paul et al. (2024), See Text
η	Approval-access elasticities	1	Agarwal et al. (2024), See Text
<i>Panel C. Financial intermediary sector</i>			
γ_B	Bank equity–payout parameter	0.126	Reference target $E_B/L_B = 0.164$
γ_{NB}	NBFI equity–payout parameter	0.014	Reference target $E_{NB}/L_{NB} = 0.249$
ξ	Capital-requirement cost scale	0.0012	Ulate (2021)
ν	Bank regulatory requirement	0.06	Regulatory capital requirement
$\epsilon_B^\ell, \epsilon_{NB}^\ell$	Within-sector loan substitution	360	Ulate (2021); Abadi et al. (2023)
ζ	Default-resolution resource cost	0.10	An and Cordell (2021)
<i>Panel D. New Keynesian block</i>			
θ_{prod}	Intermediate goods substitution	3.9	Abadi et al. (2023)
κ_π	Phillips-curve slope	0.0062	Hazell et al. (2022)
ρ_R	Taylor-rule smoothing	0.72	Hofmann et al. (2024)
ϕ_π	Taylor-rule inflation response	2.11	Hofmann et al. (2024)
ϕ_y	Taylor-rule output response	0.26	Hofmann et al. (2024)

Evidence from the Data Figure 8 provides empirical evidence on the applicant creditworthiness measure by comparing it with credit scores and also shows the shapes of the two schedules used in the quantitative model.

The first panel compares the applicant creditworthiness measure I construct using HMDA information with the credit score after merging HMDA with the GSE dataset. The measure has a strong positive linear correlation with the credit score even though the HMDA dataset does not report credit scores. The second panel of Figure 8 reports, for each value of applicant creditworthiness, the share of applicants who would be approved if the approval cutoff were set at that value. This supports the idea of modeling the approval decision using the logistic distribution in the main text. The right panel plots subsequent loan default against borrower creditworthiness using HMDA applications matched to GSE loan-performance data. Default

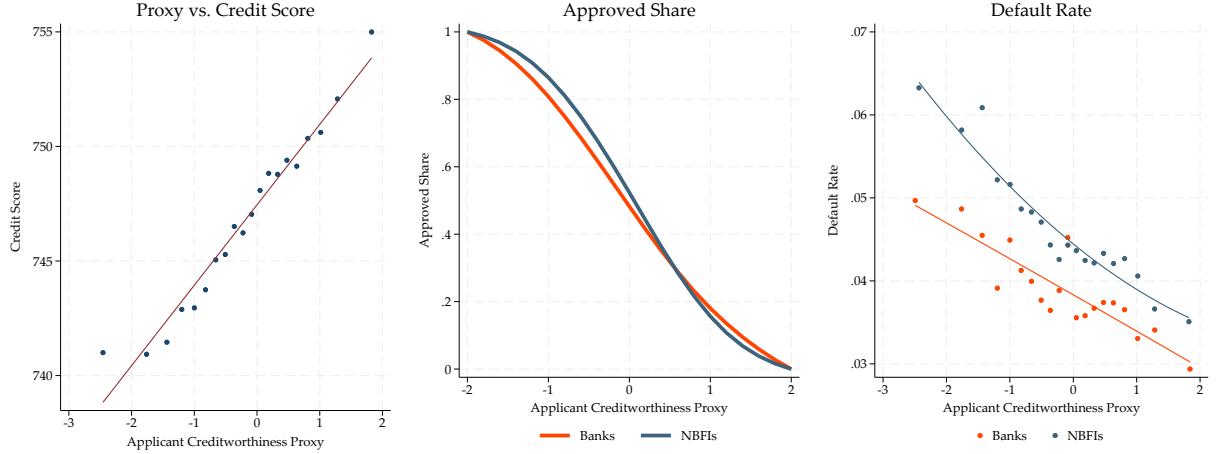


FIGURE 8. Applicant Creditworthiness Proxy and Credit Score, Approval function, Default Schedule

declines with borrower creditworthiness over the empirical support. This relationship motivates a decreasing and approximately log-linear local default schedule.

5.2.1. Approval Schedule

The approval schedule is assumed to follow

$$p_j(q) = \frac{1}{1 + b_{0,j} \exp(b_{1,j}q)}, \quad j \in \{B, NB\},$$

where $b_{1,j} > 0$. Thus, a higher q represents stricter underwriting and implies a lower approval probability. This specification follows from the assumption that applicant creditworthiness is logistically distributed, consistent with Ouazad and Rancière (2016).

Because underwriting stringency q is not directly observed, I first use the applicant quality index x and observed approval decisions to construct its empirical counterpart. Given applicant creditworthiness x , I define each lender-county-year combination as a cell g and, within each cell, estimate the level of applicant creditworthiness that best separates approved from denied applications. I rank applications by x and choose the cutoff $\hat{q}_g$ that minimizes the misclassification error, following Manski (1975). The cutoff $\hat{q}_g$ therefore measures the threshold used by a lender in a given county and year, and a higher $\hat{q}_g$ indicates stricter underwriting. I use this common policy measure as the empirical counterpart of the underwriting choice q in the model for each sector.²⁴

²⁴The object $\hat{q}_g$ summarizes the location of a lender's approval policy at the application stage using the common applicant-quality index x . Although $\hat{q}_g$ does not capture all information used by lenders in actual approval decisions,

I then use $\hat{q}_g$ and observed approval probabilities to calibrate $b_{0,j}$ and $b_{1,j}$ for each sector. The basic idea is to use the common underwriting measure constructed above to estimate the relationship between underwriting stringency and approval probabilities. I first estimate the log-odds relationship implied by the logistic approval schedule using the cell-level approval rate and the estimated underwriting stringency

$$\log\left(\frac{P_g}{1-P_g}\right) = \gamma_j \hat{q}_g + \mu_{\ell,m} + \tau_{m,t} + \varepsilon_g,$$

where $\hat{q}_g$ is the estimated underwriting-policy location in cell g and P_g is the approval rate in the same cell. I include lender-by-county fixed effects $\mu_{\ell,m}$ and county-by-year fixed effects $\tau_{m,t}$. Under this common-scale normalization, the model implies $\gamma_j = -b_{1,j}$.

I calculate the sector-level underwriting standard q_j as the simple average of the estimated cutoffs $\hat{q}_g$ within each sector, $q_j = \frac{1}{N_j} \sum_{g \in \mathcal{G}_j} \hat{q}_g$, and use this average as the empirical target for the model's steady-state underwriting standard.

Finally, I calibrate $b_{0,j}$ so that the approval schedule evaluated at q_j matches the observed approval rate in each sector. The target approval rates are 80.13 percent for banks and 88.09 percent for NBFIs. I focus here on the main intuition, while Appendix G provides the full estimation procedure and additional details.

5.2.2. Default Schedule

I specify the default rate given underwriting stringency as

$$d_j(q) = \delta_{0,j} \exp(-\delta_{1,j}q), \quad d'_j(q) < 0. \quad (29)$$

There are two reasons for using this exponential function specification. First, the loan-level data show that subsequent default declines approximately exponentially with borrower creditworthiness. I measure borrower creditworthiness using the first principal component of borrower characteristics and examine its relationship with subsequent default using HMDA applications matched to GSE loan-level performance data, following Buchak and Jørring (2026). Since underwriting stringency changes the creditworthiness of borrowers who receive loans, this relationship provides a direct link between underwriting standards and the default rate of originated loans.

Second, the borrower default problem provides a microfoundation for the negative relationship between underwriting stringency and portfolio default risk. Borrower repayment capacity

its advantage is that it provides a common measure that can be compared across banks and NBFIs.

increases with creditworthiness, and default occurs when realized repayment capacity falls below a fixed threshold. Under the income shock process that follows a fat-tailed distribution as in Toda (2012), default probability declines exponentially with the underwriting tightness. The logic is that tighter underwriting raises the creditworthiness cutoff for approval and therefore lowers the average default probability among originated loans. Around the steady state, this relationship is summarized by the exponential schedule in equation (29). Appendix F provides the exact portfolio-default expression and motivates this parameterization.

I calibrate the slope of the schedule using loan-level performance data and the same applicant-quality index used in the approval calibration. I match HMDA applications to GSE loan performance data and estimate the relationship between borrower quality and subsequent default separately for banks and NBFIs. Appendix G shows the details of the procedure for estimating $\delta_{1,j}$.

I then choose $\delta_{0,j}$ so that the default schedule matches the observed sectoral default rate at q_j . The annual default rates are 1.76 percent for banks and 4.51 percent for NBFIs, based on GSE loans matched to HMDA applications. I convert these rates to quarterly rates and use them as the calibration targets to match the frequency of the model.

5.2.3. Screening Cost and Effective Default Loss Retention

Screening uses real resources, and its cost is assumed to increase with underwriting stringency according to

$$s_j(q) = c_j \exp(q).$$

I use the exponential form instead of the quadratic screening cost used in Dell’Ariccia et al. (2014). The reasons are twofold. First, the underwriting index q is measured using a standardized principal component and can take both positive and negative values. Under a quadratic specification, moving q further below zero raises screening costs even though underwriting becomes less stringent. This is not consistent with the idea that less screening requires fewer resources. Second, because q is measured using a principal component, its origin is a normalization. Under the exponential specification, a shift in the origin of q is absorbed by c_j without changing the shape of the cost function.

Given the calibrated approval and default schedules, I jointly calibrate the screening cost parameter c_j and the effective default loss retention parameter κ_j from the lender’s pricing and underwriting conditions. More specifically, I choose c_j and κ_j so that both conditions hold at the observed steady state, given the calibrated approval and default schedules.

The recovered values of κ_j are 0.91 for banks and 0.75 for NBFIs, and both sectors yield interior values, with $\kappa_B > \kappa_{NB}$. While these values are model-implied parameters conditional

on the steady-state targets, the recovered ordering is consistent with the difference in default loss internalization between banks and NBFIs discussed in Section 3.4.2. I also additionally show that interest rates increase less steeply with observable borrower risk for NBFIs than for banks, while the probability of default increases more steeply using GSE data in Appendix G.3. Thus, the weaker pricing response of NBFIs is accompanied by a stronger deterioration in realized loan performance, rather than a smaller increase in default risk. This joint pattern is consistent with NBFIs internalizing a smaller share of default losses and supports the calibrated ordering of effective default loss retention across the two sectors.

Internally calibrated parameters are summarized in Table 3.

TABLE 3. Internally Calibrated Parameter Values

Parameter	Value	Description	Target or Source
<i>Panel A. Approval schedule</i>			
$b_{0,B}$	0.2410	Bank approval-schedule level	$p_B(q_B) = 0.8013$
$b_{0,NB}$	0.1311	NBFI approval-schedule level	$p_{NB}(q_{NB}) = 0.8809$
$b_{1,B}$	0.0936	Bank approval-schedule slope	Estimation, See Text
$b_{1,NB}$	0.1407	NBFI approval-schedule slope	Estimation, See Text
<i>Panel B. Default schedule</i>			
$\delta_{0,B}$	0.0045	Bank default-schedule level	$d_B(q_B) = 1.76\%$ annually
$\delta_{0,NB}$	0.0116	NBFI default-schedule level	$d_{NB}(q_{NB}) = 4.51\%$ annually
$\delta_{1,B}$	0.0806	Bank default-schedule slope	Estimation, See Text
$\delta_{1,NB}$	0.1353	NBFI default-schedule slope	Estimation, See Text
<i>Panel C. Verification costs and effective default loss retention</i>			
c_B	0.0002	Bank verification-cost parameter	Bank loan-rate and underwriting-stringency FOCs, equations 10 and 11
κ_B	0.9079	Bank effective default loss retention	Bank loan-rate and underwriting-stringency FOCs, equations 10 and 11
c_{NB}	0.0009	NBFI verification-cost parameter	NBFI loan-rate and underwriting-stringency FOCs, equations 15 and 16
κ_{NB}	0.7490	NBFI effective default loss retention	NBFI loan-rate and underwriting-stringency FOCs, equations 15 and 16

6. Does Bank Regulation Reduce Credit Market Risk?

I first examine how the economy and aggregate default risk change as capital requirements are tightened to varying degrees in the steady state. This allows me to show the mechanism through which the model generates the pattern observed in the data.

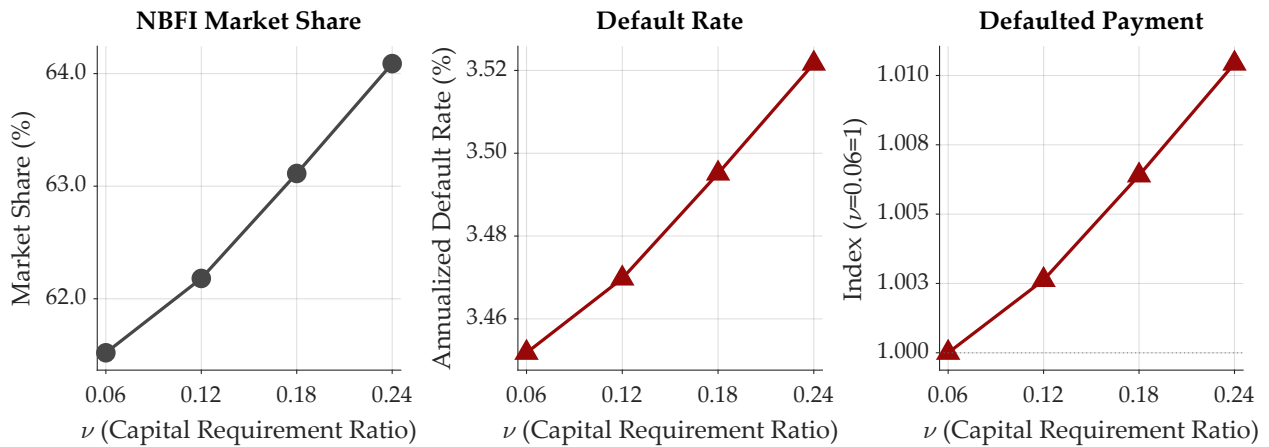
6.1. Steady-State Effects of Tighter Bank Capital Requirements

I recompute the steady-state equilibrium for alternative values of the bank capital requirement ν , holding all other parameters at their calibrated values. The analysis is useful in that it shows through which channels and to what extent those channels contribute to the changes.

I vary ν from 6% to 24%. The lowest value is the model's benchmark value and corresponds to the Basel III minimum Tier 1 capital requirement, while the upper limit represents a stringent counterfactual suggested in the literature, yielding a narrower range than that considered in Begenau and Landvoigt (2022), with an upper bound of 30%.²⁵

Figure 9 shows NBFI share of loan originations, the aggregate default rate, and the aggregate defaulted principal.

FIGURE 9. Intermediation response to higher bank capital requirements



Notes. Each point is computed by re-solving the steady-state equilibrium at the indicated value of ν , holding all other parameters at their benchmark values.

The first panel of Figure 9 shows the regulation generates leakage from banks to NBFIs, consistent with the model in Buchak et al. (2018) and Begenau and Landvoigt (2022). The NBFI share of loan originations rises from 61.5% to 64.1%, an increase of approximately 2.6 percentage points. In my model, this reallocation occurs because a tighter capital requirement raises the cost of bank lending and induces banks to apply stricter underwriting standards with higher loan rates. Bank credit consequently becomes more expensive and less accessible to borrowers, in particular those near the margin between the two lender types. Therefore, given their application preferences, NBFI offers become more attractive, and these borrowers instead apply to NBFIs, whose more permissive underwriting standards allow a larger fraction of them to obtain credit.

The remaining two panels show the effect of tighter capital regulation on the credit risk, which I decompose in greater detail in Section 6.2. The middle panel shows that the aggregate default rate rises from 3.45% to 3.52% as ν increases from 6% to 24%. Banks improve the quality of the loans remaining on their balance sheets by tightening their underwriting standards. However,

²⁵Quantitative studies report a wide range of optimal risk-weighted bank capital ratios, including 16–20% in Miles et al. (2013), 15–23% in Dagher et al. (2016), 13–26% in Firestone et al. (2019), and 16–31%, with a baseline estimate of approximately 23%, in Soederhuizen et al. (2023). Pancost and Robatto (2023) obtain 21.9% in their baseline model and values up to 24.5% in extensions.

this improvement is more than offset by the reallocation of borrowers toward NBFIs, where more permissive underwriting allows riskier marginal borrowers to obtain credit. These are borrowers who apply to banks when capital regulation is relatively loose and the bank offer is marginally better given their application preferences. However, because their observable creditworthiness is below the banks' underwriting cutoff, they are rejected, and their credit risk, which is realized once they receive loans and may result in default, does not enter the aggregate loan pool. Once capital regulations are tightened, they apply to NBFIs and receive funding because they can pass the NBFIs' underwriting cutoff.

The right panel shows the changes made in aggregate defaulted payments. I normalize aggregate defaulted payments to one at $\nu = 0.06$, measuring changes relative to the benchmark while incorporating the endogenous response of aggregate credit volume. The default level rises by approximately 1% when ν changes from 0.06 to 0.24. Total credit contracts, as will be shown in Section 6.2, because NBFI loans are imperfect substitutes for bank loans and borrowers who migrate across sectors recover only part of the credit withdrawn by banks. Nevertheless, the increase in credit risk due to looser NBFI underwriting standards, which allow riskier borrowers who would otherwise remain unfunded to obtain credit, dominates the reduction in credit volume.

6.2. Decomposition of Effect of Regulation on Total Defaults

This section decomposes and quantifies the underlying channels through which total defaults respond to tighter bank capital requirements. Let ω_{NB} denote the NBFI share of applications, as defined in equation (17). Then, originations in each sector j satisfy

$$L_j = \underbrace{p_j(q_j)}_{\text{Approval Rate}} \times \underbrace{\omega_j}_{\text{Application Share}} \times \underbrace{L^{app}}_{\text{Total Applications}}.$$

and aggregate originated lending is therefore

$$L = L_B + L_{NB} = [p_{NB}(q_{NB})\omega_{NB} + p_B(q_B)(1 - \omega_{NB})] L^{app}.$$

Define π_{NB} as the NBFI share of aggregate *originations*. Its relation to the application share ω_{NB} is expressed in odds form as

$$\frac{\pi_{NB}}{1 - \pi_{NB}} = \frac{\omega_{NB}}{1 - \omega_{NB}} \frac{p_{NB}(q_{NB})}{p_B(q_B)}. \quad (30)$$

6.2.1. Default Measures

Given the definitions of π_{NB} and $\pi_B = 1 - \pi_{NB}$, the origination-weighted aggregate default rate is

$$d^{\text{agg}} = \pi_{NB}d_{NB}(q_{NB}) + (1 - \pi_{NB})d_B(q_B), \quad (31)$$

and total defaulted principal can then be written as

$$\begin{aligned} \mathcal{D} &= d_{NB}(q_{NB})L_{NB} + d_B(q_B)L_B \\ &= L \times [\pi_{NB}d_{NB}(q_{NB}) + (1 - \pi_{NB})d_B(q_B)] \\ &= L \times d^{\text{agg}}. \end{aligned} \quad (32)$$

6.2.2. Decomposition

Differentiating total defaulted principal with respect to ν gives

$$\frac{d\mathcal{D}}{d\nu} = L \frac{dd^{\text{agg}}}{d\nu} + d^{\text{agg}} \frac{dL}{d\nu}. \quad (33)$$

The first component captures changes in the default rate for a given amount of aggregate originated credit, while the second captures changes in total defaults arising from the response of aggregate lending.

The response of the aggregate default rate can be decomposed by taking the total derivative with respect to capital regulation tightness ν as

$$\frac{dd^{\text{agg}}}{d\nu} = \underbrace{[d_{NB}(q_{NB}) - d_B(q_B)] \frac{d\pi_{NB}}{d\nu}}_{\text{1. Origination reallocation}} + \underbrace{\pi_{NB}d'_{NB}(q_{NB}) \frac{dq_{NB}}{d\nu}}_{\text{2. Within-NBFI underwriting}} + \underbrace{(1 - \pi_{NB})d'_B(q_B) \frac{dq_B}{d\nu}}_{\text{3. Within-bank underwriting}}. \quad (34)$$

Equation (34) shows a decomposition of the aggregate default rate response into one between-sector margin and two within-sector margins. The first term holds sector-specific default rates fixed and captures changes in the composition of originations across banks and NBFIs, which account for both application migration and differences in underwriting standards. Because NBFI loans have a higher default rate than bank loans in equilibrium, a difference that arises due to the looser lending standards they apply, an increase in π_{NB} raises the aggregate default rate. The second term captures changes in the default rate conditional on NBFI origination. The third term captures the corresponding change in default risk conditional on bank origination.

To further decompose the first term, origination-reallocation channel, I derive the semi-

elasticity of the NBFi origination odds as

$$\frac{d}{dv} \log \left(\frac{\pi_{NB}}{1 - \pi_{NB}} \right) = \frac{1}{\pi_{NB}(1 - \pi_{NB})} \frac{d\pi_{NB}}{dv},$$

which implies

$$\frac{d\pi_{NB}}{dv} = \pi_{NB}(1 - \pi_{NB}) \frac{d}{dv} \log \left(\frac{\pi_{NB}}{1 - \pi_{NB}} \right) \quad (35)$$

Combining equation (30) and the application-allocation equation (17) gives

$$\frac{\pi_{NB}}{1 - \pi_{NB}} = \frac{\omega_{NB}}{1 - \omega_{NB}} \frac{p_{NB}(q_{NB})}{p_B(q_B)} = \frac{\psi_{NB}}{1 - \psi_{NB}} \left(\frac{p_{NB}(q_{NB})}{p_B(q_B)} \right)^\eta \left(\frac{R_{NB}^\ell}{R_B^\ell} \right)^{-\varphi} \times \frac{p_{NB}(q_{NB})}{p_B(q_B)}. \quad (36)$$

Hence, the total derivative of the NBFi share in origination to capital requirement tightness v , which is $d\pi_{NB}/dv$ can be rewritten as

$$\begin{aligned} \frac{d\pi_{NB}}{dv} &= \pi_{NB}(1 - \pi_{NB}) \times \frac{d}{dv} \log \left(\frac{\pi_{NB}}{1 - \pi_{NB}} \right) \\ &= \pi_{NB}(1 - \pi_{NB}) \times \left[\underbrace{\varphi \left(\frac{d \log R_B^\ell}{dv} - \frac{d \log R_{NB}^\ell}{dv} \right)}_{\text{1.1 Relative Loan Rate Application Reallocation}} + \underbrace{\eta \left(\frac{d \log p_{NB}(q_{NB})}{dv} - \frac{d \log p_B(q_B)}{dv} \right)}_{\text{1.2 Approval-probability Application Reallocation}} + \underbrace{\left(\frac{d \log p_{NB}(q_{NB})}{dv} - \frac{d \log p_B(q_B)}{dv} \right)}_{\text{1.3 Approval screening}} \right]. \end{aligned} \quad (37)$$

The first two terms capture changes at the application stage. The relative loan rate component captures the reallocation of applications induced by changes in relative loan rates, while the approval-probability component captures the reallocation of applications induced by changes in relative approval probabilities. As tighter capital requirements increase the bank loan rate R_B^ℓ and lower $p_B(q_B)$, more applications are submitted to NBFIs. The final term arises when applications are processed and converted into actual originations. Holding application shares fixed, a change in relative approval probabilities changes the relative amount of credit originated by banks and NBFIs. Once the response of aggregate originated loans, L , to v is taken into account, the condition under which a tighter capital requirement increases total defaults can be derived as follows.

PROPOSITION 2. *Suppose tighter capital regulation raises bank underwriting stringency, so that*

$dq_B/d\nu > 0$, and reduces aggregate lending, so that $dL/d\nu < 0$. Then $d\mathcal{D}/d\nu > 0$ if and only if

$$\begin{aligned}
& \underbrace{L\pi_{NB}(1 - \pi_{NB}) [d_{NB}(q_{NB}) - d_B(q_B)] \mathcal{S}}_{\text{Origination reallocation}} + \underbrace{L\pi_{NB}d'_{NB}(q_{NB}) \frac{dq_{NB}}{d\nu}}_{\text{Within-NBFI underwriting}} \\
& > \underbrace{L(1 - \pi_{NB}) [-d'_B(q_B)] \frac{dq_B}{d\nu}}_{\text{Within-bank improvement}} + \underbrace{d^{\text{agg}} \left[-\frac{dL}{d\nu} \right]}_{\text{Aggregate lending contraction}}.
\end{aligned} \tag{38}$$

where $\mathcal{S} \equiv \varphi \left(\frac{d \log R_B^\ell}{d\nu} - \frac{d \log R_{NB}^\ell}{d\nu} \right) + \eta \left(\frac{d \log p_{NB}(q_{NB})}{d\nu} - \frac{d \log p_B(q_B)}{d\nu} \right) + \left(\frac{d \log p_{NB}(q_{NB})}{d\nu} - \frac{d \log p_B(q_B)}{d\nu} \right)$

The proposition shows that an increase in the aggregate default rate does not necessarily translate into an equally large increase in total defaults. More specifically, the reallocation of applications toward NBFIs with looser screening standards, together with each sector's underwriting responses, determines the change in d^{agg} , while the response of aggregate originated lending determines the amount of credit to which this default rate applies.

TABLE 4. Four-Channel Decomposition of the Total Defaulted Principal Response

Panel A. Channel contributions (bps)		Panel B. Equilibrium objects	
Channel	Contribution	Object	Value
1. Origination reallocation	5.0848	Reallocation strength $\mathcal{S}$	0.3124
2. Within-NBFI underwriting	0.0004	NBFI underwriting response $dq_{NB}/d\nu$	-0.0001
3. Within-bank underwriting	-0.0611	Bank underwriting response $dq_B/d\nu$	0.0448
4. Aggregate lending volume	-3.3782	Aggregate lending response $dL/d\nu$	-0.0061
Total	1.6458		

Notes. ω_{NB} denotes the NBFI application share and π_{NB} denotes the NBFI origination share. Channel contributions in Panel A are basis points of benchmark aggregate originated lending per unit change in ν . The four contributions sum to the response of total defaulted principal normalized by benchmark aggregate lending.

Table 4 shows how large each channel is in contributing to the increase in aggregate defaulted payments in Panel A and how each key variable responds to changes in capital regulation in Panel B. Each value is evaluated at $\nu = 0.06$.

Panel A shows that the origination reallocation channel dominates the other channels, including reduction made in aggregate lending volume. This is mainly driven by $\mathcal{S}$, which is quantitatively large enough to dominate other effects. However, the direct effects of changes in underwriting standards are relatively small. Bank underwriting responds to tighter capital requirements, with $dq_B/d\nu = 0.045$, but the resulting reduction in the bank default rate contributes only -0.061 basis points. This is because changes in underwriting standards do not translate into a large reduction in bank default rates due to the low semi-elasticity of the default rate with respect to lending

standards, $\delta_{1,B}$. The same logic applies to NBFIs. NBFI underwriting responds indirectly through the endogenous funding rate R^F . This is because, as tighter capital regulation is imposed, funding sources also migrate from deposits to money market funds as bank balance sheets shrink and more money market funds are supplied. Then R^F decreases, and therefore q_{NB} decreases due to Proposition 1, but the resulting change in q_{NB} is limited and does not generate a large change in either approval or default rates. Finally, the aggregate lending volume declines because bank loans and NBFI loans are imperfect substitutes, and the rise in the bank loan rate results in lower aggregate lending volume.

6.3. Sensitivity Analysis

I conduct sensitivity analysis by varying four key sets of parameters. First, I vary the elasticity of substitution with respect to loan rates, φ , over the range from 4 to 11 suggested in Paul et al. (2024), which also includes the estimate of around 4 in Buchak et al. (2025). The magnitude of the response changes with φ , but tighter capital requirements continue to increase aggregate credit risk.

Second, I vary the approval elasticity, η , and find only small changes in the results. The approval probabilities change only modestly as capital requirements tighten, so changing the sensitivity of applications to approval probabilities has a limited effect on the reallocation of applications across sectors. This is because approval probabilities change only modestly as capital requirements tighten, leaving the relative approval component that drives application reallocation largely unchanged.

Third, I vary ψ_B , which determines the initial preference toward bank applications. The parameter ψ_B generates a larger quantitative difference because the amount of applications that can migrate to NBFIs depends on the initial shares of banks and NBFIs. As ψ_B increases and the initial NBFI share becomes smaller, the effect of tighter capital requirements on aggregate credit risk becomes smaller. Note that the origination channel is nonmonotonic in ψ_B because the response of sectoral shares depends on the initial shares of both banks and NBFIs. The reallocation effect is strongest when both sectors have sizable initial shares and becomes weaker when either sector dominates.

Finally, I vary the default-loss retention parameters, κ_B and κ_{NB} , and the default sensitivities to underwriting, $\delta_{1,B}$ and $\delta_{1,NB}$. A higher κ_j strengthens the incentive to tighten underwriting by increasing the default loss internalized by the intermediary, but the resulting changes in underwriting are small relative to the reallocation channel. I also recalibrate $\delta_{0,j}$ for each value of $\delta_{1,j}$ so that the benchmark default rate remains unchanged. Therefore, changing $\delta_{1,j}$ affects how default risk responds to changes in underwriting rather than the initial default-rate gap across

sectors. Since the changes in underwriting induced by tighter capital requirements are limited, particularly for NBFIs, variations in $\delta_{1,j}$ generate relatively small differences in the aggregate credit-risk response. Overall, φ and ψ_B mainly affect the magnitude of the results, while variations in η , κ_j , and $\delta_{1,j}$ leave the results largely unchanged. The full sensitivity results are reported in Appendix H.

6.4. Extension

Lastly, I extend the model by relaxing the assumptions made in the baseline. The details and the results from each extension are documented in the Appendix I.

First, I allow for heterogeneous borrowers who respond differently to changes in loan rates and approval rates, while the underlying risks of each borrower type are also different. Risky borrowers are more sensitive to changes in interest rates and approval rates, while their default risks are exogenously higher than those of safer borrowers. Banks and NBFIs only know the share of risky borrowers in the economy but cannot identify borrower types, as they apply the same underwriting stringency rule based on observables. This margin incorporates the possibility of borrower sorting in terms of unobservables. The results show that introducing heterogeneity in risk and borrower sorting mildly decreases the effect of tighter regulation on aggregate credit risk. A new margin that arises is the response of NBFIs' underwriting stringency. This is because risky borrowers now migrate to NBFIs more actively, and NBFIs tighten their underwriting standards because there is a larger share of risky borrowers in their applicant pool, even though they do not identify the actual types of the borrowers.

Second, I allow banks to fund NBFIs, as documented in Acharya et al. (2024). I simplify the credit line contract by fixing the committed amount and the contractual spread over the wholesale funding rate, while NBFIs choose how much to borrow from banks. NBFIs have an incentive to borrow from banks because bank funding reduces their reliance on volatile money market funding. The marginal benefit is the greater stability of their funding source. The marginal cost is the interest paid to banks and NBFIs borrow from banks until the marginal benefit equals the marginal cost. Banks provide the credit line because the interest income and commitment fee compensate for the additional capital cost. This channel amplifies the effect of tighter regulation on aggregate credit risks as banks additionally support the NBFIs' lending.

Third, I allow banks to sell part of their loans. This extension adds a margin through which banks can reduce their capital regulatory burden, which is highlighted in Buchak et al. (2024). The marginal benefit of selling loans is the reduction in the capital cost from holding fewer loans on their balance sheets. The marginal cost is the cost of selling loans, which captures the fees charged by secondary market activity in practice. The extended setup shows that while the loan-

selling margin dampens the effect of regulation on both the aggregate default rate and defaulted repayment, credit risk still increases after tighter capital regulation.

Finally, I relax the assumption that denied applicants do not reapply within the same period. I allow applicants denied by banks to draw a search cost and make one additional application to an NBFI under the model's maintained nested approval structure. I parameterize nonpecuniary search costs using the mean additional-search cost estimated by Agarwal et al. (2024), equivalent to an annual interest-rate spread of 29.7 basis points, while holding all other parameters at their baseline values. Allowing for reapplication attenuates, but does not overturn, the effects of tighter capital requirements on aggregate credit risk. When the capital requirement rises from 6 to 24 percent, the increases in the annualized default rate and total defaulted payments are 7.2 and 24.2 percent smaller than in the benchmark, respectively.²⁶ A concern with this sensitivity analysis is that the calibration targets are pool-level statistics that may already incorporate reapplication. I therefore recalibrate the extended model to the same targets while holding the search cost fixed. The responses of the aggregate default rate and total defaulted payments remain robust to this recalibration.

7. Financial Stress Experiment

Using the calibrated model as a laboratory, I then evaluate whether bank capital regulation achieves its central objective, which is to ensure that banks are sufficiently resilient to absorb adverse shocks, while additionally taking into account credit market responses.

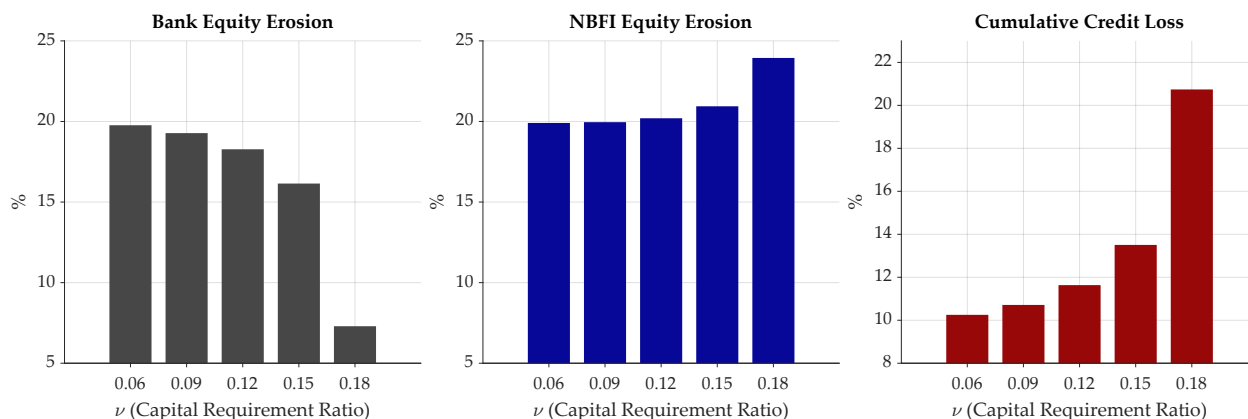
To implement the financial stress experiment, I introduce default shocks on the loans each sector originates and equity shocks for banks and NBFIs. The default shock enters each sector's default schedule multiplicatively as $d_j(q_{j,t}) \exp(\varepsilon_{j,t}^d)$, while the equity shock enters each intermediary's equity evolution equation through an exogenous innovation $\varepsilon_{j,t}^E$. The shocks occur at the beginning of the stress episode. Specifically, I impose an identical increase in the default shock to each sector's default rate function, calibrated to generate a 10 percent increase in aggregate credit losses, which is calculated as the defaulted principal, together with an exogenous 20 percent decline in the equity value of both types of intermediaries. The equity shocks are introduced into the equity evolution equations of the two intermediaries, and these shocks are approximately similar in magnitude to those experienced by the U.S. economy in 2008. The default shocks capture the

²⁶Note that reapplication also changes the response of NBFI approval to tighter capital requirements. In this exercise, NBFI approval increases, whereas the empirical analysis finds no statistically significant response. As banks tighten their underwriting standards, some borrowers who are newly rejected by banks remain eligible for NBFI credit. Their reapplications add eligible borrowers to the NBFI applicant pool and contribute to the increase in its observed approval rate.

increase in defaults that accompanies a severe recession stemming from the house price crash. The equity erosion shocks capture charge-offs and declines in equity valuations due to defaults and losses in the value of the loans.

In the experiment, I measure the troughs in bank equity and NBFI equity during the first four quarters after the shock, as well as cumulative credit losses over the same period, and examine how much credit losses increase following the shock.

FIGURE 10. Financial Stress Outcomes across Bank Capital Requirements



Notes. The figure compares economies with bank capital requirements of 6, 9, 12, 15, and 18 percent. Every economy is exposed to the same four crisis innovations. Bank equity and NBFI equity capture their respective largest declines during the first four quarters. Credit loss is the cumulative aggregate response of defaulted principal over the same period.

The first two panels of Figure 10 report the erosion troughs of bank equity and NBFI equity. The asymmetry arises as capital regulations get tighter. Bank equity erosion is mitigated given the same shock, as banks are less exposed to the credit market. However, NBFI equity erosion increases as capital regulation gets tighter. Once the capital requirement is set to 18%, the effect of the financial stress shock is dampened to below 10%, which is a reduction of more than half. This is due to the fact that, given the same shock in the credit market, as NBFIs are exposed to the credit market and to riskier borrowers, the same increase in defaults results in higher default rates and faster equity erosion. While the magnitude of the rise is less than the reduction in the banking sector, it rises to 23.9%.

The right panel of Figure 10 additionally shows the effect on credit losses. The shock is calibrated to generate a 10% increase in cumulative excess credit losses over the first four quarters following the shock, measured as defaulted principal, when $\nu = 0.06$, and this increase rises to 21% once ν is set to 0.18. This divergence highlights the asymmetric incidence of tighter bank capital regulation across intermediary types. While the regulated banking sector becomes more resilient, the same policy shifts the burden of financial stress toward NBFIs, whose equity becomes increasingly exposed as regulation tightens.

FIGURE 11. IRFs to Financial Stress Shock across Bank Capital Requirements

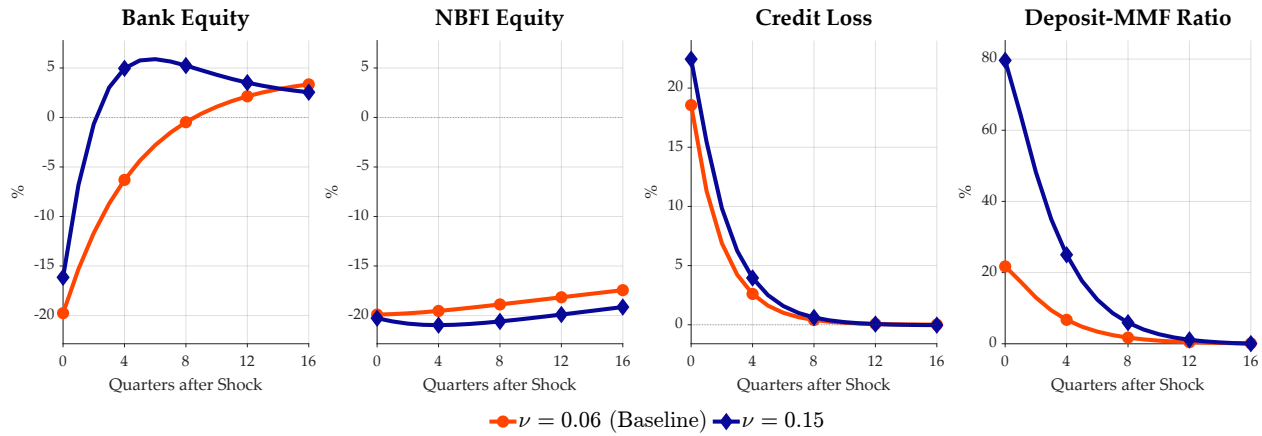


Figure 11 shows the clear differences through impulse responses. I plot the impulse responses of bank equity, NBFI equity, credit losses, and the deposit-to-money market funds ratio, and compare the cases of $\nu = 0.06$ and $\nu = 0.15$. The first two panels clearly exhibit the differences between the two equity responses from a dynamic perspective as well. The first panel shows the impulse response of bank equity and the second shows NBFI equity after the shocks for 16 quarters. Bank equity initially erodes, but after the shock it returns to the steady state under the baseline level of regulation, while it recovers faster as capital regulation becomes tighter. On the other hand, NBFI equity falls and does not return, but rather remains depressed for a prolonged period, and it erodes further as regulation becomes tighter. Credit losses also rise and remain elevated. The reason for the gap between the equity responses can be found in the funding response. The reason becomes clear once I plot the impulse response of the deposit-to-money market funds ratio. Once the credit shock arrives, the funding source migrates from money market funds to deposits, and the degree becomes larger as capital regulation becomes tighter. This implies that once the shock arrives, the banking sector experiences funding inflows while NBFIs experience funding outflows.

The results show that bank regulation can achieve its goal of making the banking sector more robust to adverse shocks. It successfully insulates the banking sector from the credit shock and keeps bank value safe. However, in the credit market, the shocks are now absorbed by NBFIs, as they provide more credit before the shock arrives under higher capital regulation regimes and are therefore more exposed to the credit market. This results in a larger decline in NBFI equity. In addition, their larger presence, combined with looser lending standards and limited banking-sector capacity in lending, generates a larger increase in credit losses. Furthermore, as the funding source migrates from NBFIs to banks, the damage to the NBFI sector is prolonged.

While the model abstracts from channels that connect banks and NBFIs, as highlighted in

Acharya et al. (2024), and NBFI assets can flow into bank assets through securitizations, once these channels are incorporated, the effects could also be amplified by weakening the protective effect of regulation on bank equity, as larger NBFI equity erosion and greater credit losses again affect the banking sector.

8. Policy Implication: Regulating NBFI Underwriting

The main force increasing aggregate credit risk, measured by default rates and default payments, is the reallocation of applications toward NBFIs as bank credit becomes less attractive due to higher loan rates and lower approval probabilities. This channel is weak, however, when the NBFI market share, ω_{NB} , is close to either zero or one, or when the default-rate gap between NBFIs and banks, $d_{NB} - d_B$, is small. In these cases, the increase in risk due to borrower migration may be outweighed by the benefits associated with safer bank lending.

As Proposition 2 shows, this reallocation affects the risk composition of credit only when NBFIs apply less stringent underwriting standards and consequently originate loans with higher default rates. If the two sectors originated loans with similar default risk for otherwise comparable borrowers, regulation would mainly shift credit provision across intermediary types rather than materially change the risk of newly originated loans.

This mechanism motivates a complementary underwriting requirement for NBFIs aimed at limiting the resulting deterioration in the risk composition of credit.

8.1. Regulating NBFI Underwriting

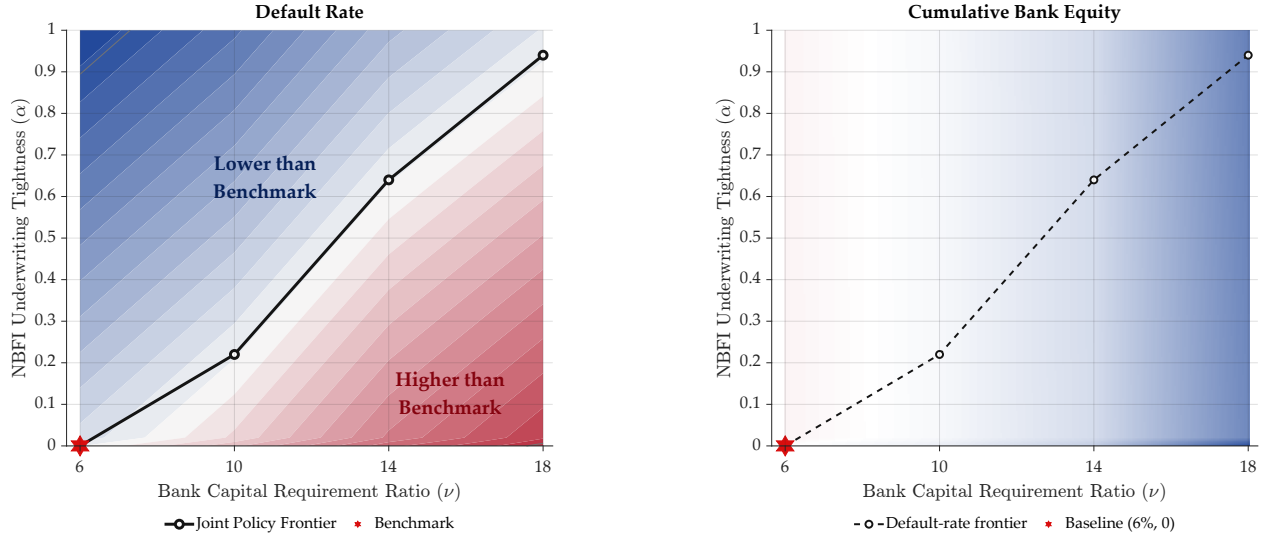
Let $\alpha_q \in [0, 1]$ denote the fraction of the baseline bank–NBFI underwriting gap closed by the requirement:

$$\begin{aligned} q_{NB}^{\text{reg}}(\alpha_q) &= q_{NB} + \alpha_q (q_B - q_{NB}) \\ &= (1 - \alpha_q)q_{NB} + \alpha_q q_B, \end{aligned} \tag{39}$$

Each level of q_B and q_{NB} here is the level at the steady state when the capital requirement is set to be $\nu = 0.06$. When $\alpha_q = 0$, the NBFI chooses q_{NB} from its own first-order condition. When $\alpha_q > 0$, the requirement replaces that condition and fixes q_{NB} at the regulated steady-state target throughout the stress response.

The policy target in this exercise is the payment-weighted default rate over the first twelve quarters after the stress shock. Let $\ell(\nu, \alpha_q)$ denote this default rate at each level of ν and α_q under the financial stress scenario in Section 7. Then, I search the minimum level of $\alpha_q \in [0, 1]$,

FIGURE 12. Joint Bank Capital and NBFU Underwriting Policies: Default Rate and Bank Equity



Notes. The horizontal axis reports the bank capital requirement, ν , and the vertical axis reports the fraction of the baseline bank–NBFU underwriting gap closed by the policy, α_q . The left panel reports the default rate over the first twelve quarters after the stress shock, weighted by scheduled loan repayments and normalized by its baseline value. Blue indicates a default rate below the baseline, while red indicates a default rate above it. The right panel reports the difference between each policy combination and the baseline in the sum of quarterly bank equity percentage deviations from their respective steady states over the first four quarters following the stress shock, measured in percentage-point-quarters. Positive values (blue) indicate less cumulative equity erosion than in the baseline, while negative values (red) indicate greater erosion. The red star marks the baseline economy, $(\nu, \alpha_q) = (0.06, 0)$. In both panels, black markers identify the minimum solved-grid α_q at each ν that keeps the twelve-quarter default rate at or below its baseline level.

intervention on NBFU underwriting stringency, to keep this default rate at or below its value in the baseline economy where $\nu = 0.06$ and $\alpha_q = 0$.

$$\alpha_q^{\text{req}}(\nu) \equiv \min \{ \alpha_q \in [0, 1] : \ell(\nu, \alpha_q) \leq \ell(0.06, 0) \}. \quad (40)$$

Figure 12 plots the twelve-quarter default rate relative to its value when $\nu = 0.06$ without intervention. The blue region captures policy mixes that lower this default rate compared to the baseline, while the red region reports higher default rates. The black line reports the minimum α_q required at a given level of ν to keep this default rate at or below the baseline level. The required adjustment rises with the bank capital requirement. On the solved grid, closing 22% of the baseline underwriting gap is needed at a 10% bank capital requirement. The corresponding values are 64% at a 14% requirement and 94% at an 18% requirement. The larger adjustment reflects the greater migration of lending toward NBFIs as bank capital regulation tightens.

Along the default-rate frontier, the right panel shows that the cumulative bank equity response improves relative to the baseline as the bank capital requirement increases, while this improvement remains almost unchanged as NBFU underwriting is tightened. In particular, under the policy mix

with $\nu = 0.18$ and $\alpha_q = 0.94$, cumulative bank equity erosion is reduced by 68.63 percentage-point-quarters relative to the baseline, which is almost the same as the reduction achieved when there is no NBFI underwriting intervention. Therefore, the two policies are complementary in that tighter NBFI underwriting lowers the default rate while largely preserving the reduction in cumulative bank equity erosion generated by tighter bank capital regulation.

9. Why Does Credit Risk Matter If Loans are Securitized?

This paper highlights how capital regulation can change aggregate credit risk at the origination stage in the first place, focusing on heterogeneous lending standards across financial intermediaries. One could argue that aggregate credit risk generated at origination is of limited concern as long as the loans are subsequently sold and are no longer retained by the originators, especially banks. This concern may appear even less important when the purchasers are the government or government-related institutions, such as GSEs.

However, from the policymakers' perspective, securitization changes who bears credit losses but does not eliminate the losses associated with credit risk. In particular, transferring credit risk to government-related institutions such as the GSEs does not necessarily make an increase in aggregate credit risk less relevant. This was evident during the COVID-19 period, when mortgage delinquency required public measures to support the functioning of the securitization system, ultimately relying on public resources and taxpayer support. The experience of the financial crisis also illustrates this point, as Fannie Mae and Freddie Mac incurred large losses and were ultimately placed into conservatorship with government support. What matters is aggregate credit risk itself rather than the institution that ultimately bears the losses. More fundamentally, defaults generate real resource costs through foreclosure, legal resolution, and property disposition, which arise regardless of which institution ultimately holds the mortgage.

From the banks' perspective, selling loans also does not fully disconnect their exposure from the performance of the loans they eventually sell. Banks can still remain exposed after selling the loans. One important channel is mortgage servicing rights. When default occurs, servicing becomes more costly because banks need to manage delinquent loans and maintain payment flows. Banks can also face putback risk. Poor loan performance can trigger underwriting reviews and require banks to repurchase the loans eventually.

As long as capital requirements are introduced to make credit intermediation more resilient to adverse shocks, bank equity is only one dimension of this objective. Securitization itself therefore cannot mitigate or trivialize a possible increase in credit risk, especially if that increase is generated by the regulation. Importantly, this margin is also a current concern of regulators. [Financial Stability](#)

Oversight Council (2025) points out that bank regulation can generate unintended consequences and economic distortions, noting that increased regulatory costs have shifted credit origination from banks toward NBFIs.²⁷

10. Conclusion

This paper studies how tighter bank capital requirements affect aggregate credit risk when credit intermediation migrates from regulated banks to NBFIs following tighter capital requirements. Using loan-level evidence from the U.S. residential mortgage market, I first show that NBFIs apply more lenient underwriting standards, which can explain the gap between NBFI loans and bank loans in terms of observable risk and their ex post performance. I further show that tighter bank capital regulation contributes not only to the growth of NBFIs but also to a shift in credit toward riskier borrowers, on average, with higher subsequent default rates.

Then, to suggest a mechanism that can explain the empirical facts and quantify to what extent bank regulation can successfully respond to financial stress shocks, I build a general equilibrium model in which banks and NBFIs jointly choose loan rates and underwriting standards, calibrated to the U.S. mortgage market. The model shows that while tighter bank capital requirements induce banks to tighten their standards, they lead applicants to reallocate a larger share of applications and credit toward NBFIs. As more applicants are now processed under looser lending standards, credit risks that would have remained unfunded now materialize by entering the aggregate loan pool and create additional default risk. The model further shows that tighter regulation can successfully insulate the banking sector from financial stress while it leads to a larger deterioration in NBFI equity with larger credit losses. Banks remain stable as they are less exposed to shocks in the credit market, with an inflow of deposits from money market funds, which actually amplifies the losses and erosion that occur in the NBFI sector.

This paper suggests a complementary minimum underwriting requirement for NBFIs that minimizes these unintended costs that regulation can generate. By narrowing the underwriting stringency gap between the two sectors, the requirement limits the credit risk generated by the shift in lending. Now that NBFIs also apply underwriting standards similar to those of banks, even though applications reallocate to NBFIs, credit risk no longer increases. Bank capital requirements and NBFI underwriting regulation therefore serve different purposes and should be treated as

²⁷Financial Stability Oversight Council (2025) states that “Recent experience also suggests that bank regulations can result in unintended consequences and economic distortions. One example of such a distortion in recent years is that as banking regulations have become more complex and costly, the share of credit originated by banks has declined. Lending has been driven out of the banking system, often to nonbank intermediaries.”

complementary instruments.

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Online Appendix

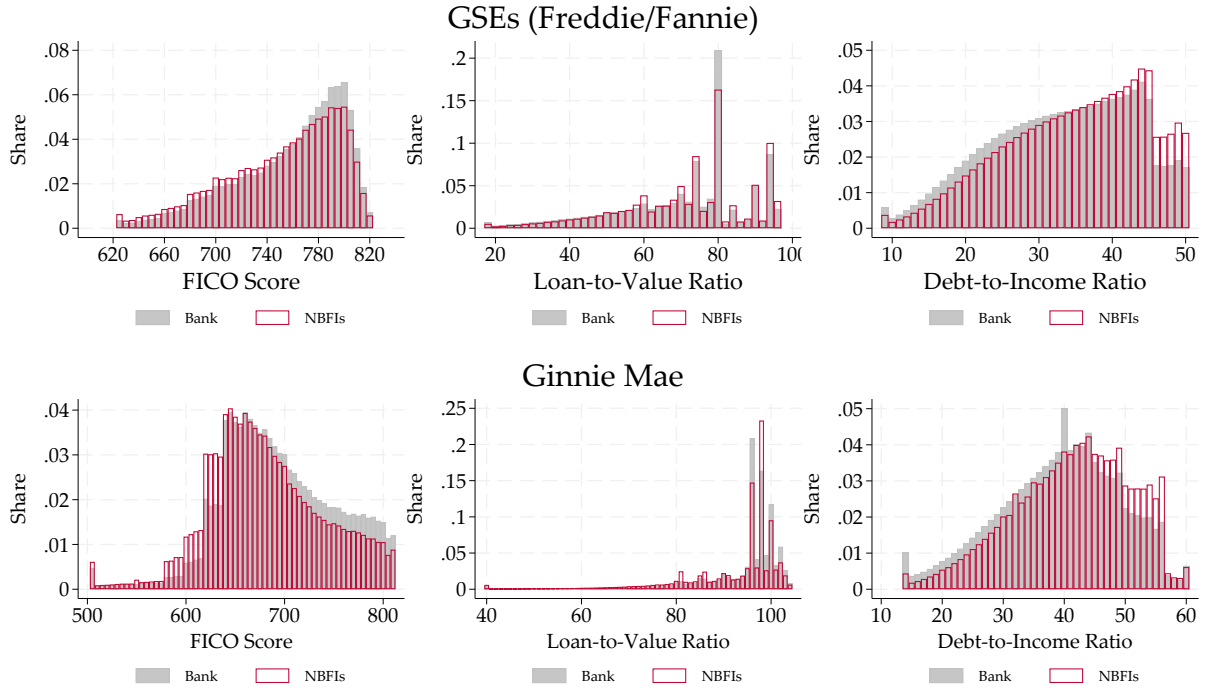
Appendix A. Additional Evidence on NBFIs' Origination

A.1. Univariate Analysis

While the main text constructs measures summarizing borrower creditworthiness, or the inverse of riskiness, using principal component analysis with HMDA, GSE, and Ginnie Mae data, this section compares bank and NBFI loans across each metric, as in Buchak et al. (2018); Fuster et al. (2019); Jiang et al. (2024).

Figure A1 compares the marginal distributions of FICO, LTV, and DTI by originator type. The upper three panels report loans sold to Fannie Mae and Freddie Mac, while the lower three panels report loans sold to Ginnie Mae.

FIGURE A1. Distribution of the Characteristics of the Borrowers



Notes. The upper panels of the figure plot the marginal distributions of FICO, LTV, and DTI for loans sold to Fannie Mae and Freddie Mac. The lower panels plot the marginal distributions for loans sold to Ginnie Mae. The red line corresponds to NBFI originations and the gray line corresponds to bank originations. The sample includes loans originated from 2010Q1 to 2024Q4. *Sources.* Single-Family Loan-Level Datasets provided by Freddie Mac, Fannie Mae, Ginnie Mae and author's calculations.

In the Fannie Mae and Freddie Mac segment, NBFI loans are riskier along the main underwriting margins. NBFI originations place more mass on lower credit scores, especially below 740. They also place more mass at very high LTV values, particularly around 100 percent and above. The DTI distribution shows a similar pattern, with NBFI loans more concentrated in the upper tail, especially above 40 percent and 45 percent. The lower three panels of Figure A1 show that the same risk pattern is stronger in the Ginnie Mae segment. Relative to banks, NBFIs originate more loans with low credit scores particularly below 650. They also originate more loans with high leverage, as reflected in LTV ratios around 90 percent and above, and more loans with high DTI ratios, especially above 40 percent. These distributional differences indicate that NBFI risk is not confined to one underwriting variable. It appears jointly across credit scores, leverage, and debt burden.

Appendix B. Robustness Checks on the Approval Difference and Distribution of Applications

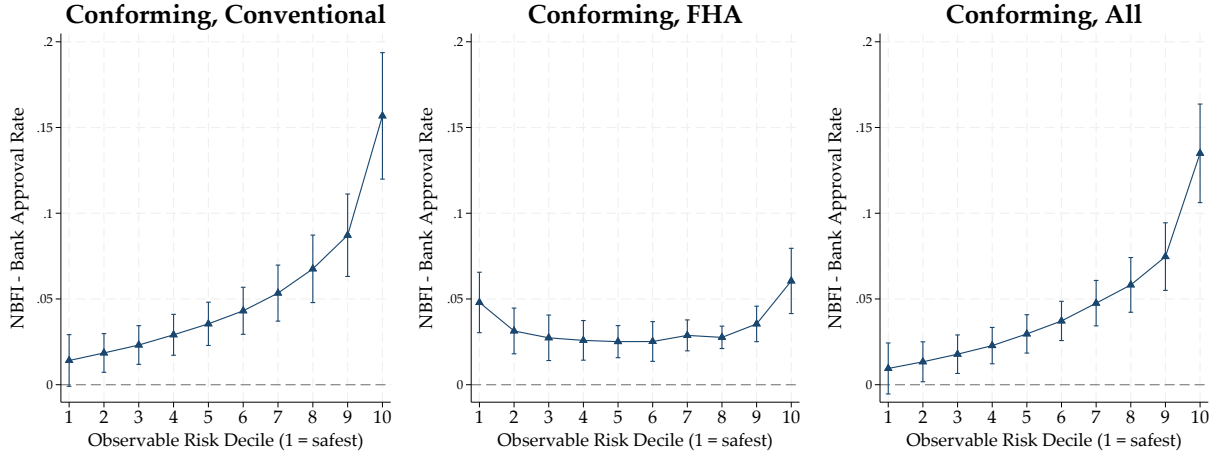
B.1. Approval Difference

To examine more formally whether this difference becomes larger as observable applicant risk increases, I construct an observable risk index using the first principal component of loan size, Debt-to-Income ratio, and income using HMDA data from 2018 to 2024. I normalize the index so that higher values indicate riskier ex ante profiles, corresponding to higher DTI, lower income, and smaller loan amounts. I then divide the index into deciles and interact the NBFI indicator with an indicator for each risk decile.

$$\mathbb{I}(\text{Approved}_{l,t}) = \sum_{d=1}^{10} \beta_d \times \mathbb{I}(\text{NBFI}_l) \times \mathbb{I}(\text{Risk}_l = d) + \sum_{d=1}^{10} \delta_d \times \mathbb{I}(\text{Risk}_l = d) + \Gamma' X_{l,t} + \alpha_{c(l),t} + \varepsilon_{l,t}. \quad (\text{A1})$$

Figure A2 shows the estimated coefficients $\{\beta_d\}_{d=1}^{10}$ for each segment. The approval-rate gap is close to zero among the safest applicants and increases steadily with observable risk, with the largest differences appearing in the highest-risk deciles. This pattern is present for conventional conforming loans and for the full conforming sample, while FHA loans exhibit a relatively constant gap across risk deciles, remaining around 3 to 4 percentage points and rising above 5 percentage points in the highest-risk decile. The results are robust to alternative risk indices, including specifications that incorporate the LTV ratio. These results imply that even for loans that can be sold to third parties, NBFIs are more likely than banks to approve applications with riskier

FIGURE A2. Estimated NBFI–Bank Approval Rate Differences by Observable Risk Decile



Notes. The figure reports estimates of β_d from Regression (A1), which measure the approval-rate difference between NBFIs and banks within each decile of the observable risk index. Higher deciles correspond to riskier observable applicant profiles. The left panel reports estimates for conventional conforming loans, and the right panel reports estimates for the full conforming sample. Vertical bars indicate 95% confidence intervals. Standard errors are two-way clustered at the county and lender levels. *Sources.* HMDA and author's calculations.

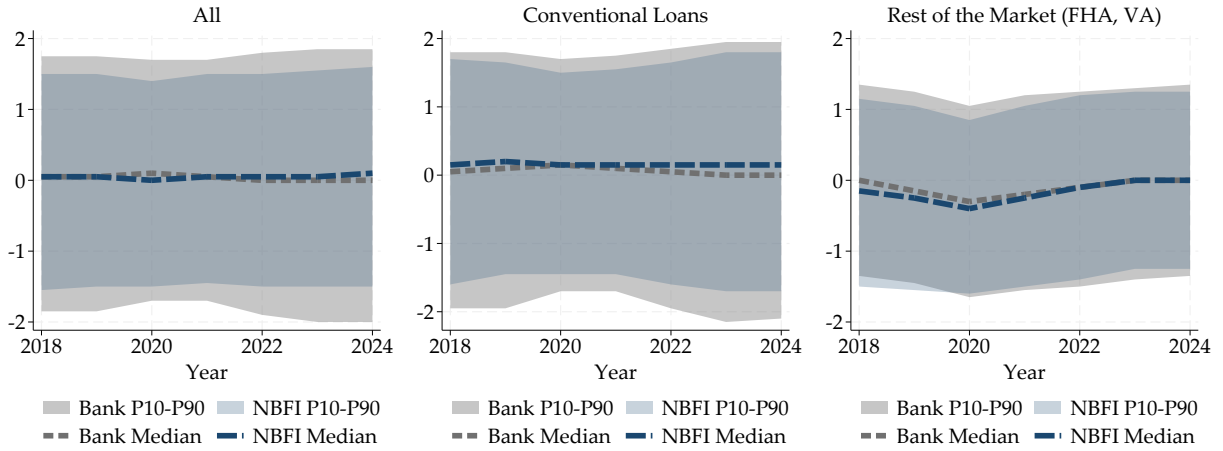
observable profiles, and that the NBFI approval advantage is not uniform across applicants. This pattern is also consistent with the composition of originated loans displayed in Figure 2, as the NBFI share increases among riskier loans while the bank share decreases.

B.2. Distribution of Applications

In this section, I compare the application characteristics at the application stage. I compare the risk measure of the first principal component of income, DTI, and size of the loans that is normalized so that a higher value corresponds to creditworthy borrowers. The measure can be constructed since 2018 and onward as the information of the borrowers on top of the income has been disclosed after 2018.

Figure A3 compares the distribution of applicant risk across lender types. In the full market, the median applicant risk proxy and the 10th-to-90th percentile range are closely aligned between banks and NBFIs. The same pattern holds when the market is divided into conventional loans, which constitute the largest mortgage segment, and government-backed loans, such as FHA and VA loans, whose borrowers generally have riskier credit profiles. Across these segments, applicants to NBFIs do not appear observably riskier than applicants to banks at the application stage.

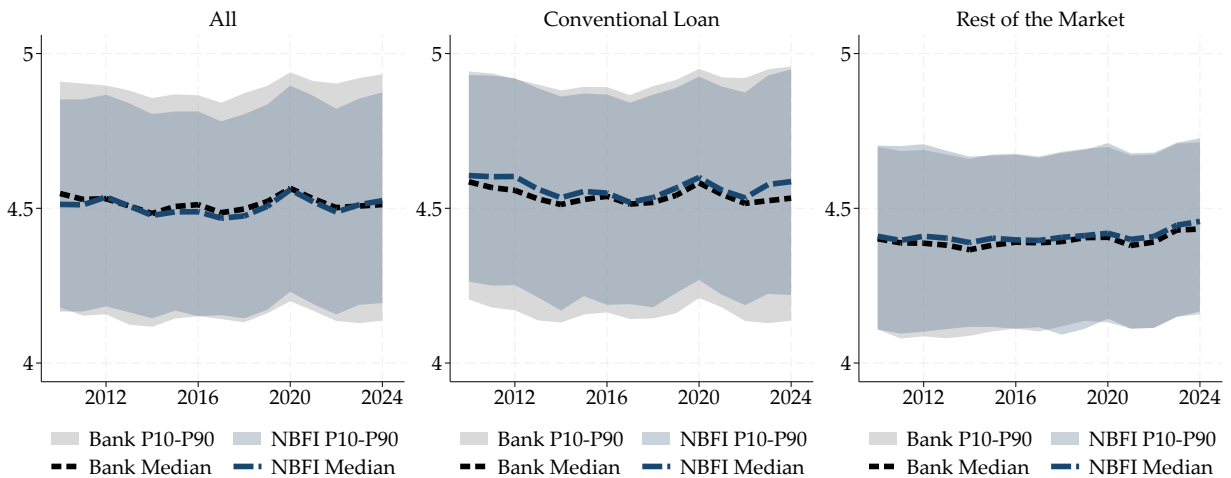
FIGURE A3. Risk Proxy Distribution for Applicants



Notes. The proxy is constructed as the first principal component of standardized income, loan amount, and DTI from 2018 onward. *Sources.* HMDA and author's calculations.

I further examine applicant sorting using real income, which is available over the full post-crisis sample beginning in 2010. Figure A4 repeats the application-stage comparison using applicants' log real income. The figure shows that the income distribution is broadly stable over time and similar across lender types. The medians and 10th–90th percentile bands are closely aligned for bank and NBFI applications throughout the sample.

FIGURE A4. Real Income Distribution of Applicants



Notes. Lines report medians and means. Shaded areas report the 10th–90th percentile range. *Sources.* HMDA and author's calculations.

The full market and conventional loan segment deliver patterns similar to the PCA-based

analysis for the 2018–2024 period. The right panel also shows that applicants in FHA, VA, and FSA/RHS loan segments have lower income distributions than applicants in the conventional loan segment, consistent with the role of these programs in serving lower-income borrowers. Within these government-related segments, however, income distributions remain similar across bank and NBFIs applicants. This pattern supports the PCA-based evidence that observable applicant risk is closely aligned across lender types at the application stage.

Appendix C. Robustness Checks

C.1. Balance in Pre-Regulation Characteristics

In this section, I compare the pre-regulation characteristics of regions with high and low exposure to regulated banks. I classify regions as high exposure if the regulated-bank market share is above the median and report the standardized differences in local characteristics between the high- and low-exposure groups. The figure shows whether regions with greater regulatory exposure systematically differ from less exposed regions in their demographic and economic characteristics before the regulatory change.

FIGURE A5. Balance in Pre-Regulation Characteristics

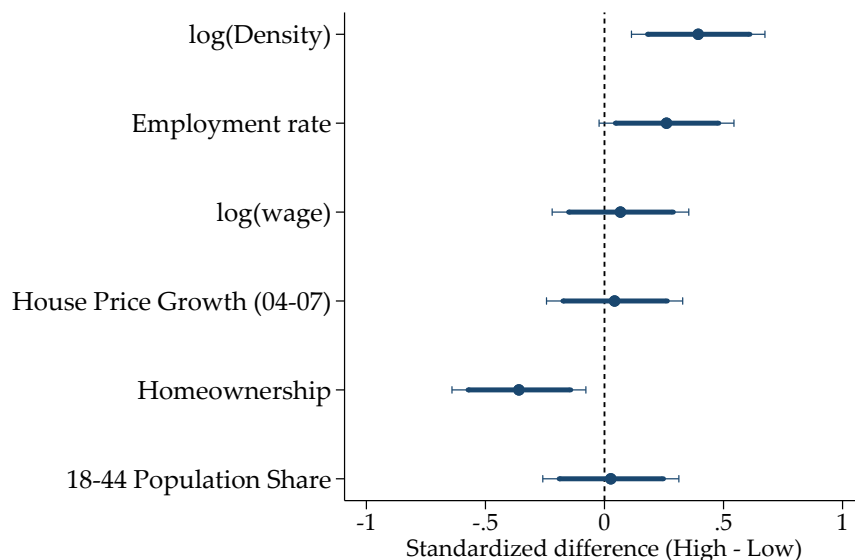


Figure A5 shows the results. Regions with higher regulatory exposure are more densely populated and have higher employment rates, while homeownership rates are lower. In contrast, differences in wages, pre-crisis house price growth, and the share of the population aged 18 to 44

are relatively small and statistically indistinguishable from zero. Overall, the figure indicates that higher-exposure regions differ along some local characteristics, particularly population density, employment, and homeownership, but not systematically across all pre-regulation characteristics.

C.2. Robustness Checks on Main Findings

In this section, I conduct robustness checks on the main findings from the regional exercise showing that greater exposure to regulation is associated not only with the rise of NBFIs but also with a credit shift toward riskier borrowers and higher subsequent default rates.

I first change the covariates by excluding regulatory measures and by excluding all covariates, as suggested in Altonji et al. (2005) and Oster (2019).

**FIGURE A6. Dynamic Effects of Bank Regulation on
NBFi Share in Origination, Borrowers' Riskiness, and Defaults
– Without Regulatory Controls**

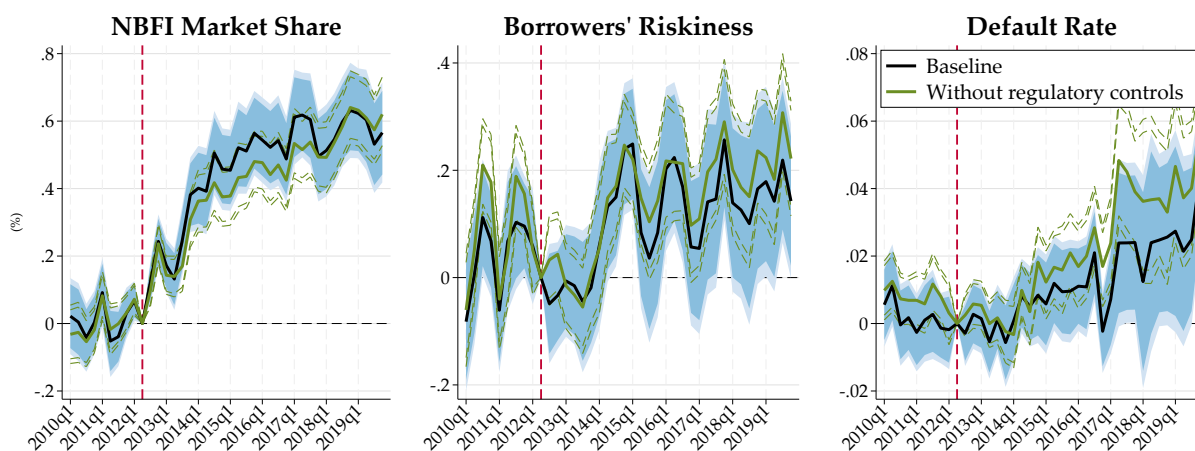
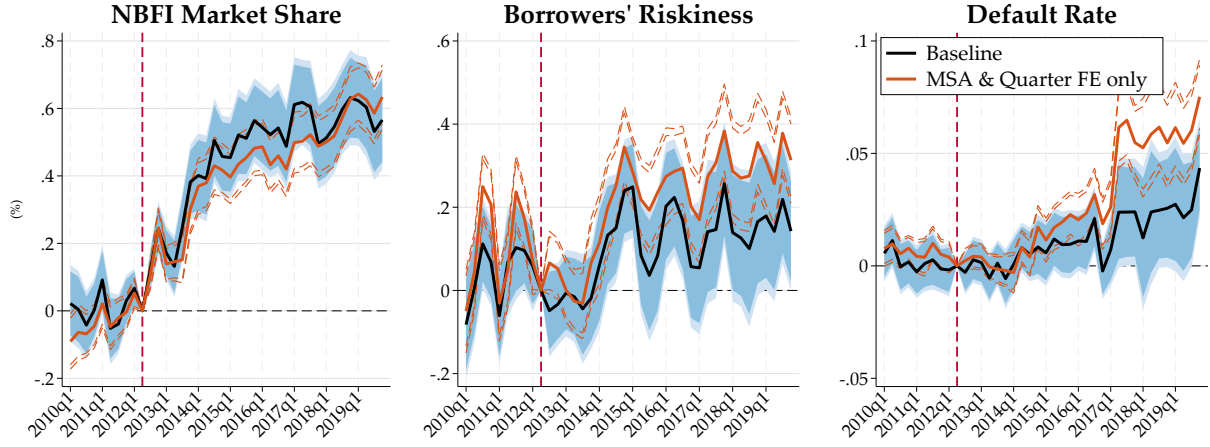


Figure A6 shows the results when regulatory measures are excluded, and Figure A7 reports the case where only fixed effects are included. The results show that in both cases, the responses become more pronounced while the pre-trends are still maintained. This implies that while the covariates are important for adjusting the magnitude of the effect of regulation on the reshaping of the credit market, the main results are not driven by the choice of covariates.

Figure A8 shows the results using the variation in market shares measured in each of these years, and the main message remains the same. Then, I change the treatment by measuring the market share of regulated banks in other years, including 2010, 2011, and 2012. I choose 2008 to avoid taking into account regulated banks' responses to the regulation announcement and to

**FIGURE A7. Dynamic Effects of Bank Regulation on
NBFI Share in Origination, Borrowers' Riskiness, and Defaults
– With only TWFE**



compare with the results of Buchak et al. (2018), but there can still be a concern that 2008 was in the middle of the crisis.

Finally, I estimate the responses of credit scores and defaulted principal. Defaulted principal is calculated as the logarithm of the amount of defaulted principal in each region and quarter. Figure A9 shows the results. Credit scores decline after the regulation, while defaulted principal also increases in regions with greater exposure to the regulation.

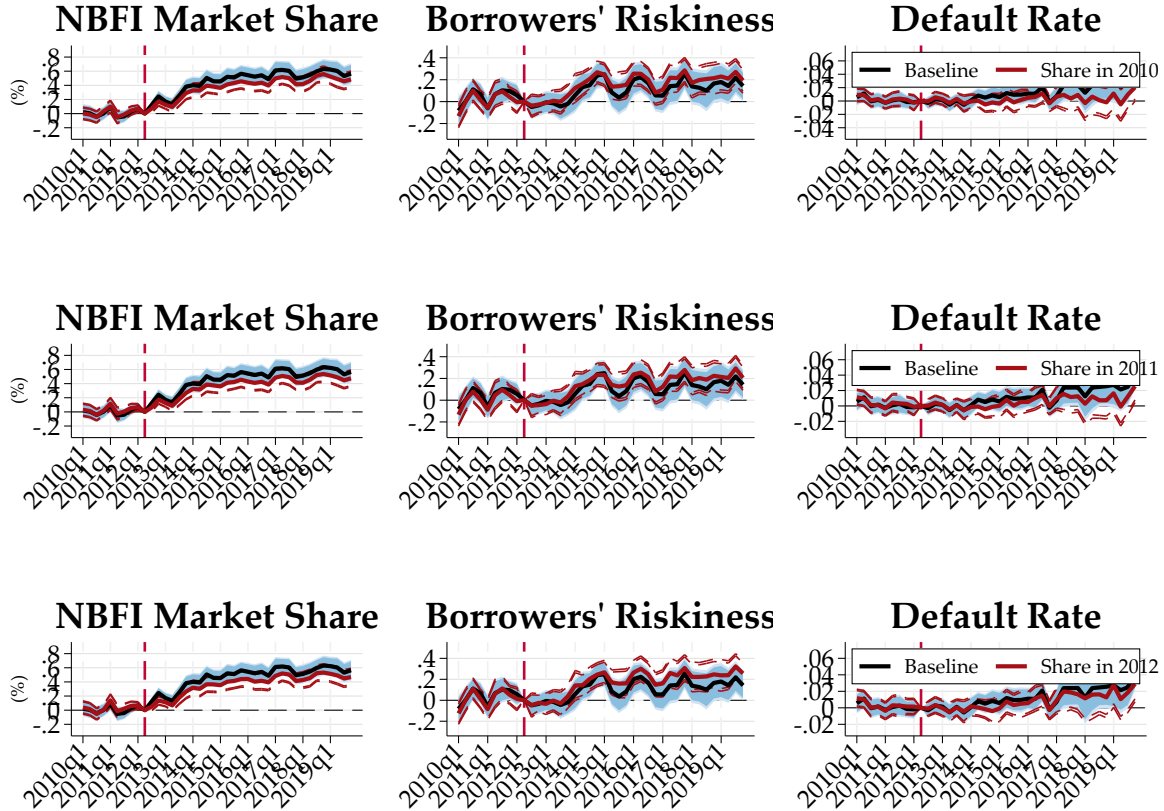
C.3. Robustness of Approval Rate Responses and NBFI Application Share Responses

In the main text, I show the results of the average approval rate responses after the regulation. The results show that regions with greater exposure to the regulation proxy exhibit lower bank approval rates, while NBFI approval rates remain stable. However, since this measures the raw average approval rate, the composition of applicants could matter.

To address this concern, I first obtain the residual of the approval decision, a dummy variable, by regressing it on observable borrower characteristics, including income, loan size, sex and ethnicity of the borrower, and loan purpose and property type. I then recalculate the average approval measure based on this residualized approval decision for each MSA and each sector.

Figure A10 shows the results. The left panel compares the results in the main text with the residualized approval rate. The initial declines from both measures are almost identical, while the

**FIGURE A8. Dynamic Effects of Bank Regulation on
NBFI Share in Origination, Borrowers' Riskiness, and Defaults
– Different Year Weight**



persistence decreases after 2016. The right panel shows the approval rate of NBFIs and shows that the residualized approval rate also remains stable after the regulation, while there is a slight increase in 2016 that vanishes in 2017.

Then, I test the response of the NBFI share of applications after the regulation.

Figure A11 shows the results. The results show that the NBFI share of applications also increases after the regulation is implemented. In terms of magnitude, the increase in the NBFI share of applications is smaller than the increase in the share of originations. The results could partially reflect changes in the approval-rate gap across the two sectors.

FIGURE A9. Dynamic Effects of Bank Regulation on Credit Score and Defaulted Principal

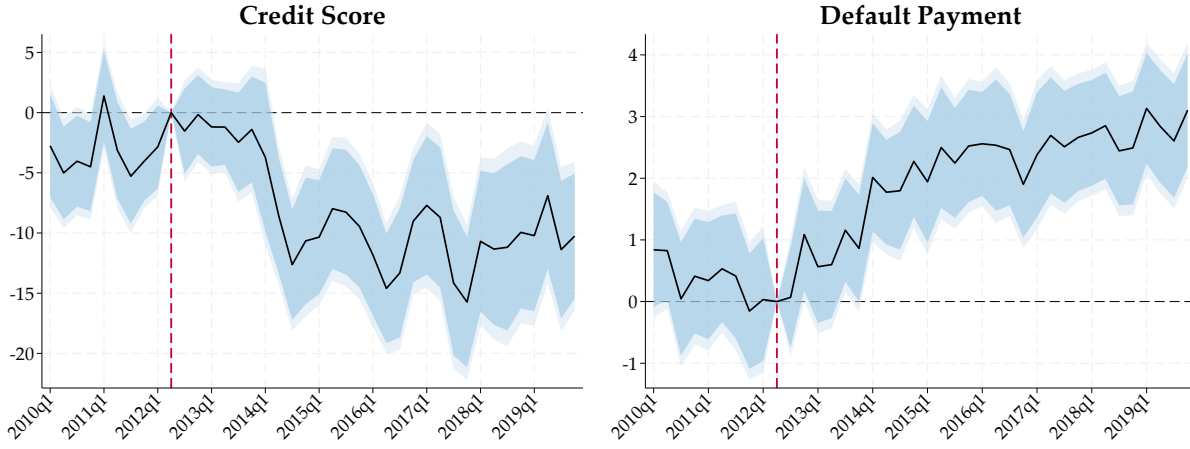
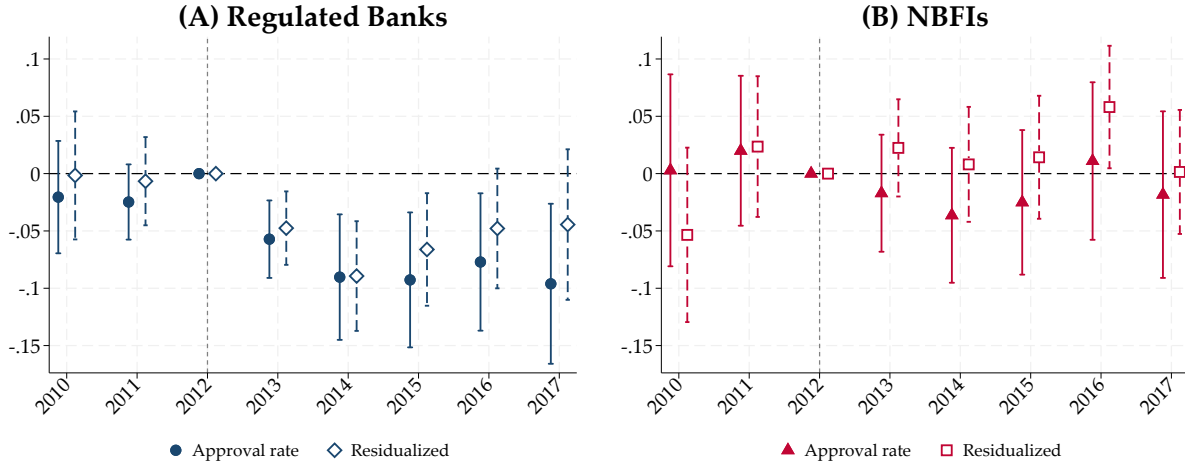


FIGURE A10. Dynamic Effects of Bank Regulation on Raw Approval Rates and Residualized Approval Rate



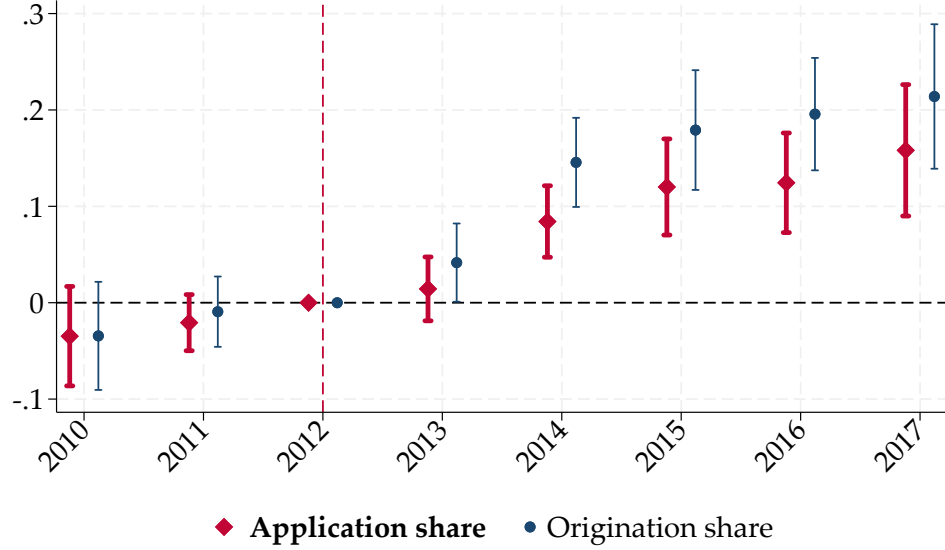
Note: Confidence interval is 95%.

Note. This figure reports the annual event-study estimates β_t from equation (5), where MSA-level regulatory exposure, $Regulation_m$, is interacted with year indicators. The omitted year is 2012. In Panel (A), the dependent variables are the approval rate and residualized approval rate of banks within each MSA. In Panel (B), the dependent variables are the corresponding approval rates of NBFIs. The specification includes MSA fixed effects, year fixed effects, and pre-Basel III local characteristics interacted with year indicators. Vertical bars denote 95% confidence intervals based on standard errors clustered at the MSA level. *Source.* HMDA

C.4. Alternative Treatment Measure

In this section, I construct an alternative measure that captures exposure to regulation. Instead of using the market-share-weighted average of an indicator variable for regulatory exposure,

**FIGURE A11. Dynamic Effects of Bank Regulation on
NBFI Shares of Applications and Originations**



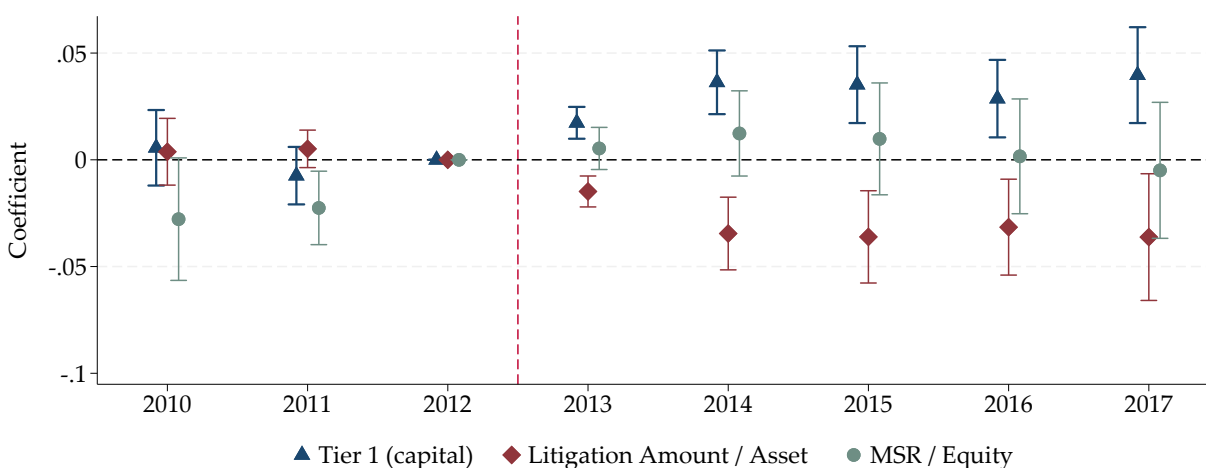
I calculate the regional-level capital ratio as the market-share-weighted average of the capital ratios of regulated banks. I first construct three bank-level measures for initial capitalization, mortgage-related legal exposure, and mortgage servicing rights. The capital measure, Cap_b , is bank b 's Tier 1 risk-based capital ratio in 2008. The legal-overhang measure, $Legal_b$, is the cumulative value of major mortgage-related settlements and enforcement actions through 2012, scaled by the bank's total assets in 2008. The MSR measure, MSR_b , is the bank's mortgage servicing assets relative to equity in 2008. The capital and MSR measures are constructed from FR Y-9C data. I aggregate these bank-level measures to the county level using banks' pre-Basel III mortgage market shares. Each county-level measure is then standardized by its cross-sectional standard deviation. For Cap_c , I reverse the sign after standardization so that a higher value indicates greater exposure to banks with weaker initial capitalization. Thus, counties with higher Cap_c were more exposed to banks facing greater capital pressure before the regulatory changes.

Also, to increase the market coverage from GSE loans to the entire mortgage market, I utilize the HMDA dataset. I then augment the baseline local event-study specification by jointly interacting Cap_c , $Legal_c$, and MSR_c with year indicators.

$$\begin{aligned}
 Y_{c,t} = & \sum_{\tau=2010, \tau \neq 2012}^{2017} \beta_{\tau} \mathbb{I}[t = \tau] \times Cap_c + \sum_{\tau=2010, \tau \neq 2012}^{2017} \theta_{\tau} \mathbb{I}[t = \tau] \times Legal_c \\
 & + \sum_{\tau=2010, \tau \neq 2012}^{2017} \phi_{\tau} \mathbb{I}[t = \tau] \times MSR_c + \mathbf{X}'_{ct} \Gamma + \alpha_c + \delta_t + \varepsilon_{c,t}.
 \end{aligned} \tag{A2}$$

The dependent variable, $Y_{c,t}$, is the NBFI share of mortgage originations by dollar volume in county c and year t . The omitted year is 2012. County fixed effects, α_c , absorb time-invariant differences across local mortgage markets, while year fixed effects, δ_t , absorb aggregate shocks common to all counties. The vector $\mathbf{X}_{ct}$ includes pre-period housing boom and bust measures, log population, homeownership, and unemployment, each interacted with year indicators. The regression is weighted by total county mortgage origination volume, and standard errors are clustered at the county level.

FIGURE A12. Dynamic Effects of Capital Requirement and Confounder on NBFI market share



Note. This figure reports the annual event-study estimates from equation (A2). The plotted coefficients correspond to county-level exposure to capital pressure, legal overhang, and mortgage servicing rights, each interacted with year indicators. The omitted year is 2012. The dependent variable is the NBFI share of mortgage originations. All exposure measures are standardized by their cross-sectional standard deviations, and the capital measure is normalized so that higher values indicate weaker initial capitalization. The specification includes county fixed effects, year fixed effects, and pre-period local characteristics interacted with year indicators. Vertical bars denote 95% confidence intervals based on standard errors clustered at the county level. *Sources.* HMDA, FR Y-9C, and public litigation records.

Figure A12 reports the joint event-study estimates. The coefficients on capital pressure become positive after 2013 and remain positive through 2017, indicating that counties initially more exposed to weakly capitalized banks experienced a larger subsequent increase in the NBFI origination share. In contrast, legal-overhang exposure is negatively associated with the NBFI share after 2013, which is inconsistent with litigation-related retrenchment explaining the baseline increase in NBFI lending. MSR exposure exhibits no persistent positive post-2013 effect and is generally estimated imprecisely. Taken together, the results indicate that the expansion of NBFIs is associated with banks' broader capital pressure rather than with mortgage-related legal overhang or the direct capital treatment of MSRs.

C.5. Within-Bank Analysis

To address the concern that regulated banks may have reduced lending in response to supervisory pressures other than the new capital requirements, I conduct a bank-level analysis that exploits cross-sectional variation in banks' pre-announcement balance-sheet characteristics. In particular, I take Tier 1 leverage capitalization as a proxy for exposure to the Basel III reform and additionally take into account mortgage servicing rights (MSR) holdings, legal exposure, and vulnerability to the supervisory stress tests.

I control for stress-test vulnerability because banks with lower initial capital ratios may also have been considered banks that would experience larger projected losses under the adverse scenario. The estimated effect of the initial capital ratio could then capture responses to stress testing rather than adjustment to the Basel III capital requirements. I therefore measure stress-test exposure using each bank's projected stress loss relative to its pre-announcement total assets.²⁸

The estimating equation is

$$\begin{aligned} y_{it} = & \beta_1 (\text{Low Capital}_i \times \text{Post}_t) + \beta_2 (\text{MSR Exposure}_i \times \text{Post}_t) \\ & + \beta_3 (\text{Legal Exposure}_i \times \text{Post}_t) + \beta_4 (\text{Stress Exposure}_i \times \text{Post}_t) \\ & + \Gamma (X_i \times \text{Post}_t) + \alpha_i + \lambda_t + \varepsilon_{it}, \end{aligned} \quad (\text{A3})$$

where y_{it} is bank i 's mortgage origination activity in year t . Bank fixed effects, α_i , absorb time-invariant differences across banks, while year fixed effects, λ_t , absorb aggregate changes in the economy. Low Capital is defined as the negative standardized value of the bank's pre-announcement Tier 1 leverage ratio, so that a higher value indicates weaker initial capitalization and greater potential adjustment pressure. MSR Exposure is the ratio of mortgage servicing rights to equity, and Legal Exposure is legal exposure scaled by total assets. Stress Exposure is projected stress losses divided by pre-announcement total assets. The vector X_i includes pre-announcement profitability and mortgage-asset exposure, respectively measured relative to total assets. All exposure measures are fixed before the Basel III announcement and interacted with Post_t , which equals one beginning in 2013.

Table A1 shows the results. Consistent with the results in the Appendix section C.4, capital requirements are an important factor that can explain the decline in originations after the implementation of the regulation. Importantly, mortgage servicing rights are also an important margin

²⁸Stress losses equal the sum of projected loan, securities, trading and counterparty, and other losses reported in the 2012 CCAR, scaled by total assets. More specifically, stress losses comprise projected losses on first-lien mortgages, junior liens and HELOCs, C&I loans, CRE loans, credit cards, other consumer and other loans, realized losses on AFS/HTM securities, trading and counterparty losses, and fair-value and goodwill-impairment losses, scaled by total assets.

TABLE A1. Within-Bank Analysis of Mortgage Originations

	Log Mortgage Originations		
	(1)	(2)	(3)
Low Capital $\times$ Post	-0.080*** (0.025)	-0.101*** (0.030)	-0.066** (0.022)
MSR Exposure $\times$ Post	-0.111*** (0.030)	-0.116** (0.040)	-0.078*** (0.025)
Legal Exposure $\times$ Post	0.619 (0.442)	0.902* (0.495)	0.800** (0.352)
Stress Exposure $\times$ Post		0.084* (0.046)	0.183** (0.066)
Profitability $\times$ Post			0.256** (0.112)
Mortgage-Asset Exposure $\times$ Post			0.222 (0.360)
Bank fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	107	107	107
R^2	0.969	0.972	0.974
Within R^2	0.197	0.270	0.318

Notes. The dependent variable is the log of bank-level mortgage originations. Low Capital is the negative standardized value of the bank's pre-announcement Tier 1 leverage ratio. MSR Exposure is mortgage servicing rights relative to equity. Legal Exposure is legal exposure relative to total assets. Stress Exposure is projected stress losses in the 2012 CCAR relative to pre-announcement total assets. Profitability and Mortgage-Asset Exposure are measured relative to total assets. All exposure measures are fixed at their pre-announcement values and interacted with Post, which equals one beginning in 2013. All specifications include bank and year fixed effects. Standard errors clustered at the bank level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

that can explain the reduction, through which tighter capital requirements also operate. While I do not explicitly model the secondary market with servicing rights, regulation of mortgage servicing rights through capital requirements still operates as an important margin.

Appendix D. Proof of Proposition 1

The first-order condition with respect to the loan rate is

$$R^L = \frac{\varepsilon}{\varepsilon - 1} \frac{1}{1 - d(q)} \left(R^f + s(q) \right). \quad (\text{A4})$$

and it follows that the per-dollar margin at the equilibrium can be rewritten as

$$(1 - d(q))R^L - R^f - s(q) = \frac{R^f + s(q)}{\varepsilon - 1}. \quad (\text{A5})$$

The first-order condition with respect to underwriting stringency is

$$-d'(q)R^L - s'(q) + \left((1 - d(q))R^L - R^f - s(q) \right) \left(\frac{p'(q)}{p(q)} + \frac{A'(q)}{A(q)} \right) = 0. \quad (\text{A6})$$

where

$$\frac{A'(q)}{A(q)} = \frac{\partial \log L^{app}(R^L, q)}{\partial q}$$

under the assumption of constant elasticity of substitution with respect to loan rates. Substituting (A4) and (A5) into (A6) gives the reduced underwriting condition

$$\Phi(q, R^f) \equiv -\frac{\varepsilon}{\varepsilon - 1} (R^f + s(q)) \frac{d'(q)}{1 - d(q)} - s'(q) + \frac{R^f + s(q)}{\varepsilon - 1} \left(\frac{p'(q)}{p(q)} + \frac{A'(q)}{A(q)} \right) = 0. \quad (\text{A7})$$

Let $\widehat{\Pi}(q, R^f)$ denote profit *after* optimizing over the loan rate. By the envelope theorem,

$$\frac{\partial \widehat{\Pi}(q, R^f)}{\partial q} = p(q) L^{app} \left(R^{L*}(q, R^f), q \right) \Phi(q, R^f).$$

At an interior optimum, $\Phi(q^*, R^f) = 0$ holds. Therefore,

$$\frac{\partial^2 \widehat{\Pi}(q^*, R^f)}{\partial q^2} = p(q^*) L^{app} \left(R^{L*}(q^*, R^f), q^* \right) \Phi_q(q^*, R^f).$$

Since originated lending is positive, the second-order condition implies

$$\Phi_q(q^*, R^f) < 0. \quad (\text{A8})$$

Differentiating (A7) with respect to R^f , holding q fixed, gives

$$\Phi_{R^f}(q, R^f) = -\frac{\varepsilon}{\varepsilon - 1} \frac{d'(q)}{1 - d(q)} + \frac{1}{\varepsilon - 1} \left(\frac{p'(q)}{p(q)} + \frac{A'(q)}{A(q)} \right). \quad (\text{A9})$$

Equations (A7) and (A9) imply

$$\Phi(q, R^f) = (R^f + s(q))\Phi_{R^f}(q, R^f) - s'(q).$$

Evaluating this expression at the optimum gives

$$(R^f + s(q^*))\Phi_{R^f}(q^*, R^f) = s'(q^*).$$

Since $R^f + s(q^*) > 0$ and $s'(q^*) > 0$,

$$\Phi_{R^f}(q^*, R^f) > 0. \quad (\text{A10})$$

Applying the implicit function theorem to (A7) yields

$$\frac{dq^*(R^f)}{dR^f} = -\frac{\Phi_{R^f}(q^*(R^f), R^f)}{\Phi_q(q^*(R^f), R^f)}.$$

Combining (A8) and (A10), we obtain

$$\frac{dq^*(R^f)}{dR^f} > 0.$$

Thus, a higher funding cost raises the intermediary's optimal underwriting stringency.

Appendix E. Borrower Application Allocation, Default, and Aggregation

E.1. Applications and timing

The representative borrower household consists of a unit measure of members indexed by $i \in [0, 1]$ who pool idiosyncratic application, approval, and income risk. Member i submits an application

$L_{i,t}^{app}$, and total applications satisfy

$$L_t^{app} = \int_0^1 L_{i,t}^{app} di. \quad (A11)$$

Each application's observable creditworthiness is measured by $x_{i,t}$, where higher values indicate safer applicants. At time t , the household chooses C_t^b , N_t^b , and L_t^{app} . Each member submits one application to one lender and cannot reapply after rejection within the same period. Applicants observe posted loan rates and lender-level approval probabilities before applying, but they do not observe lenders' underwriting stringency or their individual approval realizations. At date $t + 1$, borrowers who obtained loans at t decide whether to repay or default given the realization of the income shock.

E.2. Application Allocation Decision

In this subsection, I explain how each member i decides where to apply. Time subscripts are omitted throughout this subsection for simplicity. There are two lending sectors $j \in \{B, NB\}$, which represent banks and NBFIs, respectively, with each sector consisting of a unit measure of differentiated lenders $k \in \mathcal{K}_j$. Lender k in sector j posts a loan rate $R_{k,j}^\ell$ and chooses an underwriting standard $q_{k,j}$, which determines its approval probability $p_j(q_{k,j})$.

Applicant i chooses the lender offering the highest effective application value,

$$(J_i, K_i) \in \arg \max_{j \in \{B, NB\}, k \in \mathcal{K}_j} \left\{ \frac{e_{i,k,j} (\bar{p}_j)^{\eta/\varphi} \left(\frac{p_j(q_{k,j})}{\bar{p}_j} \right)^{\eta/\epsilon_j^\ell}}{R_{k,j}^\ell} \right\}. \quad (A12)$$

This captures the idea that member i seeks an offer with a lower effective loan rate and a higher chance of obtaining the loan.

First, $e_{i,k,j}$ captures application preference shocks across offers, which are reduced-form matching heterogeneity terms used to generate application allocation across lenders. The parameter ϵ_j^ℓ captures substitution across lenders within sector j with respect to loan rates, while φ governs substitution across lending sectors.

The parameter η governs the response of applications to approval probabilities. I impose the same approval elasticity on within-sector and across-sector competitiveness. To do so, let $\bar{p}_j$ denote the sector-level approval environment perceived by applicants. An individual financial intermediary takes $\bar{p}_j$ as given when choosing its own loan rate and underwriting standard. The borrower then chooses based not only on the sector-level approval rate but also, in principle,

on within-sector approval differences. In a sector-symmetric equilibrium, $\bar{p}_j = p_j(q_j)$, and the borrower ultimately responds to sector-level differences.

I assume that the vector of application preference shocks is drawn from the nested Fréchet distribution following Herreño (2023).

$$\Pr(e_{i,k,j} \leq z_{kj} \text{ for all } k, j) = \exp \left\{ - \sum_{j \in \{B, NB\}} \psi_j \left(\int_{\mathcal{K}_j} \mathcal{T}_{kj} z_{kj}^{-\epsilon_j^\ell} dk \right)^{\frac{\varphi}{\epsilon_j^\ell}} \right\}, \quad (\text{A13})$$

where $\psi_j > 0$, with $\psi_B + \psi_{NB} = 1$, captures sector-level non-price attractiveness. The term $\mathcal{T}_{kj} > 0$ captures each lender's specific non-price attractiveness, such as accessibility or prior relationships, and is normalized according to

$$\int_{\mathcal{K}_j} \mathcal{T}_{kj} dk = 1.$$

The nesting parameters satisfy $0 < \varphi \leq \epsilon_j^\ell$ and $\varphi > 1$.

E.2.1. Application allocation

For a given sector j , substituting the deterministic component of (A12) into (A13) gives sector competitiveness

$$s_j = \psi_j (\bar{p}_j)^\eta \left[\int_{\mathcal{K}_j} \mathcal{T}_{kj} \left(\frac{p_j(q_{k,j})}{\bar{p}_j} \right)^\eta \left(R_{k,j}^\ell \right)^{-\epsilon_j^\ell} dk \right]^{\frac{\varphi}{\epsilon_j^\ell}}, \quad (\text{A14})$$

Applications are allocated across sectors according to

$$\omega_j = \frac{s_j}{\sum_{m \in \{B, NB\}} s_m}. \quad (\text{A15})$$

Conditional on applying to sector j , the share of applications received by lender k is

$$v_{k|j} = \frac{\mathcal{T}_{kj} \left(\frac{p_j(q_{k,j})}{\bar{p}_j} \right)^\eta \left(R_{k,j}^\ell \right)^{-\epsilon_j^\ell}}{\int_{\mathcal{K}_j} \mathcal{T}_{k'j} \left(\frac{p_j(q_{k',j})}{\bar{p}_j} \right)^\eta \left(R_{k',j}^\ell \right)^{-\epsilon_j^\ell} dk'}. \quad (\text{A16})$$

Application size is assumed to be common across household members. Choice probabilities therefore also determine the shares of aggregate application face value,

$$L_{k,j}^{app} = v_{k|j} \times \omega_j \times L^{app}. \quad (\text{A17})$$

Because each lender is atomistic, it takes $\bar{p}_j$, ω_j , and the sector aggregates as given when choosing its own loan rate and underwriting standard. Equation (A16) therefore implies

$$\frac{\partial \log L_{k,j}^{app}}{\partial \log R_{k,j}^\ell} = -\epsilon_j^\ell \quad \text{and} \quad \frac{\partial \log L_{k,j}^{app}}{\partial \log p_j(q_{k,j})} = \eta. \quad (\text{A18})$$

Lender-level originations are

$$L_{k,j}^{orig} = p_j(q_{k,j}) L_{k,j}^{app}. \quad (\text{A19})$$

It follows that

$$\frac{\partial \log L_{k,j}^{orig}}{\partial q_{k,j}} = (1 + \eta) \frac{p'_j(q_{k,j})}{p_j(q_{k,j})}. \quad (\text{A20})$$

Thus, the individual lender internalizes both the direct effect of underwriting on the fraction of applications approved and the induced response of its application inflow. This is the origin of the $1 + \eta$ term in the underwriting first-order conditions.

E.2.2. Allocation in the Symmetric Equilibrium

I consider a symmetric equilibrium in which

$$q_{k,j} = q_j, \quad R_{k,j}^\ell = R_j^\ell \quad (\text{A21})$$

for each sector j . In the symmetric equilibrium, the sectoral application share is given by

$$\omega_j = \frac{\psi_j [p_j(q_j)]^\eta (R_j^\ell)^{-\varphi}}{\sum_{m \in \{B, NB\}} \psi_m [p_m(q_m)]^\eta (R_m^\ell)^{-\varphi}}, \quad L_j^{app} = \omega_j L^{app}. \quad (\text{A22})$$

E.2.3. Composition of applicant pools

As I discuss in the main text, the baseline does not assume sorting, in that application preferences are independent of members' creditworthiness.

$$\{e_{i,k,j}\}_{k \in \mathcal{K}_j, j \in \{B, NB\}} \perp x_i, \quad x_i \sim F_x. \quad (\text{A23})$$

This condition implies that the underlying creditworthiness distribution of applications processed by each lender k in sector j is invariant, while the volume changes proportionally at each

level of creditworthiness.

$$\Pr(x_i \leq x \mid (J_i, K_i) = (j, k)) = F_x(x). \quad (\text{A24})$$

E.3. Approval Decision

The underwriting standard of lender k in sector j induces the approval probability

$$p_{k,j}(q_{k,j,t}) = 1 - F_x(\tau_{k,j}(q_{k,j,t})), \quad (\text{A25})$$

where $\tau'_{k,j}(q) > 0$ and therefore $p'_{k,j}(q) < 0$. $\tau(q)$ can be understood as a cutoff that determines whether applicants' creditworthiness is high enough to be approved for loans, and it is assumed to increase with the tightness of the underwriting standard. Equation (A24) implies that the applicant creditworthiness distribution is invariant, and therefore the approval function $p_{k,j}(\cdot)$ is a fixed function of the underwriting standard.

Applicants observe $p_{k,j}$ before applying, but they do not observe $q_{k,j,t}$, the approval threshold $\tau_j(q_{k,j,t})$, or the decisions on their applications. Lenders internalize applicants' behavior when choosing $q_{k,j,t}$.

Once applications are allocated to lender k in sector j , it applies its own underwriting standard and approves an application if $x_{i,t} \geq \tau_j(q_{k,j,t})$. In the symmetric equilibrium, each sector's origination can then be computed as

$$L_{j,t}^{orig} = \omega_{j,t} p_j(q_{j,t}) L_t^{app}. \quad (\text{A26})$$

E.4. Repayment and borrower-family aggregation

Given the no-sorting assumption on borrowers, the distribution of creditworthiness among approved loans is determined by the underwriting stringency that the lender applies. Given the approved borrowers, they then decide whether to default, and the default rate of the given loans can be derived as $d_{j,t} = d_j(q_{j,t})$. The detailed derivation is provided in Appendix F.

The repayment at date $t + 1$ from sector- j loans originated at date t is then

$$(1 - d_j(q_{j,t})) R_{j,t}^\ell L_{j,t}^{orig}. \quad (\text{A27})$$

Under the definition of the aggregate loan rate below

$$R_t^\ell = \left(\frac{\sum_{j \in \{B, NB\}} \psi_j [p_j(q_{j,t})]^\eta (R_{j,t}^\ell)^{-\varphi}}{\sum_{j \in \{B, NB\}} \psi_j [p_j(q_{j,t})]^\eta} \right)^{-1/\varphi}, \quad (\text{A28})$$

the aggregate repayment wedge d_t is defined as

$$\sum_{j \in \{B, NB\}} (1 - d_{j,t}) R_{j,t}^\ell L_{j,t}^{orig} = (1 - d_t) R_t^\ell L_t^{orig}. \quad (\text{A29})$$

The object d_t is the wedge that makes (R_t^ℓ, L_t^{orig}) sufficient for aggregate repayment.

After member-level repayment and default outcomes have been settled, members pool labor income, loan proceeds, and remaining resources. I assume perfect risk sharing across household members, so members whose applications are rejected share household resources with those whose applications are approved. Given the definition of the aggregate loan rate and default wedge, aggregating the member-level settlement outcomes therefore gives

$$P_t C_t^b + (1 - d_{t-1}) R_{t-1}^\ell L_{t-1}^{orig} = W_t N_t^b + L_t^{orig}. \quad (\text{A30})$$

where loans are in nominal terms. Let $c_{i,t}^b$ and $n_{i,t}^b$ denote member-level consumption and labor, and define

$$C_t^b = \int_0^1 c_{i,t}^b di, \quad N_t^b = \int_0^1 n_{i,t}^b di. \quad (\text{A31})$$

Household welfare before imposing the symmetric risk-sharing allocation is

$$\mathbb{E}_0 \sum_{t=0}^{\infty} (\beta^b)^t \int_0^1 \mathcal{U}(c_{i,t}^b, n_{i,t}^b) di. \quad (\text{A32})$$

Because members have identical preferences, face the same wage, and fully share resources after member-level repayment and default have been settled, the symmetric allocation satisfies $c_{i,t}^b = C_t^b$ and $n_{i,t}^b = N_t^b$ for almost every i .

$$\int_0^1 \mathcal{U}(c_{i,t}^b, n_{i,t}^b) di = \mathcal{U}(C_t^b, N_t^b). \quad (\text{A33})$$

The same aggregation applied to the application-stage loan-to-income restriction gives $L_t^{app} \leq$

$\theta W_t N_t^b$. The representative borrower household therefore solves

$$\max_{\{C_t^b, N_t^b, L_t^{app}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} (\beta^b)^t \mathcal{U}(C_t^b, N_t^b) \quad (\text{A34})$$

subject to equation (A30), $L_t^{orig} = \bar{p}_t L_t^{app}$, and $L_t^{app} \leq \theta W_t N_t^b$, where $\bar{p}_t$ is the average approval rate.

Appendix F. Microfoundation of Approval and Default Schedules

F.1. Approval Function

Assume that applicant creditworthiness x follows a logistic distribution with location μ_x and scale $s_x > 0$,

$$F_x(x) = \frac{1}{1 + \exp\left(-\frac{x - \mu_x}{s_x}\right)}. \quad (\text{A35})$$

The distribution F_x is the distribution of creditworthiness used in the baseline application block. Suppose that underwriting stringency q shifts the latent approval threshold according to

$$\tau_j(q) = \bar{\tau}_j + \gamma_j q, \quad \gamma_j > 0. \quad (\text{A36})$$

An application is approved when $x \geq \tau_j(q)$. The approval probability is therefore

$$\begin{aligned} p_j(q) &= 1 - F_x(\tau_j(q)) \\ &= \frac{1}{1 + \exp\left(\frac{\bar{\tau}_j - \mu_x}{s_x}\right) \exp\left(\frac{\gamma_j}{s_x} q\right)} = \frac{1}{1 + b_{0,j} \exp(b_{1,j} q)}. \end{aligned} \quad (\text{A37})$$

The reduced-form parameters satisfy

$$b_{0,j} = \exp\left(\frac{\bar{\tau}_j - \mu_x}{s_x}\right), \quad b_{1,j} = \frac{\gamma_j}{s_x} > 0. \quad (\text{A38})$$

Hence, the logistic creditworthiness distribution and the affine approval threshold deliver exactly the approval function used in the main text,

$$p_j(q) = \frac{1}{1 + b_{0,j} \exp(b_{1,j} q)}, \quad p'_j(q) = -b_{1,j} p_j(q) [1 - p_j(q)] < 0. \quad (\text{A39})$$

Default Condition and Portfolio Default Consider a borrower approved by sector j . Repayment capacity in the following period is

$$Y_{i,t+1} = \bar{Y}_j \exp(\rho_j x_{i,t}) \varepsilon_{i,t+1}, \quad \rho_j > 0, \quad (\text{A40})$$

where $x_{i,t}$ denotes borrower creditworthiness and $\varepsilon_{i,t+1}$ is an idiosyncratic income shock. A borrower defaults when realized repayment capacity falls below a fixed threshold $\underline{Y}_j^D$. I assume that this threshold is independent of the contractual loan rate, consistent with the evidence in Bosshardt et al. (2025).

Under the lower-tail assumption for the income shock, the borrower-level default probability can be written as

$$g_j(x) = \bar{d}_j \exp(-\lambda_j x), \quad \lambda_j > 0. \quad (\text{A41})$$

The underlying repayment-capacity parameters, the distress threshold, and the income shock distribution are summarized by $\bar{d}_j$ and λ_j .²⁹ Since $\lambda_j > 0$, borrowers with higher creditworthiness have lower default risk.

Underwriting affects the default rate by changing the composition of approved loans. In the baseline model, application preference shocks are independent of borrower creditworthiness, so changes in application volume do not change the underlying distribution of x faced by each sector. Underwriting stringency instead determines which borrowers in that distribution receive credit. If sector j approves borrowers with $x \geq \tau_j(q)$, the default rate among originated loans is

$$d_j^{\text{exact}}(q) = \mathbb{E} [g_j(x) \mid x \geq \tau_j(q)]. \quad (\text{A42})$$

A higher underwriting standard raises the approval threshold and removes borrowers with lower creditworthiness from the originated pool. Since borrower-level default risk decreases with creditworthiness, the portfolio default rate also decreases with underwriting stringency.

The exact portfolio default rate determines how borrower selection translates underwriting into default risk, but its curvature depends on the full distribution of approved borrowers. The calibration disciplines the sectoral default level and the local default gradient rather than this curvature. I therefore parameterize the default schedule with a constant semi-elasticity, which gives

$$d_j(q) = \delta_{0,j} \exp(-\delta_{1,j} q), \quad \delta_{1,j} > 0. \quad (\text{A43})$$

²⁹Over the relevant range, assume $\Pr(\varepsilon_{i,t+1} \leq z) = \pi_j z^{\xi_j}$ as in Toda (2012). Combining this condition with the repayment-capacity equation and the default threshold gives $\bar{d}_j = \pi_j (\underline{Y}_j^D / \bar{Y}_j)^{\xi_j}$ and $\lambda_j = \xi_j \rho_j$. Thus the primitive parameters enter borrower default through these two objects.

Here, $\delta_{0,j}$ determines the default level and $\delta_{1,j}$ determines the response of default to underwriting around the steady state.

Appendix G. Calibration Details

G.1. Calibration of Underwriting, Approval, Default, and Screening Costs

This appendix describes how the underwriting index and the approval and default schedules are disciplined quantitatively. The calibration uses a common observable applicant-quality scale for the approval and default blocks. The approval block measures the location and slope of the underwriting standard from application decisions. The default block measures how subsequent default risk varies with borrower quality and maps that gradient into the local portfolio-default schedule. Screening costs and effective default-loss retention are then recovered from the lender optimality conditions.

G.1.1. Applicant Quality Index

Let r_i denote the observable applicant-risk index used in the application analysis in Section 2.3.2, where higher values indicate riskier observable applicant profiles. The index is constructed as the first principal component of applicant income, debt-to-income ratio, and loan amount. The underlying variables are standardized in the pooled sample of bank and NBFI applications, including approved and denied applications, and a single set of principal-component loadings is estimated in that pooled sample and applied to every observation.

For the structural calibration, it is convenient to express the same underlying index as applicant quality rather than applicant risk. I therefore define

$$x_i = -r_i, \tag{A44}$$

so that higher x_i denotes a safer observable applicant profile. No lender-specific, market-specific, year-specific, or approval-specific restandardization is performed. The approval and default calibration blocks therefore use the same underlying principal-component score and differ from the empirical application-risk index only by the sign normalization in (A44).

This applicant-quality index should not be confused with the separate originated-loan risk index used in the loan-performance decomposition in Section 2.1, which is constructed from credit score, debt-to-income ratio, and loan-to-value ratio.

G.1.2. Classification-Based underwriting standard

Let g index a lender, county, and year cell. Let A_{ig} equal one when application i is approved and zero otherwise. For any candidate policy location c , define the classification score

$$Q_g(c) = \sum_{i \in g} [A_{ig} \mathbf{1}\{x_i \geq c\} + (1 - A_{ig}) \mathbf{1}\{x_i < c\}]. \quad (\text{A45})$$

The empirical underwriting-policy location in cell g is

$$\hat{q}_g \in \arg \max_c Q_g(c). \quad (\text{A46})$$

When several adjacent values attain the maximum, I use the midpoint of the maximizing interval. Cells in which the maximum corresponds to approving every applicant or denying every applicant are treated as boundary solutions and are retained only in the diagnostic calculations described below.

The object $\hat{q}_g$ is a classification-based summary of the location of the observed approval policy. It is not interpreted as a literal legal cutoff, and approval decisions do not need to follow a deterministic one-dimensional rule.

To connect $\hat{q}_g$ to the structural underwriting variable, let q_g^* denote the latent structural policy location. I impose the following maintained measurement restriction. In a neighborhood of the calibrated steady state, there exists a sector-specific mapping h_j such that

$$\hat{q}_g = a_j + h_j(q_g^*) + u_g, \quad h_j'(q_g^*) > 0, \quad (\text{A47})$$

where u_g is cell-level measurement error. The mapping is assumed to be locally stable across lender-county-year cells within a sector. The monotonicity restriction means that a cell classified as having a higher policy location is also interpreted as having tighter structural underwriting locally.

Because the level and units of the latent policy variable are otherwise arbitrary, I normalize

$$a_j = 0, \quad h_j'(q_j^*) = 1 \quad (\text{A48})$$

at the sectoral steady state. The model underwriting variable q is therefore measured locally in the same units as the classification-based policy index. This is a maintained measurement assumption rather than a consequence of the one-dimensional classifier.

To match the definition used in the main text, the sectoral steady-state underwriting standard

is the simple mean of the cell-level policy locations

$$q_j = \frac{1}{G_j} \sum_{g: j(g)=j} \widehat{q}_g, \quad j \in \{B, NB\}, \quad (\text{A49})$$

where G_j is the number of admissible lender-county-year cells in sector j . The same equal-weighted definition is used in the quantitative implementation.

The estimated $\widehat{q}_g$ is a simple summary of the location of approval decisions in each lender, county, and year cell. It should not be read as a literal rule that determines every approval decision. I use it as the empirical measure of the underwriting standard, with a higher value indicating tighter underwriting. The model variable q is normalized to use the same local scale.

G.1.3. Approval Schedule

The model approval schedule is

$$p_j(q) = \frac{1}{1 + b_{0,j} \exp(b_{1,j}q)}, \quad b_{1,j} > 0. \quad (\text{A50})$$

I estimate the slope $b_{1,j}$ from the relationship between approval rates and underwriting standards across lender-county-year cells. The regression is estimated separately for banks and NBFIs

$$\log\left(\frac{P_g}{1 - P_g}\right) = \alpha_j - b_{1,j}\widehat{q}_g + \lambda_{\ell(g),c(g)} + \tau_{c(g),t(g)} + \varepsilon_g. \quad (\text{A51})$$

Here, P_g is the approval rate in cell g . The regression includes lender-by-county and county-by-year fixed effects and is weighted by the number of applications in the cell. In the actual estimation, the approval rate and the underwriting measure are constructed from different random halves of the applications in each cell.³⁰

Let P_j denote the application-weighted approval rate in sector j . Once $b_{1,j}$ is estimated, $b_{0,j}$ is chosen so that the model matches the observed sectoral approval rate at q_j

$$p_j(q_j) = P_j. \quad (\text{A52})$$

³⁰The split sample avoids using the same applications to construct both the approval rate and the underwriting measure. It also addresses sampling error in the cell-level cutoff. We pair the cutoff from one half with the approval rate from the other half, use the cross-half cutoff covariance to correct attenuation, repeat the split twenty times, and use the median valid estimate.

This gives

$$b_{0,j} = \left(\frac{1}{P_j} - 1 \right) \exp(-b_{1,j}q_j). \quad (\text{A53})$$

G.1.4. Default Schedule

The model needs a measure of how default risk changes when underwriting becomes tighter. I obtain this slope from loan-level performance. The applicant-quality index is observed for originated loans, so the data directly show how subsequent default risk varies with borrower quality. A tighter underwriting standard shifts originations toward safer applicants. I therefore use the loan-level relationship between borrower quality and default as the empirical slope of the default schedule. In the benchmark, one unit of tighter underwriting is assumed to shift average originated borrower quality by one unit on the same standardized scale.

I estimate the loan-level default gradient using originated loans matched between HMDA and the GSE loan-performance data. The outcome is serious delinquency within eight quarters of origination. For each sector, I estimate

$$\mathbb{E} [\mathbf{1} \{\text{serious delinquency within eight quarters}\} \mid x_i, Z_i, j] = \exp(\alpha_j - \beta_j x_i + Z_i' \Gamma_j). \quad (\text{A54})$$

The baseline specification is estimated by Poisson quasi-maximum likelihood. The controls in Z_i are the same controls used in the loan-performance analysis. The coefficient β_j is estimated separately for banks and NBFIs. A positive β_j means that safer borrowers have lower subsequent default risk.

The model default schedule is

$$d_j(q) = \delta_{0,j} \exp(-\delta_{1,j}q). \quad (\text{A55})$$

Under the benchmark unit pass-through described above, the estimated loan-level gradient gives the slope of the model default schedule.³¹

$$\delta_{1,j} = \beta_j. \quad (\text{A56})$$

Appendix H varies this pass-through assumption and recalibrates the default schedule under

³¹To see the mapping, write originated borrower quality locally as $X_{ij}(q) = \mu_j + \vartheta_j q + U_{ij}$. Combining this with the estimated loan-level relation $\Pr(\text{serious delinquency} \mid X_{ij}, j) \propto \exp(-\beta_j X_{ij})$ gives $d_j(q) \propto \exp(-\vartheta_j \beta_j q)$. Hence $\delta_{1,j} = \vartheta_j \beta_j$. The benchmark sets $\vartheta_j = 1$, so $\delta_{1,j} = \beta_j$. Because β_j is estimated separately for banks and NBFIs, the mapping is sector specific.

the alternative values.

The level of the default schedule is set to match the observed sectoral default rate. The empirical outcome is measured over eight quarters, while the model is quarterly, so I divide the observed eight-quarter serious-delinquency rate by eight to obtain the quarterly target. The corresponding annualized default rates are 1.76 percent for banks and 4.51 percent for NBFIs.

Let $D_{j,q}$ denote the quarterly default target. I choose $\delta_{0,j}$ so that

$$d_j(q_j) = D_{j,q}. \quad (\text{A57})$$

Therefore,

$$\delta_{0,j} = D_{j,q} \exp(\delta_{1,j} q_j). \quad (\text{A58})$$

The default response used in the underwriting condition is

$$d'_j(q_j) = -\delta_{1,j} D_{j,q}. \quad (\text{A59})$$

G.2. Estimating the Bank–NBFI Substitution Elasticity

I estimate the substitution elasticity φ using the allocation of mortgage applications between banks and NBFIs in HMDA. I first construct county–year observations for each sector.

The estimating equation follows from the ratio of sectoral application shares in equation (17), while setting the approval elasticity to $\eta = 1$. I estimate

$$\log\left(\frac{L_{B,c,t}^{app}}{L_{NB,c,t}^{app}}\right) - \log\left(\frac{p_{B,c,t}}{p_{NB,c,t}}\right) = \alpha_c + \tau_t - \varphi \log\left(\frac{R_{B,c,t}^\ell}{R_{NB,c,t}^\ell}\right) + u_{c,t}, \quad (\text{A60})$$

where α_c and τ_t denote county and year fixed effects. I interpret the negative of the coefficient on the relative loan rate as a conditional calibration estimate of φ under the restriction $\eta = 1$. It is not intended to provide a causal estimate of the loan-demand.

HMDA reports interest rates for originated loans. I adjust these rates for borrower and loan characteristics using a hedonic regression that controls for income, loan amount, LTV, DTI, loan maturity, and loan purpose. I then calculate average adjusted interest rates within each sector, county, and year. The estimated results show that $\varphi = 6.744$, with a standard error of 2.114, clustered at the county level. Without fixed effects and with only county fixed effects, the estimated values of φ are 11.15 and 9.33, respectively.

I additionally examine sensitivity by repeating the same fixed-effects OLS regression for alternative fixed values of η between zero and two. The resulting estimates range from 5.28 to

8.20, showing that the estimated values remain within the range suggested by Paul et al. (2024).

G.3. Additional Evidence on Default Loss Internalization

I examine whether differences in loan pricing and subsequent loan performance support the calibrated ordering of default loss internalization between banks and NBFIs, κ_B and κ_{NB} . Using GSE loan-level data, I compare how interest rates and subsequent defaults vary with observable borrower risk across the two sectors.

I use the riskiness proxy that I constructed in Section 2.2, which is an observable borrower-risk index from credit scores, loan-to-value ratios, and debt-to-income ratios. The index loads negatively on credit scores and positively on LTV and DTI ratios, so that a higher value corresponds to a riskier observable borrower profile relative to the local origination cohort.

The estimated regression model is

$$Y_{i,t} = \beta_1 \text{Risk}_i + \beta_2 \mathbb{I}[\text{NBFI}_i] \times \text{Risk}_i + \alpha_{m_i,t} + \lambda_{s_i,t} + \phi_{c_i} + \varepsilon_{i,t},$$

where $Y_{i,t}$ is either the interest rate or an indicator for default for loan i originated in quarter t . Default is defined as a loan becoming 60 or more days past due within this horizon, as in the main empirical analysis. The variable Risk_i is the borrower-risk index, and $\mathbb{I}[\text{NBFI}_i]$ equals one for loans originated by NBFIs. The coefficient β_1 measures the response to borrower risk for banks, while β_2 measures the difference in this response between NBFIs and banks. The corresponding NBFI slope is $\beta_1 + \beta_2$.

MSA-by-quarter fixed effects, $\alpha_{m_i,t}$, absorb local demand conditions and market-level shocks common to loans originated in the same MSA and quarter. Sector-by-quarter fixed effects, $\lambda_{s_i,t}$, absorb sector-specific differences in average pricing and default levels across origination quarters. In the pricing regressions, these controls account for sector level shifts in funding conditions and interest-rate levels. They also absorb the NBFI main effect, so that the comparison concerns risk sensitivity rather than the average difference between sectors. FICO-by-LTV cell fixed effects, ϕ_{c_i} , are included as a proxy for G-fee following the loan-level price adjustment (LLPA) grid. This helps distinguish differences in risk pricing between banks and NBFIs from differences associated with where their loans fall in that grid. I include the same cell fixed effects in the default regression to compare pricing and loan performance conditional on the same underwriting categories. Standard errors are clustered at the MSA level.

Table A2 shows the results. Columns (1) and (2) indicate that interest rates increase less steeply with borrower risk for NBFIs than for banks. The interaction coefficient is negative and statistically significant at the 1 percent level in both specifications, and its magnitude increases after including

TABLE A2. Borrower Risk, Loan Pricing, and Subsequent Default

	Interest Rate		Default
	(1) Basis points	(2) Basis points	(3) Percentage points
Borrower Risk	12.739*** (0.155)	6.162*** (0.148)	0.616*** (0.022)
NBFI $\times$ Borrower Risk	-0.641*** (0.145)	-0.916*** (0.152)	0.035** (0.017)
MSA-by-quarter fixed effects	✓	✓	✓
Sector-by-quarter fixed effects	✓	✓	✓
FICO-by-LTV (g-fee) fixed effects	–	✓	✓
Observations	34,371,223	34,371,223	34,371,223
MSA clusters	468	468	468
R^2	0.840	0.845	0.046

Notes. The dependent variable is the mortgage interest rate in columns (1) and (2) and an indicator for default within two years of origination in column (3). Interest-rate coefficients and standard errors are reported in basis points, and default coefficients and standard errors in percentage points. Higher values of Borrower Risk indicate riskier observable borrower profiles. Standard errors clustered at the MSA level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. *Sources.* GSE loan-level data and author's calculations.

FICO-by-LTV cell fixed effects. In column (2), a one-unit increase in the risk index is associated with an interest-rate increase of 6.16 basis points for banks and 5.25 basis points for NBFIs. The NBFI pricing slope is therefore 14.87 percent smaller.

Column (3) shows the opposite pattern for default. A one-unit increase in the risk index is associated with an increase in the default probability of 0.616 percentage points for banks and 0.651 percentage points for NBFIs. The NBFI default slope is 5.60 percent larger, and the interaction coefficient is statistically significant at the 5 percent level. Thus, the weaker pricing response of NBFIs is accompanied by a stronger increase in subsequent defaults, rather than a smaller increase in default risk.

Appendix H. Sensitivity Analysis

In this section, I conduct sensitivity analysis by changing key parameter values. First, I vary the elasticity of substitution with respect to the loan rate, φ . The range is borrowed from Paul et al. (2024), which suggests a range from 1 to 12. While Paul et al. (2024) identifies the substitution between bank loans and corporate bonds, it also draws on Buchak et al. (2025), which estimates the elasticity of substitution between information-sensitive loans (bank-retained loans) and information-insensitive loans (securitized loans) at around 4.05. When the elasticity of sub-

stitution is 4, the aggregate default rate still increases, while the increase in total defaults is mild.

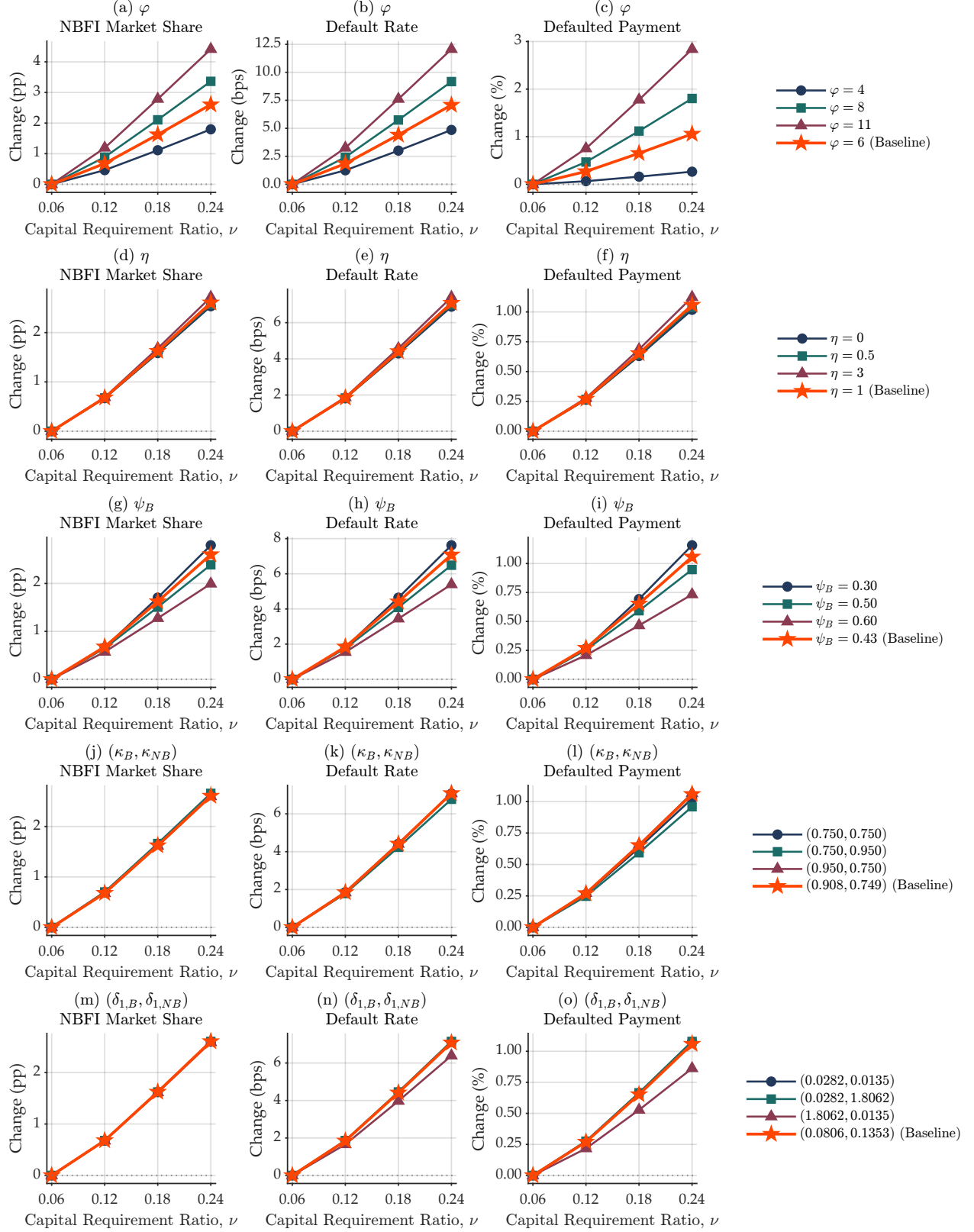
Second, I vary η , which determines how strongly approval probabilities affect borrowers' choice between bank and NBFI credit and, through loan originations, lenders' incentives to screen applicants. The results remain similar across the alternative values of η . Approval probabilities change only modestly as capital requirements increase, and η changes symmetrically across the two sectors. Therefore, the relative approval component that determines the allocation of applications changes little. In addition, each specification is recalibrated to match the same benchmark targets, and the responses are measured relative to its own baseline. Thus, the main results are not sensitive to the strength of the approval-rate channel.

Third, I vary ψ_B , which summarizes the preference toward the banking sector. The effect of capital requirements on aggregate credit risk depends on the initial market shares. As the share of banks increases, the effect of regulation on aggregate credit risk becomes smaller. This is because the amount of applications that migrate to NBFIs depends on the initial shares of NBFIs and banks. If the share of banks is large, in other words, if the share of NBFIs is small, fewer applications migrate to NBFIs, so the effect is smaller.

Fourth, I vary κ_B and κ_{NB} , which determine how much of each sector's default loss is internalized in intermediary profits. The results remain similar across the alternative parameter pairs. Aggregate default rates and defaulted payments are measured from the borrower side and therefore do not mechanically depend on κ_j . Instead, the retention parameters affect aggregate credit risk indirectly through loan pricing, underwriting, balance-sheet adjustment, and the allocation of credit across sectors. In addition, the retained-default-loss term is only one component of lenders' optimality conditions, while loan rates and underwriting adjust endogenously in response to changes in κ_j . Therefore, changes in default retention have only limited effects on the aggregate credit-risk response.

Finally, I vary $\delta_{1,B}$ and $\delta_{1,NB}$, which determine how default probabilities respond to underwriting in each sector. For each parameter pair, I recalibrate $\delta_{0,j}$ so that the benchmark default probability remains unchanged at the calibrated underwriting level. Therefore, this exercise changes the responsiveness of default risk to underwriting rather than its initial level. The resulting differences are generally small because policy-induced changes in underwriting remain close to the calibration point, particularly for NBFIs. As a result, even substantially different default-schedule slopes operate on relatively small changes in underwriting, while the effects of capital requirements on sectoral migration and aggregate credit risk remain qualitatively the same.

FIGURE A13. Sensitivity Analysis



Appendix I. Extensions

I.1. Two-Type Borrowers

I relax the baseline no-sorting assumption by introducing a latent safe type S and a risky type R . Borrowers observe their own type, but lenders only know the population share of each type and apply the same approval schedule to both types.³²

The two borrower types differ in their responses to loan rates and approval probabilities. Given the loan rate and approval probability offered by each lender, a type $T \in \{S, R\}$ borrower allocates applications according to

$$\omega_{T,j} = \frac{\psi_j p_j(q_j)^{\eta_T} (R_j^\ell)^{-\varphi_T}}{\sum_{k \in \{B, NB\}} \psi_k p_k(q_k)^{\eta_T} (R_k^\ell)^{-\varphi_T}}, \quad (\text{A61})$$

where η_T determines the response to approval probabilities and φ_T determines the response to loan rates. Therefore, risky and safe borrowers can sort differently across banks and NBFIs even though lenders cannot observe their types.

Because lenders apply the same approval schedule to both types, the share of type T among lender j 's originated loans is

$$\pi_{T,j} = \frac{\mu_T \omega_{T,j}}{\sum_{T' \in \{S, R\}} \mu_{T'} \omega_{T',j}}. \quad (\text{A62})$$

Thus, $\pi_{T,j}$ can differ from the population share μ_T because the two types can choose lenders differently. The default rate also differs across borrower types. I assume

$$d_{T,j}(q_j) = \delta_{0,T,j} \exp(-\delta_{1,j} q_j), \quad \delta_{0,R,j} = \lambda_R \delta_{0,S,j}, \quad (\text{A63})$$

where $\lambda_R > 1$ captures the higher underlying default risk of risky borrowers. The portfolio default rate faced by lender j is then

$$\bar{d}_j = \pi_{S,j} d_{S,j}(q_j) + \pi_{R,j} d_{R,j}(q_j). \quad (\text{A64})$$

This extension allows changes in loan rates and approval probabilities to affect not only the allocation of applications across lenders but also the unobserved risk composition of each lender's

³²I set the share of risky borrowers μ_R to be 0.732 using the complement of the high-creditworthiness population share reported in Agarwal et al. (2024, Table II).

originated loans.

Because borrower type is unobservable, lender j offers one loan rate and one underwriting stringency and uses the endogenous origination weights $\pi_{T,j}$ in both conditions

$$R_j^\ell = \frac{\epsilon}{\epsilon - 1} \frac{C_j}{1 - \kappa_j \bar{d}_j},$$

$$0 = \sum_{T \in \{S, R\}} \pi_{T,j} \left[\underbrace{-\kappa_j d'_{T,j}(q_j) R_j^\ell}_{\text{reduction in expected default losses}} - \underbrace{s'_j(q_j)}_{\text{marginal underwriting cost}} + \underbrace{\left((1 - \kappa_j d_{T,j}) R_j^\ell - C_j \right) (1 + \eta) \frac{p'_j(q_j)}{p_j(q_j)}}_{\text{change in approved lending}} \right].$$

For banks, marginal cost is $C_B = R + s_B + \frac{\partial c_B(L_B, E; \nu)}{\partial L_B}$, and for NBFIs, marginal cost is $C_{NB} = R^F + s_{NB}$.

With two types of borrowers, risky borrowers migrate more to the NBFI sector. NBFIs therefore also have a direct incentive to adjust their underwriting stringency because their underwriting choice now depends on the share of risky borrowers in their applicant pool.

FIGURE A14. Effect of higher capital requirements with Two-Type Borrowers

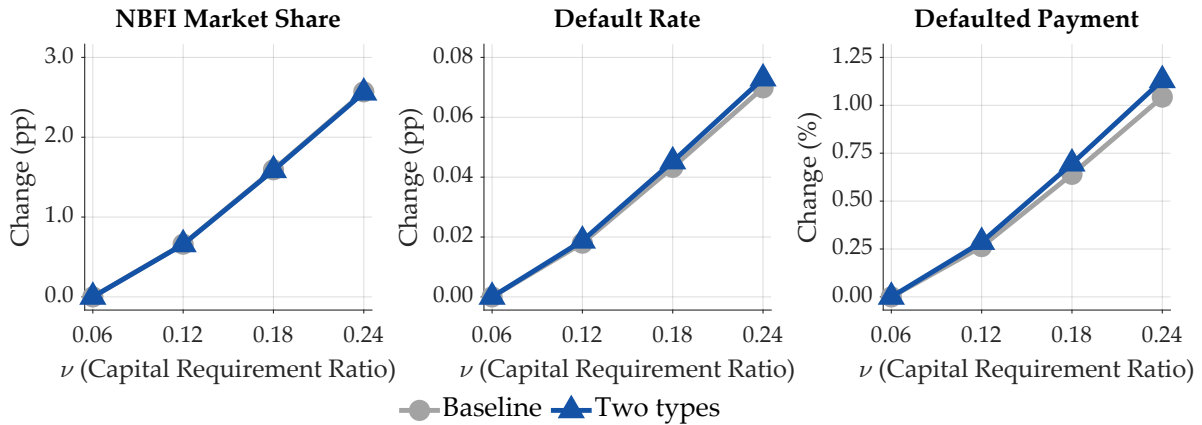


Figure A14 compares the baseline results with those in the two-type borrower economy. Two findings are noteworthy. First, the increases in market share are almost identical across the two economies. However, aggregate credit risk, in terms of both the default rate and defaulted payments, increases slightly more in the two-type economy. This difference comes from changes in borrower composition and the underwriting response of NBFIs. As more risky borrowers, who are more sensitive to lending rates and have exogenously higher default rates, migrate to

NBFIs, NBFIs respond to the change in the composition of their applications and tighten their underwriting stringency. However, this response is not strong enough to offset the increase in the share of risky borrowers. As a result, the average default rate of NBFI loans rises despite tighter underwriting. The market share remains almost identical, but the similar market-share responses conceal different changes in borrower composition and credit risk. Thus, tighter capital requirements increase aggregate credit risk in both economies, with a slightly larger increase in the two-type economy.

I.2. Bank-NBFI Funding Channel

I add a committed bank credit line to the baseline NBFI funding structure, as documented in Jiang (2023); Acharya et al. (2025); Xu (2025). Let $\bar{H}$ denote the total commitment and let $H \leq \bar{H}$ denote the amount drawn by the NBFI. The interest rate on the drawn amount is $R_H = R^F + s_H$, where R^F is the NBFI wholesale funding rate and s_H is the contractual spread. The draw changes the balance sheets of banks and NBFIs and the bank's regulatory exposure according to

$$L_{NB} = F_{NB} + H + E_{NB}, \quad L_B + H + S_B = D_B + E_B, \quad RWA_B = L_B + \omega_D H + c_U (\bar{H} - H).$$

I set $\omega_D = 1$ and use $c_U = 0.50$ for commitments with an original maturity greater than one year.

The credit line allows NBFIs to diversify their funding and reduce their reliance on wholesale funding. I represent this benefit as

$$\mathcal{B}_H(H, L_{NB}) = \theta_H H - \frac{\eta_H}{2} \frac{H^2}{L_{NB}}.$$

Because one dollar of H replaces one dollar of F_{NB} , the common funding rate R^F cancels from the draw decision. The NBFI therefore draws the line until the marginal funding benefit equals the additional spread

$$\underbrace{\theta_H - \eta_H \frac{H}{L_{NB}}}_{\text{marginal funding benefit}} = \underbrace{s_H}_{\text{marginal funding cost}}.$$

For an interior draw, this condition gives

$$\frac{H}{L_{NB}} = \frac{\theta_H - s_H}{\eta_H}.$$

I calibrate the facility at $\nu = 0.06$ using three separate moments reported by the Federal Reserve. First, I set $H/L_{NB} = 0.05$ to match the finding that outstanding bank loans to private-credit vehicles are approximately 5 percent of the U.S. private-credit industry. Conditional on η_H and s_H , this moment determines θ_H . Second, I set $H/\bar{H} = 0.56$ to match the 44 billion drawn from 79 billion of revolving commitments to private-credit vehicles. This moment determines the commitment $\bar{H}$ and implies $\bar{H}/L_{NB} = 0.0893$ at the benchmark. Third, I set $R_H^{ann} - 1 = 0.066$ to match the average interest rate on bank facilities to private debt funds. In quarterly units, this gives $R_H = (1.066)^{1/4}$ and determines $s_H = R_H - R^F$. I set $\eta_H = 0.010$ as the curvature governing how the marginal funding benefit declines with the draw share.

For the bank, the drawn amount earns R_H but replaces a safe asset earning R and increases the capital cost through RWA_B . The calibrated facility produces a positive net margin for the bank at $\nu = 0.06$. I therefore treat $\bar{H}$ and s_H as the terms of an existing contract. After ν changes, the NBFI chooses H , while the bank honors any draw up to $\bar{H}$. Thus, H is the only additional endogenous choice after the contract date.

FIGURE A15. Effect of higher capital requirements with Bank-NBFI Credit Line

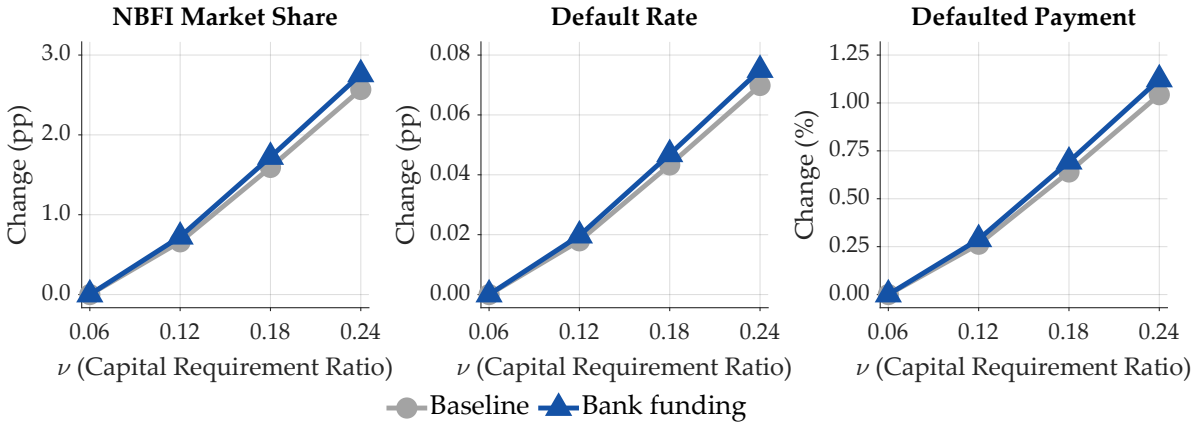


Figure A15 compares the baseline with the committed bank credit line. When the capital requirement rises, banks reduce lending and borrowers shift toward NBFIs. The committed line lowers the effective funding cost of NBFIs by replacing part of their wholesale funding with a more stable funding source. NBFIs can therefore absorb more of the lending displaced from banks. As a result, the increases in their market share, the aggregate default rate, and total defaulted payments become larger.

I.3. Loan Sales and Balance-Sheet Retention

I relax the baseline assumption that banks retain all newly originated loans. Let O_B denote total bank originations, H_B retained loans, and X_B sold loans. The allocation of originated loans and the bank's regulatory exposure are

$$O_B = H_B + X_B, \quad \text{RWA}_B = H_B + \rho_X X_B.$$

A retained loan carries full regulatory exposure, whereas a sold loan leaves the fraction ρ_X of that exposure with the bank.

I represent the cost of selling loans as

$$\Phi_X(X_B) = m_X X_B + \frac{\zeta_X}{2\bar{O}_B} (X_B - \bar{X}_B)^2, \quad \bar{X}_B = \bar{x}_B \bar{O}_B.$$

The model does not determine a separate secondary-market price or investor demand. Instead, $\Phi_X(X_B)$ summarizes the cost of placing loans outside the bank's balance sheet. The marginal sale cost is

$$\Phi'_X(X_B) = m_X + \frac{\zeta_X}{\bar{O}_B} (X_B - \bar{X}_B).$$

Retaining one additional loan requires funding at the return R and creates one unit of regulatory exposure. Selling the loan incurs the marginal sale cost and leaves the residual exposure ρ_X . The bank therefore chooses X_B according to

$$R + c_B^{\text{RWA}} = \Phi'_X(X_B) + \rho_X c_B^{\text{RWA}},$$

or equivalently,

$$\Phi'_X(X_B) = R + (1 - \rho_X) c_B^{\text{RWA}}.$$

The bank sells loans until the marginal sale cost equals the funding cost and capital cost avoided by not retaining them.

At $\nu = 0.06$, I impose the benchmark sale share $\bar{X}_B/\bar{O}_B = 0.40$ and choose $m_X = R + (1 - \rho_X) c_B^{\text{RWA}}$ so that the benchmark allocation satisfies the sale condition. I set $\rho_X = 0.20$ as a maintained assumption about the residual regulatory exposure of sold loans. I set $\zeta_X = 0.154$ to govern how rapidly the marginal sale cost increases when sales move away from their benchmark level.

FIGURE A16. Effect of higher capital requirements when Banks can sell Loans

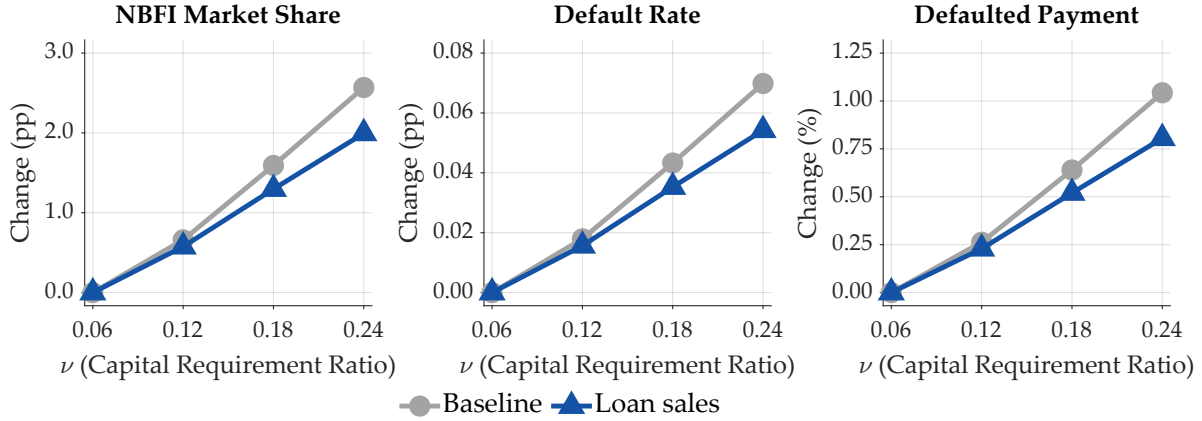


Figure A16 shows the results. The effects of capital regulation on market share and aggregate credit risk are mitigated. The reason is that when the capital requirement rises, the marginal capital cost of retaining loans increases. Banks therefore sell a larger share of their originations and reduce their regulatory exposure. This margin limits the contraction in bank lending and weakens the reallocation of credit toward NBFIs. The aggregate default rate and defaulted payments still increase, but the increases are smaller than in the baseline without loan sales.

I.4. Reapplication after Bank Rejection

I relax the baseline assumption that denied applicants do not submit another application. Applicants who are denied, especially by banks, draw a search cost and make one additional application to an NBFI. Some borrowers rejected by banks still satisfy the NBFI underwriting cutoff and can obtain the loans under the NBFI rules. I assume that borrowers who fail the NBFI cutoff cannot obtain bank credit, so reapplication occurs only from banks to NBFIs. This is because applicants who are rejected by NBFIs cannot get the loans from the banks in the model. Finally, I do not allow additional applications.

Let c denote the search cost and V the expected benefit from an NBFI loan conditional on bank rejection. An applicant reapplies when this expected benefit exceeds the search cost. When $V > 0$, the fraction of bank-denied applicants who reapply is

$$\rho = \Pr(c \leq V). \quad (\text{A65})$$

When $V \leq 0$, there is no reapplication. Search costs are independent of creditworthiness and application preferences. Applicants observe that they were rejected by a bank but do not observe

their exact creditworthiness when deciding whether to reapply. The search cost is incurred upon submitting the second application even if the NBFi rejects it. I treat this cost as nonpecuniary and include it in borrower utility.

I parameterize search costs using the mean additional-search cost estimated by Agarwal et al. (2024), which corresponds to an annual interest-rate spread of 29.7 basis points. I map these costs into the present value of additional mortgage payments. This conversion is used only to express the search cost in model units and does not add a spread to the loan rate paid by borrowers.³³ The reapplication rate is determined in equilibrium. As the search cost becomes sufficiently large, the model becomes isomorphic to the baseline where the reapplication is absent.

Because reapplication is an additional decision made only after a bank rejection in the model, reapplication changes the composition of the NBFi applicant pool. Let $p_B = p_B(q_B)$ and $p_{NB} = p_{NB}(q_{NB})$ denote approval probabilities for initial applicants. Then, the probability that an applicant rejected by a bank still satisfies the NBFi cutoff is

$$r = \frac{p_{NB} - p_B}{1 - p_B}. \quad (\text{A66})$$

Then, let s_R denote the share of NBFi applications coming from bank-denied borrowers. The observed NBFi approval rate is then

$$\tilde{p}_{NB} = (1 - s_R)p_{NB} + s_R r. \quad (\text{A67})$$

The first term comes from initial NBFi applicants and the second captures the contribution from reapplicants. The NBFi default rate is also determined by the composition of originated loans. NBFis therefore choose loan rates and underwriting standards taking both groups into account. A reapplication creates an additional application record but does not increase the amount the borrower is allowed to borrow.

I introduce reapplication while holding all other parameters at the baseline values. I recompute the steady-state equilibrium as the capital requirement increases from 6 to 24 percent. Each curve is measured relative to the steady state of the same economy at $\nu = 0.06$.

³³I convert the interest-rate spread into the present value of the extra payments a borrower would make on a thirty-year mortgage. I use the baseline NBFi loan rate and the borrower discount factor, assuming no prepayment or refinancing. Search costs are set so that their interest-rate equivalents average 29.7 basis points. This calculation is only used to measure the search cost and loans in the model still last one quarter.

FIGURE A17. Effect of Higher Capital Requirements with Reapplication: Baseline Parameters Held Fixed

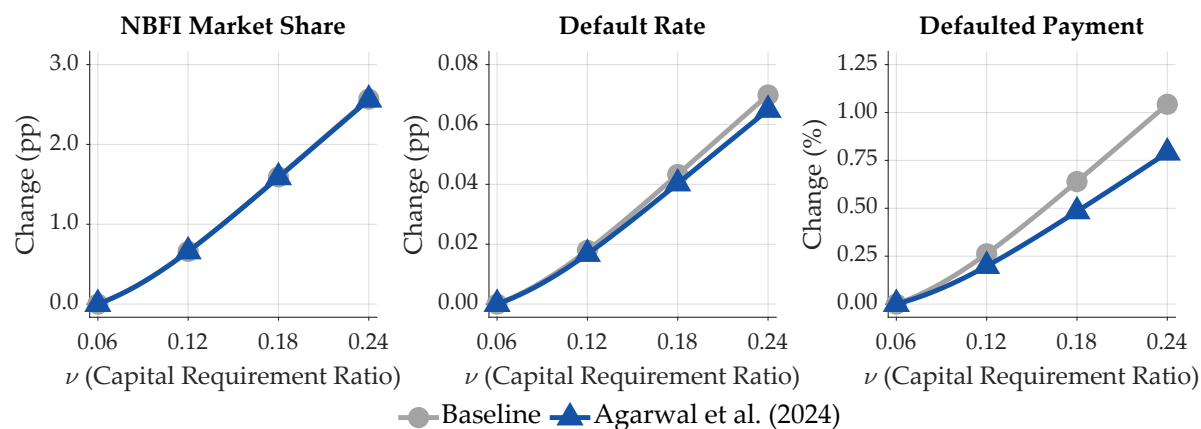


Figure A17 shows that reapplication reduces the aggregate credit risk responses, which does not overturn the results. When the capital requirement increases from 6 to 24 percent, the annualized default rate rises by 0.065 percentage points and total defaulted payments rise by 0.79 percent. These increases are 7.2 and 24.2 percent smaller than in the baseline, respectively.

The reason is that some borrowers who would otherwise remain unfunded after the first application can obtain NBFI credit after a bank rejection, even before regulation tightens. A subsequent shift in initial applications toward NBFIs therefore brings *fewer* previously unfunded borrowers into the originated loan pool compared to the baseline. Under this setting, the model generates a quarterly reapplication rate among bank-denied applicants of 25.5 percent at $\nu = 0.06$, rising to 26.1 percent at $\nu = 0.24$. Applying each quarterly rate over a year gives annual probabilities of 69.3 and 70.2 percent, broadly comparable to the 64.1 percent SCF-based proxy. This comparison supports the quantitative plausibility of the search-cost calibration and shows that the main results survive even with substantial reapplication.

A concern with this exercise is that the observed approval and default rates used as calibration targets may already include reapplication. I therefore recalibrate the model to the same targets while allowing for reapplication with the search cost introduced above. NBFI moments now include both initial applicants and reapplicants, and the application allocation counts both initial applications and reapplications. I then vary the capital requirement while holding the recalibrated parameters fixed.

FIGURE A18. Effect of Higher Capital Requirements with Reapplication: Recalibrated to the Benchmark Targets

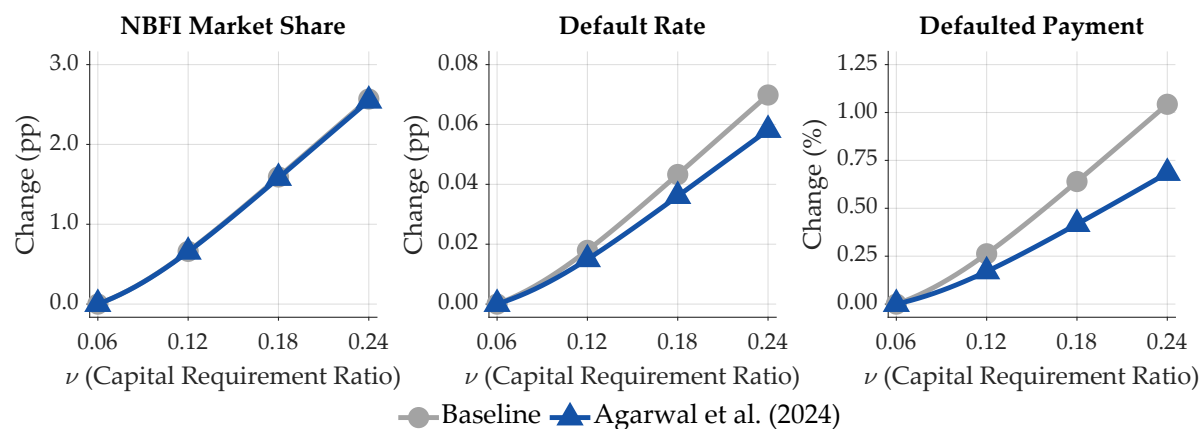


Figure A18 shows the results after recalibration. When the capital requirement increases from 6 to 24 percent, the annualized default rate rises by 0.0581 percentage points and total defaulted payments rise by 0.6839 percent. These increases are 16.8 and 34.4 percent smaller than in the baseline, respectively. They are also smaller than in the fixed-parameter exercise.